

1. INTRODUCTION TO MANAGEMENT

INTRODUCTION TO MANAGEMENT:

Management acts as a guide to a group of people working in the organization and coordinating their efforts, towards the attainment of the common objective.

In other words, it is concerned with optimally using 5M's, i.e. men, machine, material, money and methods and, this is possible only when there proper direction, coordination and integration of the processes and activities, to achieve the desired results.

CONCEPT OF MANAGEMENT:

"Management is the coordination of all resources through the process of planning, organizing, directing and controlling in order to attain stated goals."

1. "To manage is to forecast, to plan, to organize, to command, to co-ordinate and to control." — Henry Fayol. It attempts to describe management in terms of what a manager does, and not what management is.

2. "Management is a multipurpose organ that manages a business and manages manager, and manages worker and work."— P. Drucker: The Practice of Management

(i) Managing a business;

(ii) Managing manager; and

(iii) Managing workers and work.

Even if one is omitted, we would not have management anymore and we also would not have a business enterprise or an industrial society. According to P. Drucker, the manager has to balance and harmonies three major functions of the business enterprise.

Hence, a manager is a dynamic and life-giving element in every business. Without efficient management we cannot secure the best allocation and utilization of human, material and financial resources.

Meaning:

Management is a process of planning, decision making, organizing, leading, motivation and controlling the human resources, financial, physical, and information resources of an organization to reach its goals efficiently and effectively.

Definition: "Management can be defined as the process of administering and controlling the affairs of the organization, irrespective of its nature, type, structure and size".

It is an act of creating and maintaining such a business environment wherein the members of the organization can work together, and achieve business objectives efficiently and effectively.

“Management is the art of getting things done through others.” -Mary Parker Follett

“Management is the art of getting things done through and with people in formally organized groups.” -Koontz

Nature of Management:

- Universal: All the organizations, whether it is profit-making or not, they require management, for managing their activities. Hence it is universal in nature.
- Goal-Oriented: Every organization is set up with a predetermined objective and management helps in reaching those goals timely, and smoothly.
- Continuous Process: It is an ongoing process which tends to persist as long as the organization exists. It is required in every sphere of the organization whether it is production, human resource, finance or marketing.
- Multi-dimensional: Management is not confined to the administration of people only, but it also manages work, processes and operations, which makes it a multi-disciplinary activity.
- Group activity: An organization consists of various members who have different needs, expectations and beliefs. Every person joins the organization with a different motive, but after becoming a part of the organization they work for achieving the same goal. It requires supervision, teamwork and coordination, and in this way, management comes into the picture.
- Dynamic function: An organization exists in a business environment that has various factors like social, political, legal, technological and economic. A slight change in any of these factors will affect the organization's growth and performance. So, to overcome these changes management formulates strategies and implements them.
- Intangible force: Management can neither be seen nor touched but one can feel its existence, in the way the organization functions.
- Management is an art as well as science: Management is a science because it has developed certain principles which are of universal application. But the results of management depend upon the personal skill of managers and in this sense management is an art.
- Management is a social process: Management consists of getting things done through others. This involves dealings with people. The efforts of the human beings have to be directed, coordinated and regulated by management in order to achieve the desired results.

Importance of Management:

It helps in Achieving Group Goals: It arranges the factors of production, assembles and organizes the resources, integrates the resources in effective manner to achieve goals. It directs group efforts towards achievement of pre-determined goals.

Management Increases Efficiency: A manager increases efficiency through the optimum utilisation of all the resources such as man, machine, material and money.

Management Creates a Dynamic Organisation: Every organisation works in an ever changing environment. To face the changing environment, many changes need to be made in the organisation as well. But people resist changes. Manager creates a favorable environment through introducing employees to the benefits of adapting to changes.

Management Helps in Achieving Personal Objectives: Every employee wants to get suitable remuneration, a share in profit, participation in management, promotion, etc. in the form of his personal objectives. This objective can only be achieved, if they work while using their full abilities.

Managers make employees able through motivation, good leadership and open communication. As a result, they attain their individual objectives.

Management Helps in the Development of Society: Management has some responsibility towards society. Managers by fulfilling their social responsibilities help in the development of society. These responsibilities include providing employment opportunities, preventing environment from getting polluted, making available good quality products at a reasonable price, etc.

Industrial peace: Management helps to develop a healthy environment within the organisation by promoting a two-way communication between superiors and subordinates. This develops cordial relations between them and promotes industrial peace and harmony.

Builds competitive strength: In today's competitive environment quality of performance determinants the profitability of the organisation. Sound management enables the enterprise to achieve higher levels of productivity and profitability, which in turn build competitive strength.

Improves standard of living: A sound management helps to improve standard of living of both-the workers and consumers by improving its productivity and profitability. Due to increased profitability, workers get higher wages and consumers get better quality products at lower prices.

Initiative and innovation: Management is a group activity. A good manager works with a common consensus and creates an environment in which everyone get as an opportunity to express his/her views. This gives an opportunity to subordinates to come forward with innovative ideas.

Minimizes wastages: Along with ensuring judicious use of resources, a good management also aims at cutting down wastage, both human as well as non-human management plans, organises, direct and controls the activities of the organisation and thereby helps in controlling wastage.

FUNCTIONS OF MANAGEMENT:

Planning: It is the first and foremost function of management, i.e. to decide beforehand what is to be done in future. It encompasses formulating policies, establishing targets, scheduling actions and so forth.

Organizing: Once the plans are formulated, the next step is to organise the activities and resources, as in identifying the tasks, classifying them, assigning duties to subordinates and allocating the resources.

Staffing: It involves hiring personnel for carrying out various activities of the organization. It is to ensure that the right person is appointed to the right job.

Directing: It is the task of the manager to guide, supervise, lead and motivate the subordinates, to ensure that they work in the right direction, so far as the objectives of the organization are concerned.

Controlling: The controlling function of management involves a number of steps to be taken to make sure that the performance of the employees is as per the plans. It involves establishing performance standards and comparing them with the actual performance. In case of any variations, necessary steps are to be taken for its correction.

Coordination is an important feature of management which means the integration of the activities, processes and operations of the organization and synchronization of efforts, to ensure that every element of the organization contributes to its success.

EVALUATION OF SCIENTIFIC MANAGEMENT

Scientific management is a theory of management that analyzes and synthesizes workflows. Its main objective is improving economic efficiency, especially labour productivity. It was one of the earliest attempts to apply science to the engineering of processes and to management. Scientific management is sometimes known as Taylorism after its founder, Frederick Winslow Taylor.

Taylor began the theory's development in the United States during the 1880s and '90s within manufacturing industries, especially steel. Its peak of influence came in the 1910s;[2] Taylor died in 1915 and by the 1920s, scientific management was still influential but had entered into competition and syncretism with opposing or complementary ideas.

Although scientific management as a distinct theory or school of thought was obsolete by the 1930s, most of its themes are still important parts of industrial engineering and management today. These include: analysis; synthesis; logic; rationality; empiricism; work ethic; efficiency

and elimination of waste; standardization of best practices; disdain for tradition preserved merely for its own sake or to protect the social status of particular workers with particular skill sets; the transformation of craft production into mass production; and knowledge transfer between workers and from workers into tools, processes, and documentation.

The main elements of the Scientific Management are : "Time studies Functional or specialized supervision Standardization of tools and implements Standardization of work methods Separate Planning function Management by exception principle The use of "slide-rules and similar time-saving devices" Instruction cards for workmen Task allocation and large bonus for successful performance The use of the 'differential rate' Mnemonic systems for classifying products and implements A routing system A modern costing system etc. etc. " Taylor called these elements "merely the elements or details of the mechanisms of management" He saw them as extensions of the four principles of management.

1. The development of a true science
2. The scientific selection of the workman
3. The scientific education and development of the workman
4. Intimate and friendly cooperation between the management and the men.

PRINCIPLES OF SCIENTIFIC MANAGEMENT

Principles of scientific management propounded by Taylor are: 1. Science, Not Rule of Thumb 2. Harmony, Not Discord 3. Mental Revolution 4. Cooperation, Not Individualism 5. Development of each and every person to his or her greatest efficiency and prosperity.

1. Science, Not Rule of Thumb:

In order to increase organisational efficiency, the 'Rule of Thumb' method should be substituted by the methods developed through scientific analysis of work.

Rule of Thumb means decisions taken by manager as per their personal judgments. According to Taylor, even a small production activity like loading iron sheets into box cars can be scientifically planned. This will help in saving time as well as human energy. Decisions should be based on scientific enquiry with cause and effect relationships.

This principle is concerned with selecting the best way of performing a job through the application of scientific analysis and not by intuition or hit and trial methods.

The work assigned to any employee should be observed and analyzed with respect to each element or part thereof and the time involved therein so as to decide the best way of performing that the work and to determine the standard output for same.

2. Harmony, Not Discord:

Taylor emphasized that there should be complete harmony between the workers and the management since if there is any conflict between the two, it will not be beneficial either for the workers or the management.

Both the management and the workers should realize the importance of each other. In order to achieve this state, Taylor suggested complete mental revolution on the part of both management and workers.

It means that there should be complete change in the attitude and outlook of workers and management towards each other. It should always be kept in mind that prosperity for an employer cannot exist for a long time unless it is accompanied by the prosperity of the employees of that organisation and vice versa.

It becomes possible by (a) sharing a part of surplus with workers (b) training of employees, (c) division of work (d) team spirit (e) positive attitude (f) sense of discipline (g) sincerity etc.

Management should always be ready to share the gains of the company with the workers and the latter should provide their full cooperation and hard work for achieving organizational goals. Group action with mutual-trust and understanding should be perfect understanding the focus of working.

This principle requires that there should be perfect understanding between the management and workers and both should feel that they are part of same family. It helps to produce synergy effect since both management and workers work in unison.

For example, in most of the Japanese companies, paternalistic style of management is in practice and there is complete openness between workers and the management. Usually, workers don't go on the strike but, if at all they do so, they just wear a black badge and work even more than the normal hours just to impress upon the management that their focus is on their demands as well as organisational objectives.

3. Mental Revolution:

The technique of Mental Revolution involves a change in the attitude of workers and management towards each other. Both should realize the importance of each other and should work with full cooperation. Management as well as the workers should aim to increase the profits of the organization.

For this the workers should put in their best efforts so that the company makes profit and on the other hand management should share part of profits with the workers. Thus, mental revolution requires a complete change in the outlook of both management and workers. There should be a spirit of togetherness between workers and management.

4. Cooperation, Not Individualism:

This principle is an extension of principle of 'Harmony, not discord' and lays stress on mutual cooperation between workers and the management. Cooperation, mutual confidence, sense of goodwill administration that is often called Fayolism. Mining engineering should prevail among both, managers as well as workers. The intention is to replace internal competition with cooperation.

Both 'Management' and 'Workers' should realize the importance of each other. Workers should be considered as part of management and should be allowed to take part in decision making process of the management. Management should always welcome their suggestions and should also reward them if their suggestions prove to be beneficial for the organisation viz. reduction of costs or increase in production etc.

At the same time, workers should also resist from going on strike or making unnecessary demands from management. Workers should be treated as integral part of organisation and all important decisions should be taken after due consultation with workers. Both of them should visualize themselves as two pillars whose soundness alone can ensure achievement of common goals of the organisation.

Taylor also suggested that there should be proper division of work and responsibility between the two. Management should always guide, encourage and help the workers.

5. Development of each and every person to his or her greatest efficiency and prosperity:

Efficiency of any organisation also depends on the skills and capabilities of its employees to a great extent. Thus, providing training to the workers was considered essential in order to learn the best method developed through the use of scientific approach. To attain the efficiency, steps should be taken right from the process of selection of employees. Employees should be scientifically selected.

The work assigned to each employee should suit his/her physical, mental and intellectual capabilities. Efficient employees produce more to earn more. This ultimately helps to attain efficiency and prosperity for both organisation and the employees.

EVOLUTION OF MODERN MANAGEMENT (OR) HENRY FAYOL'S ADMINISTRATIVE THEORY

Henri Fayol (Istanbul, 29 July 1841 – Paris, 19 November 1925) was a French mining engineer, mining executive, author and director of mines who developed a general theory of business

He and his colleagues developed this theory independently of scientific management but roughly contemporaneously. Like his contemporary, Frederick Winslow Taylor, he is widely acknowledged as a founder of modern management methods.

14 PRINCIPLES OF MANAGEMENT:

The fourteen principles of management created by Henri Fayol are explained below.

1. Division of Work-

Henri believed that segregating work in the workforce amongst the worker will enhance the quality of the product. Similarly, he also concluded that the division of work improves the productivity, efficiency, accuracy and speed of the workers. This principle is appropriate for both the managerial as well as a technical work level.

2. Authority and Responsibility-

These are the two key aspects of management. Authority facilitates the management to work efficiently, and responsibility makes them responsible for the work done under their guidance or leadership.

3. Discipline-

Without discipline, nothing can be accomplished. It is the core value for any project or any management. Good performance and sensible interrelation make the management job easy and comprehensive. Employees good behaviour also helps them smoothly build and progress in their professional careers.

4. Unity of Command-

This means an employee should have only one boss and follow his command. If an employee has to follow more than one boss, there begins a conflict of interest and can create confusion.

5. Unity of Direction-

Whoever is engaged in the same activity should have a unified goal. This means all the person working in a company should have one goal and motive which will make the work easier and achieve the set goal easily.

6. Subordination of Individual Interest-

This indicates a company should work unitedly towards the interest of a company rather than personal interest. Be subordinate to the purposes of an organization. This refers to the whole chain of command in a company.

7. Remuneration-

This plays an important role in motivating the workers of a company. Remuneration can be monetary or non-monetary. However, it should be according to an individual's efforts they have made.

8. Centralization-

In any company, the management or any authority responsible for the decision-making process should be neutral. However, this depends on the size of an organization. Henri Fayol stressed on the point that there should be a balance between the hierarchy and division of power.

9. Scalar Chain-

Fayol on this principle highlights that the hierarchy steps should be from the top to the lowest. This is necessary so that every employee knows their immediate senior also they should be able to contact any, if needed.

10. Order-

A company should maintain a well-defined work order to have a favourable work culture. The positive atmosphere in the workplace will boost more positive productivity.

11. Equity-

All employees should be treated equally and respectfully. It's the responsibility of a manager that no employees face discrimination.

12. Stability-

An employee delivers the best if they feel secure in their job. It is the duty of the management to offer job security to their employees.

13. Initiative-

The management should support and encourage the employees to take initiatives in an organization. It will help them to increase their interest and make them worth.

14. Esprit de Corps-

It is the responsibility of the management to motivate their employees and be supportive of each other regularly. Developing trust and mutual understanding will lead to a positive outcome and work environment.

MOTIVATION THEORIES

1. Maslow's hierarchy of needs

Abraham Maslow postulated that a person will be motivated when his needs are fulfilled. The need starts from the lowest level basic needs and keeps moving up as a lower level need is fulfilled. Below is the hierarchy of needs:

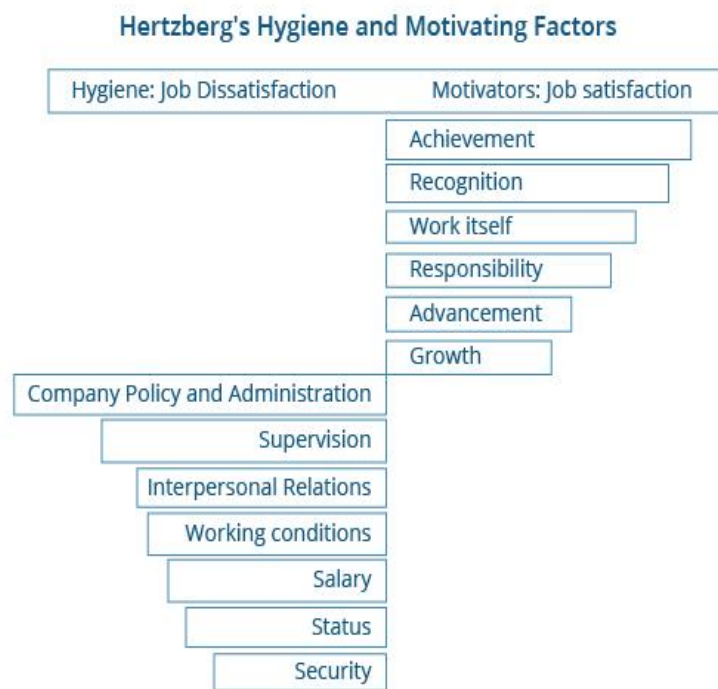


- Physiological: Physical survival necessities such as food, water, and shelter.
- Safety: Protection from threats, deprivation, and other dangers.
- Social (belongingness and love): The need for association, affiliation, friendship, and so on.
- Self-esteem: The need for respect and recognition.
- Self-actualization: The opportunity for personal development, learning, and fun/ creative/ challenging work. Self-actualization is the highest level need to which a human being can aspire.

The leader will have to understand the specific need of every individual in the team and accordingly work to help fulfil their needs.

2. Herzberg's two factor theory

Herzberg classified the needs into two broad categories namely hygiene factors and motivating factors.



Hygiene factors are needed to make sure that an employee is not dissatisfied. Motivation factors are needed for ensuring employee's satisfaction and employee's motivation for higher performance. Mere presence of hygiene factors does not guarantee motivation, and presence of motivation factors in the absence of hygiene factors also does not work.

3. McClelland's theory of needs

McClelland affirms that we all have three motivating drivers, and it does not depend on our gender or age. One of these drives will be dominant in our behaviour. The dominant drive depends on our life experiences.

The three motivators are:

- **Achievement:** a need to accomplish and demonstrate own competence People with a high need for achievement prefer tasks that provide for personal responsibility and results based on their own efforts. They also prefer quick acknowledgement of their progress.
- **Affiliation:** a need for love, belonging and social acceptance People with a high need for affiliation are motivated by being liked and accepted by others. They tend to participate in social gatherings and may be uncomfortable with conflict.
- **Power:** a need for control own work or the work of others People with a high need for power desire situations in which they exercise power and influence over others. They

aspire for positions with status and authority and tend to be more concerned about their level of influence than about effective work performance.

4. Vroom's theory of expectancy

Victor Vroom stated that people will be highly productive and motivated if two conditions are met:

- 1) people believe it is likely that their efforts will lead to successful results and
- 2) those people also believe they will be rewarded for their success.

People will be motivated to exert a high level of effort when they believe there are relationships between the efforts they put forth, the performance they achieve, and the outcomes/ rewards they receive.

5. McGregor's theory X and theory Y

Douglas McGregor formulated two distinct views of human being based on participation of workers. The first is basically negative, labelled as Theory X, and the other is basically positive, labelled as Theory Y. Both kinds of people exist. Based on their nature they need to be managed accordingly.

- **Theory X:** The traditional view of the work force holds that workers are inherently lazy, self-centred, and lacking ambition. Therefore, an appropriate management style is strong, top-down control.
- **Theory Y:** This view postulates that workers are inherently motivated and eager to accept responsibility. An appropriate management style is to focus on creating a productive work environment coupled with positive rewards and reinforcement.

6. ALDERFER'S ERG Theory:

Alderfer further developed Maslow's hierarchy of needs by categorizing the hierarchy into his ERG theory (Existence, Relatedness and Growth). The existence group is concerned with providing the basic material existence requirements of humans. They include the items that Maslow considered to be physiological and safety needs. The second group of needs is those of relatedness – the desire people have for maintaining important interpersonal relationships. These social and status desires require interaction with others if they are to be satisfied, and they align with Maslow's social need and the external component of Maslow's esteem classification. Finally, Alderfer isolates growth needs: an intrinsic desire for personal development. These include the intrinsic component from Maslow's esteem category and the characteristics included under self-actualization. Alderfer categorized the lower order needs (Physiological and Safety) into the Existence category. He fit Maslow's interpersonal love and esteem needs into the Relatedness category. The Growth category contained the self-actualization and self-esteem needs. Alderfer also proposed a regression theory to go along with the ERG theory. He said that

when needs in a higher category are not met then individuals redouble the efforts invested in a lower category need. For example if self-actualization or self-esteem is not met then individuals will invest more effort in the relatedness category in the hopes of achieving the higher need.

LEADERSHIP AND STYLES OF LEADERSHIP

LEADERSHIP

Leadership is both a research area and a practical skill encompassing the ability of an individual or organization to "lead" or guide other individuals, teams, or entire organizations.

Definition

Leadership can be described as the ability of an individual to influence, motivate, and enable others to contribute toward the effectiveness and success of an organization or group of which they are members. A person who can bring about change, therefore, is one who has this ability to be a **leader**

QUALITIES OF SUCCESSFUL LEADERS

1.HONESTY	2.DELEGATE	3.COMMUNICATION	4.CONFIDENCE	5.COMMITMENT
6.POSITIVE	7.CREATIVITY	8.INTUITION	9.INSPIRE	10.APPROACH
ATTITUDE				

Honesty

Whatever ethical plane you hold yourself to, when you are responsible for a team of people, it's important to raise the bar even higher. Your business and its employees are a reflection of yourself, and if you make honest and ethical behaviour a key value, your team will follow suit.

Delegate

Finessing your brand vision is essential to creating an organized and efficient business, but if you don't learn to trust your team with that vision, you might never progress to the next stage. It's important to remember that trusting your team with your idea is a sign of strength, not weakness.

Communication

Knowing what you want accomplished may seem clear in your head, but if you try to explain it to someone else and are met with a blank expression, you know there is a problem. If this has been your experience, then you may want to focus on honing your communication skills.

Confidence There may be days where the future of your brand is worrisome and things aren't going according to plan. This is true with any business, large or small, and the most important thing is not to panic. Part of your job as a leader is to put out fires and maintain the team morale.

Commitment

If you expect your team to work hard and produce quality content, you're going to need to lead by example. There is no greater motivation than seeing the boss down in the trenches working alongside everyone else, showing that hard work is being done on every level.

Positive Attitude

You want to keep your team motivated towards the continued success of the company, and keep the energy levels up. Whether that means providing snacks, coffee, relationship advice, or even just an occasional beer in the office, remember that everyone on your team is a person. Keep the office mood a fine balance between productivity and playfulness.

Creativity

Some decisions will not always be so clear-cut. You may be forced at times to deviate from your set course and make an on the fly decision. This is where your creativity will prove to be vital. It is during these critical situations that your team will look to you for guidance and you may be forced to make a quick decision. As a leader, it's important to learn to think outside the box and to choose which of two bad choices is the best option.

Intuition

When leading a team through uncharted waters, there is no roadmap on what to do. Everything is uncertain, and the higher the risk, the higher the pressure. That is where your natural intuition has to kick in. Guiding your team through the process of your day-to-day tasks can be honed down to a science.

Inspire

Creating a business often involves a bit of forecasting. Especially in the beginning stages of a startup, inspiring your team to see the vision of the successes to come is vital. Make your team feel invested in the accomplishments of the company.

LEADERSHIP STYLES

Leadership Styles are philosophies of leadership that guide the actions of the leader. There are different types of Leadership Styles based on culture, values, and the leader's own personal character traits.

There are several commonly accepted leadership styles, which include Transformational, Transactional, Democratic, Bureaucratic, Servant, Authoritarian, and Laissez-Faire.

Some elements of each type of Leadership styles are often combined in the same person as it is rare for a leader to neatly fit into only one category. Situational Leadership is the idea that the best leaders will use a blend of different Leadership styles to fit the situation. Types of Leadership Styles

(1) Transformational Leaders are also known as Charismatic Leaders, who lead by inspiring others to create and innovate. These leaders are generally well-liked by staff but may think too broadly about their vision to successfully implement their ideas. Charismatic leaders, in particular, tend to have an almost cult-like following, which can be detrimental when they leave their position since their position of power is tied to the individual.

(2) Transactional Leaders assert their authority using exchanges of punishments and rewards with their staff. These leaders rely upon clearly established roles and boundaries in a hierarchy. This type of leader is typically seen as fair, but is usually too conventional to allow much creative innovation.

(3) Democratic Leaders or participative leaders tend to lead by consensus, asking subordinates for their opinions and information to come to more collective decisions. These processes ensure that staff voices are heard, but may spend too much time in a discussion when action is needed.

(4) Bureaucratic Leaders take policy and protocol seriously, following rules by the book and adhering to strict hierarchies in an established chain of command. These types of leaders tend to exist in environments that are already bureaucratic, like government or military, and though they are predictable and methodical they are generally not well-liked or fast moving.

(5) Servant Leaders do not like to assert authority over their subordinates and instead model leadership through self-sacrifice and prioritization of the needs of their staff. It can also be described as altruistic leadership. These leaders may be seen as generous and kind but can also be viewed as weak in authority and easily overcome by adversaries.

(6) Authoritarian Leaders are an extreme version of transactional leaders, with an emphasis on power and control over their staff. Authoritarian leaders are dictatorial, strict, and rarely take staff input into consideration. These leaders can be effective in military environments but tend to cause workers to leave organizations when this style is used in the office.

(7) Laissez-Faire Leaders are laid-back, allowing workers to lead themselves and rarely getting involved in day-to-day decision making. Deriving their name from the French phrase “laissez-faire” which means “to leave alone”, these leaders may give employees too little direction to be able to work effectively. This leadership style may function well in an environment where employees are highly skilled and self-motivated so little supervision is needed.

Decision Making

Decision-making is an essential aspect of modern management. It is a primary function of management. A manager's major job is sound/rational decision-making. He takes hundreds of decisions consciously and subconsciously. Decision-making is the key part of manager's activities. Decisions are important as they determine both managerial and organizational actions. A decision may be defined as "a course of action which is consciously chosen from among a set

of alternatives to achieve a desired result." It represents a well-balanced judgment and a commitment to action.

Definitions of Decision-making

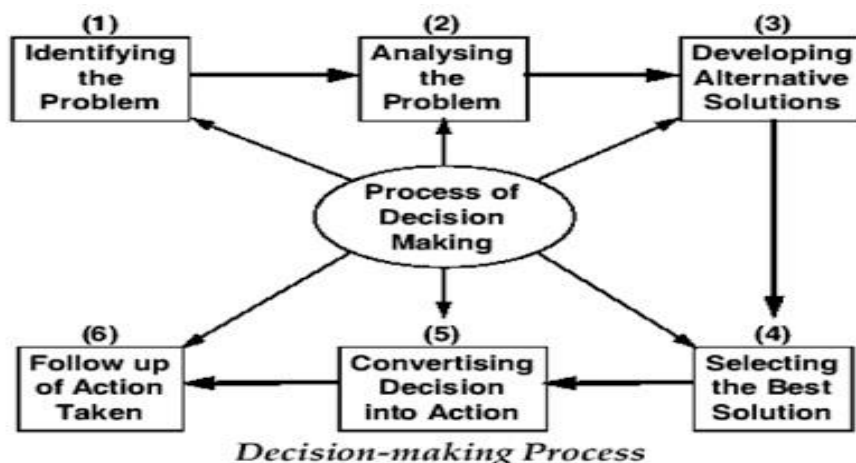
1. The Oxford Dictionary defines the term decision-making as "the action of carrying out or carrying into effect".
2. According to Trewatha & Newport, "Decision-making involves the selection of a course of action from among two or more possible alternatives in order to arrive at a solution for a given problem".

Steps Involved In Decision Making Process

Decision-making involves a number of steps which need to be taken in a logical manner. This is treated as a rational or scientific 'decision-making process' which is lengthy and time consuming. Such lengthy process needs to be followed in order to take rational/scientific/result oriented decisions. Decision-making process prescribes some rules and guidelines as to how a decision should be taken / made. This involves many steps logically arranged. It was Peter Drucker who first strongly advocated the scientific method of decision-making in his world famous book 'The Practice of Management' published in 1955. Drucker recommended the scientific method of decision-making which, according to him, involves the following six steps:

1. Defining / Identifying the managerial problem,
2. Analyzing the problem,
3. Developing alternative solutions,
4. Selecting the best solution out of the available alternatives,
5. Converting the decision into action, and
6. Ensuring feedback for follow-up.

The figure given below suggests the steps in the decision-making process:-



- 1. Identifying the Problem:** Identification of the real problem before a business enterprise is the first step in the process of decision-making. It is rightly said that a problem well-defined is a problem half-solved. Information relevant to the problem should be gathered so that critical analysis of the problem is possible. This is how the problem can be diagnosed. Clear distinction should be made between the problem and the symptoms which may cloud the real issue. In brief, the manager should search the 'critical factor' at work. It is the point at which the choice applies. Similarly, while diagnosing the real problem the manager should consider causes and find out whether they are controllable or uncontrollable.
- 2. Analyzing the Problem:** After defining the problem, the next step in the decision-making process is to analyze the problem in depth. This is necessary to classify the problem in order to know who must take the decision and who must be informed about the decision taken. Here, the following four factors should be kept in mind:
 1. Futurity of the decision,
 2. The scope of its impact,
 3. Number of qualitative considerations involved, and
 4. Uniqueness of the decision.
- 3. Collecting Relevant Data:** After defining the problem and analyzing its nature, the next step is to obtain the relevant information/ data about it. There is information flood in the business world due to new developments in the field of information technology. All available information should be utilised fully for analysis of the problem. This brings clarity to all aspects of the problem.
- 4. Developing Alternative Solutions:** After the problem has been defined, diagnosed on the basis of relevant information, the manager has to determine available alternative courses of action that could be used to solve the problem at hand. Only realistic alternatives should be considered. It is equally important to take into account time and cost constraints and psychological barriers that will restrict that number of alternatives. If necessary, group participation techniques may be used while developing alternative solutions as depending on one solution is undesirable.
- 5. Selecting the Best Solution:** After preparing alternative solutions, the next step in the decision-making process is to select an alternative that seems to be most rational for solving the problem. The alternative thus selected must be communicated to those who are likely to be affected by it. Acceptance of the decision by group members is always desirable and useful for its effective implementation.
- 6. Converting Decision into Action:** After the selection of the best decision, the next step is to convert the selected decision into an effective action. Without such action, the

decision will remain merely a declaration of good intentions. Here, the manager has to convert 'his decision into 'their decision' through his leadership. For this, the subordinates should be taken in confidence and they should be convinced about the correctness of the decision. Thereafter, the manager has to take follow-up steps for the execution of decision taken.

- 7. Ensuring Feedback:** Feedback is the last step in the decision-making process. Here, the manager has to make built-in arrangements to ensure feedback for continuously testing actual developments against the expectations. It is like checking the effectiveness of follow-up measures. Feedback is possible in the form of organised information, reports and personal observations. Feedback is necessary to decide whether the decision already taken should be continued or be modified in the light of changed conditions.

Every step in the decision-making process is important and needs proper consideration by managers. This facilitates accurate decision-making. Even quantitative techniques such as CPM, PERT/OR, linear programming, etc. are useful for accurate decision-making. Decision-making is important as it facilitates entire management process. Management activities are just not possible without decision-making as it is an integral aspect of management process itself. However, the quality of decision-making should be always superior as faulty/irrational decisions are always dangerous.

DESIGNING ORGANISATION STRUCTURE – PRINCIPLES AND TYPES OF ORGANISATION

Organization:

Organization is form of organizing which is a part of management process.

Organization defined as collectivity of people for achieving common objectives.

“Organization means the determination and assignment of duties to people, and also the establishment and the maintenance of authority relationships among these grouped activities it is the structural frame work with in which the various efforts are coordinated and related to each other”.

Definitions: “Organization are collectivities of people that have been established for the pursuit of relatively specific objectives on a more or less continuous basis”.

William Scott

“Organization is the form of every human association for the attainment of a common purpose”.

-Mooney and Reilly

“Organization involves the grouping of activities necessary to accomplish goals and plans assignment and these activities to appropriate departments and positions to appropriate departments and positions for authority delegation and coordination”.

-Koontz and O'Donnell

Organization is used in the following ways with or without prefix or suffix

1. as entity
2. as group of people
3. as structure
4. as process

Process of Organization:

Determination of objectives, strategies, plans and policies: Objectives should be clear and precise, because the entire organization is to be built around the objectives of the enterprises.

Determination of activities: Determine activities needed to execute these plans and policies and accomplish the objectives. The work load is broken into component activities that are to be performed by all the employees. The activities are so split to determine the job which can be performed by an individual.

Separation and grouping of activities: To attain the benefits of specialization and division of labour, every company, will separate its activities on the basis of primary functions like finance, engineering, purchasing, production, sales and industrial relations. All the similar or directly related activities are grouped together in the form of departments.

Delegation of authority: Authority is necessary for the performance of the job and therefore authority is delegated to the subordinates for enabling them to carry out their work smoothly and efficiently.

Delegation of responsibility: Responsibility may be described as the obligation and accountability for the performance of delegated duties. A superior is always accountable for the acts of his subordinate. Therefore, responsibility always flows from subordinates to superiors. Establish inter-relationships: The grouped activities are placed in the overall organization structure at appropriate level. It is necessary to integrate these groups of activities through.

- a) Authority relationship horizontally, vertically and diagonally
- b) Organized information or communication system i.e., with the help of effective coordination and communication.

Providing physical facilities and proper environment: Physical facilities means provide machinery, tools equipments, infrastructure etc., environment means providing proper lighting,

ventilation, heating, cooling arrangement at the work place, reasonable hours of work, safety devices, job security etc.

PRINCIPLES OF ORGANIZATION:

Principle of unity of objectives: An organization structure is effective if it enables individuals to contribute to entire objectives.

Principle of co-ordination: The aim of the objective can be achieved if proper coordination exists for different activities.

Principles of organizational efficiency: An organization is efficient if it is structured to aid the accomplishment of enterprise objective with a minimum of unwanted consequences or costs.

Span of management principle: In each managerial position, there is a limit to the number of persons an individual can effectively manage but the exact number will depend on the impact of underlying variables.

Scalar principle: The clearer the line of authority from the ultimate management position in an enterprise to every subordinate position, the clearer will be the responsibility for decision making the more effective will be organization communication.

Principle of delegation by results expected: Authority delegated to all individual managers should be adequate enough to ensure their ability to accomplish the results expected.

Principle of responsibility: The responsibility of subordinates to their superiors for performance is absolute, and superiors cannot escape responsibility for the organization activity of their subordinates.

Principle of parity of authority and responsibility: The responsibility for actions can not be greater than that implied by the authority delegated, nor should it be less.

Principle of unity command: The more complete an individual's reporting relationships to a single superior, the smaller the problem of conflicting instructions and the greater the feeling of personal responsibility for results.

Authority level principle: Maintenance of intended delegation requires that decisions within the authority of individual managers should be made by them and not be referred upward in the organization structure.

Principle of balance: The application of principles or technique must be balanced to ensure the overall effectiveness of the structure in meeting enterprise objectives.

Principle of flexibility: The more that provisions are made for building flexibility in to an organization structure can fulfill its purpose.

Principle of leadership facilitation: The more an organization structure and its delegations of authority enable managers to design and maintain an environment for performance, the more they will help the leadership abilities of those managers.

Design of Organization structure:

The main objective of an organization structure is to ensure that efforts of all the people working in various sections are co-ordinate and integrated for achieving the task in the most efficient effective way with minimum consumption of resources i.e. economical ways

1) Formal organization structure 2) Informal organization structure

Formal organization structure: According to classical theorists the formal organization is built on four pillars

- 1) Division of labour
- 2) Scalar functional processes
- 3) Structure
- 4) Span of control

Definition: An organization is formal when the activities of two or more persons are consciously coordinated towards common objectives.

Informal organization: Informal organization arises spontaneously based on friendship or some common interest and not based on rules, regulation and procedures. It is developed by the employees themselves and not by the formal authority.

Definition: Informal organization brings cohesiveness to a formal organization, it brings to the members of formal organization a feeling of belonging of status, of self-respect and of gregarious satisfaction.

Comparison between formal and informal organization: Basis of comparison	Formal	Informal
Formation	Planned	Spontaneous & deliberated
Purpose	Well-set goals	Social interaction
Structure	Well structured	Unstructured
Focus	Positions	Persons

Nature	Official	Unofficial
Leadership	Superior	Any one
Source of power	Delegated	Given by group
Guidelines for behaviour	Rules procedures	Group norms
Source of control	Rewards/Punishment	Sanctions

TYPES OF ORGANIZATION STRUCTURE

The following are the various types of organization structure:

1. Line Organization
2. Functional Organization
3. Line and Staff Organization
4. Committee Organization
5. Project Organization
6. Matrix Organization

The following analysis gives the nature, merits and demerits of these structures:

Type # 1. Line Organization:

It is also known as Scalar organization or Military form of organization. It is the oldest form of organization. Under this method, authority flows in a vertical manner from top to bottom. Each position in the structure has an authority over a lower position. Line executives are directly involved in the performance of the activities of an enterprise and are called the 'doers'.

Line organization is of two types, viz., Pure Line Organization and Departmental Line Organization. Under pure line organization all perform same type of work at any level. On the other hand, under departmental line organization, each department performs different type of work.

Merits of Line Organization:

1. It is simple to understand and easy to establish.
2. Each individual knows clearly to whom he is responsible.
3. It provides unit of command.

4. It helps in taking quick and effective decisions.
5. It ensures personal contact between workers and manager.
6. It ensures discipline between the employees and employer.

Demerits of Line Organization:

1. There is concentration of authority at the top level. It leads to autocratic control.
2. Communication flows from top to bottom. There is no participation of workers.
3. It lacks specialization as a line manager is responsible for both planning and execution of work.
4. Overburdens the top executives and they cannot perform their work properly.

Type # 2. Functional Organization:

This concept was devised by F.W. Taylor. It is based on the functional foremanship. Under this method, the structure of the enterprise is classified into different functional areas. Each functional area is headed by a specialist who has full control of that function over the organization and gives instructions direct to the personnel, rather than through the chain of command.

Each chief manager is in charge of a particular activity. For example, chief finance manager is in charge of finance function of all the three units. Thus, he enjoys functional authority over the employees of other departments also.

Merits of Functional Authority:

1. It helps to reap the benefits of specialization
2. Subordinates can make use of the expertise of functional experts
3. The burden of top executive is reduced as each expert looks after only one function.
4. It leads to joint supervision by different experts
5. There is expert and better control can be exercised by functional experts
6. It provides better scope for expansion and diversification

Demerits of functional organization:

1. It is against the principle of unity of command as the subordinate is responsible to a number of superiors
2. Lower level people are not given an opportunity for all round experience and hence they do not fit into top level positions.

3. The complex nature of functional organization with its cross relationship creates a confusion among workers
4. The decision making is very slow as involvement of many experts is required
5. There is a scope of lack of coordination as each specialist thinks of only his function and ignores other functions.

Type # 3. Line and Staff Organization:

Line and staff organization is a combination of functional and line structure. Line authority flows from top to bottom and the line executive is directly concerned with the accomplishment of primary objectives. They are actual doers and generally do not possess specialized knowledge to solve complex problems.

To provide specialized assistance to line managers, staff positions are created. Staff means a stick in the hand for support. Thus, staff helps the line executives in their work. They play the role of an advisor.

The other approach views that line and staff are two kinds of authority. According to this approach, line authority is defined as a direct authority which a superior exercises over his subordinates to carry out orders and instructions. The exercise of this authority is always downwards, that is, from a superior to a subordinate. Staff authority involves giving advice to line managers to carry on the operation. The flow of this authority may be in any direction depending on the need of such an advice. It is common that in actual practice, some variations may exist. The variations are more pronounced in the case of staff authority.

The distinction between line and staff though not rigid, is important because staff must be provided if the growing organization is to accomplish its goals. The differentiation between line and staff is necessary for the following reasons.

In line and staff organization, the line authority remains the same in the organization. But staff executives are attached with line executives who help them by providing necessary advice on important matters. Staff executives have no power to command subordinates in other departments.

It is clear that PA to Managing Director, Personnel Manager and Budget Manager play role of staff executives. In most business units, staff executives are used for collecting data required for taking decisions and to provide expert advice to line managers.

Merits of Line and Staff Organization:

- a. This combination provides for specialized knowledge where staff executives guide and advise line executives.

- b. It reduces the burden of line executives because staff carries on detailed investigation of each activity.
- c. Better decisions are possible as expertise of the staff is used.
- d. It is flexible in the sense that staff can be added to line executives.

Demerits of Line and Staff Organization:

- I. Always it creates confusion as it is very difficult to define the authority relationship between line and staff.
- II. Staff executives are not accountable and they may not take the tasks seriously.
- III. There is always constant conflict between line and staff because the nature of their functions differs from each other.

Type # 4. Committee Organization:

A committee is a group of persons who consciously and deliberately formed to discuss a problem and provide solutions to overcome it. They meet on an organized basis to discuss and deal with the matter placed before it.

The following are the characteristics of a committee:

- a. A committee is a group of people. There must be a minimum of two people and there is no limit for maximum.
- b. The scope of committee is limited to the extent of task assigned to it.
- c. Democratic principle of 'One man one vote' is followed
- d. It may be executive committee or non-executive committee.

Merits of Committee Organization:

- 1. Pooling of knowledge and experience- Knowledge and experience are pooled together to take joint decisions on the principle of two heads are better than one.
- 2. Motivation through participation- Members are allowed to participate in the decision making process. It makes them more committed to the decision.
- 3. Effective and proper coordination- Through coordination of experts bringing together diverse viewpoints many problems could be solved effectively.
- 4. Tool for management development- Every member has an opportunity for learning from the experience and mistakes of others. Thus it helps for management development.

5. Representation of diverse interest groups- When different interest groups are represented in a committee, conflicting views can be easily settled through motivation and participation.
6. Effective communication- By convening a committee any information can be easily passed on to all members of the organization.
7. Consolidation of authority- It enables management to consolidate splintered authority. This is helpful to a manager who has limited authority with which he cannot solve the problem without simultaneous exercise of authority by other managers related to the problem.

Demerits of Committee Organization:

The committee is one made of the unfit, selected by the unwilling to do the unnecessary.

The excessive or lack of use of committees creates problems to the management as follows:

- a. It is costly and time consuming activity. The cost incurred is far below its benefits.
- b. Committees do not help in taking quick decisions.
- c. To arrive at unanimous decisions agreements are arrived at on the basis of a compromise which may not be really a right decision, Compromise is not the solution.
- d. When unanimous decisions are taken by dominant members, a few persons may accept them under pressure.
- e. Committees fix group responsibility where no individual can be made accountable. This leads to inefficiency.
- f. Committee is a tool to misuse to delay or avoid or to take unpleasant decision.

Type # 5. Project Organization:

The use of the project organization has increased in the last few years. It is currently being employed in numerous undertakings engaged in the execution of construction activities, turnkey projects and research and development projects. The project organization can take various forms, but the important characteristic that distinguishes it from other forms is once the project is completed the organization is disbanded or phased out.

By definition, project management involves, “the gathering of the best available talent to accomplish a specific and complex undertaking within time, cost and quality parameters, following by the disbanding of the team upon completion of the undertaking”.

The group members then go on to another project, return to their permanent home department in the organization, are given jobs elsewhere in the organization, or, in some cases, are phased entirely out of the firm.

Merits of Project Organization:

- a. It allows maximum utilization of specialization
- b. It is tailor-made to meet specific needs of a particular project
- c. It provides more flexibility in handling resources by allocating them when they are needed.

Demerits of Project Organization:

- I. It has a limited time and creates a feeling of uncertainty and insecurity among people in the organization.
- II. Lack of proper vertical authority makes the job of a project manager difficult. He is responsible for the completion of the project but without authority over people in the project.

Type # 6. Matrix Organization:

It is a combination of project organization and functional organization.

“Any organization that employs a multiple command system that includes not only multiple command structure but also related support mechanism and an associated organization culture and behaviour pattern.” — Paul R. Lawrence and Stanley M. Davis

It is clear from the above that matrix organization leads in overlapping of command, control and behaviour pattern.

Merits of Matrix Organization:

1. It focuses attention on a single project and facilitates better control.
2. It is more flexible than traditional organization
3. It provides motivation for the people engaged in a project
4. It leads to better utilization of services of professionals.

Demerits of Matrix Organization:

1. People receive instructions and order both from functional and project managers. Thus, it violates unity of command.
2. The complex relationship leads to confusion and makes coordination difficult. Subordinates are not in a position to identify their superiors.
3. In the absence of mutual trust and confidence, in sharing the resources and in taking joint decisions, conflicts arise.
4. As people are drawn from various departments, there is a lack of commitment and morale is low.

2.1 OPERATIONS MANAGEMENT

OPERATIONS MANAGEMENT

* Operations management is the administration of business practices to create the highest level of efficiency possible within an organization.

* Operations management is concerned with converting materials and labour into goods and services as efficiently as possible

“Operation management refers to the direction and control of the process that transforms inputs into products and services”.

PLANT LOCATION

“Plant location refers to the choice of region and the selection of a particular site for setting up a business or factory”.

Plant is a place where the cost of the product is kept to low in order to maximize gains. But the choice is made only after considering cost and benefits of different alternative sites. It is a strategic decision that cannot be changed once taken.

Identifying an ideal location is very crucial, it should always maximize the net advantage, must minimize the unit cost of production and distribution.

FACTORS AFFECTING THE PLANT LOCATION

Decisions regarding selecting a location need a balance of several factors. These are divided into primary factors and secondary factors; here both the factors can influence the business in the long run.

PRIMARY FACTORS

Availability of raw materials

Availability of raw materials is the most important factor in plant location decisions. Usually, manufacturing units where there is the conversion of raw materials into finished goods is the main task then such organizations should be located in a place where the raw materials availability is maximum and cheap.

Nearness to the market

Nearness of market for the finished goods not only reduces the transportation costs, but it can render quick services to the customers. If the plant is located far away from the markets then the chances of spoiling and breakage become high during transport. If the industry is nearer to the market then it can grasp the market share by offering quick services.

Availability of labour

Another most important factor which influences the plant location decisions is the availability of labour. The combination of the adequate number of labor with suitable skills and reasonable labor wages can highly benefit the firm. However, labor-intensive firms should select the plant location which is nearer to the source of manpower.

Transport facilities

In order to bring the raw materials to the firm or to carrying the finished goods to the market, transport facilities are very important. Depending on the size of the finished goods or raw materials a suitable transportation is necessary such as roads, water, rail, and air. Here the transportation costs highly increase the cost of production, such organizations can not complete with the rival firms. Here the point considered is transportation costs must be kept low.

Availability of fuel and power

Unavailability of fuel and power is the major drawback in selecting a location for firms. Fuel and power are necessary for all most all the manufacturing units, so locating firms nearer to the coal beds and power industries can highly reduce the wastage of efforts, money and time due to the unavailability of fuel and power.

Availability of water

Depending on the nature of the plant firms should give importance to the locations where water is available.

For example, power plants where use water to produce power should be located near the water bodies.

SECONDARY FACTORS

Suitability of climate

Climate is really an influencing factor for industries such as agriculture, leather, and textile, etc. For such industries extreme humid or dry conditions are not suitable for plant location. Climate can affect the labour efficiency and productivity.

Government policies

While selecting a location for the plant, it is very important to know the local existed Government policies such as licensing policies, institutional finance, Government subsidies, Government benefits associated with establishing a unit in the urban areas or rural areas, etc.

Availability of finance

Finance is the most important factor for the smooth running of any business; it should not be far away from the plant location. However, in the case of decisions regarding plant location, it is the

secondary important factor because financial needs can be fulfilled easily if the firm is running smoothly. But it should be located nearer to the areas to get the working capital and other financial needs easily.

Competition between states

In order to attract the investment and large scale industries various states offer subsidies, benefits, and sales tax exemptions to the new units. However, the incentives may not be big but it can help the firms during its startup stages.

Availability of facilities

Availability of basic facilities such as schools, hospitals, housing and recreation clubs, etc. can motivate the workers to stick to the jobs. On the other hand, these facilities must be provided by the organization, but here most of the employees give preference to work in the locations where all these benefits/facilities are available outside also. So while selecting plant location, organizations must give preference to the location where it is suitable for providing other facilities also.

Disposal of waste

Disposal of waste is a major problem particularly for industries such as chemical, sugar, and leather, etc. So that the selected plant location should have provision for the disposal of waste.

PLANT LAYOUT

“Plant layout refers to the arrangement of physical facilities such as machinery, equipment, furniture etc. within the factory building in such a manner so as to have quickest flow of material at the lowest cost and with the least amount of handling in processing the product from the receipt of material to the shipment of the finished product”.

According to Riggs, “the overall objective of plant layout is to design a physical arrangement that most economically meets the required output – quantity and quality.”}

According to J. L. Zundi, “Plant layout ideally involves allocation of space and arrangement of equipment in such a manner that overall operating costs are minimized.

OBJECTIVES OF PLANT LAYOUT

- } Proper and efficient utilization of available floor space
- } To ensure that work proceeds from one point to another point without any delay
- } Provide enough production capacity.
- } Reduce material handling costs
- } Reduce hazards to personnel
- } Utilize labour efficiently
- } Increase employee morale

- } Reduce accidents
- } Provide for volume and product flexibility
- } Provide ease of supervision and control
- } Provide for employee safety and health
- } Allow ease of maintenance
- } Allow high machine or equipment utilization
- } Improve productivity

TYPES OF LAYOUT

In case of manufacturing unit, plant layout may be of four types:

- Product or line layout
- Process or functional layout
- Fixed position or location layout
- Combined or group layout

Product or line layout

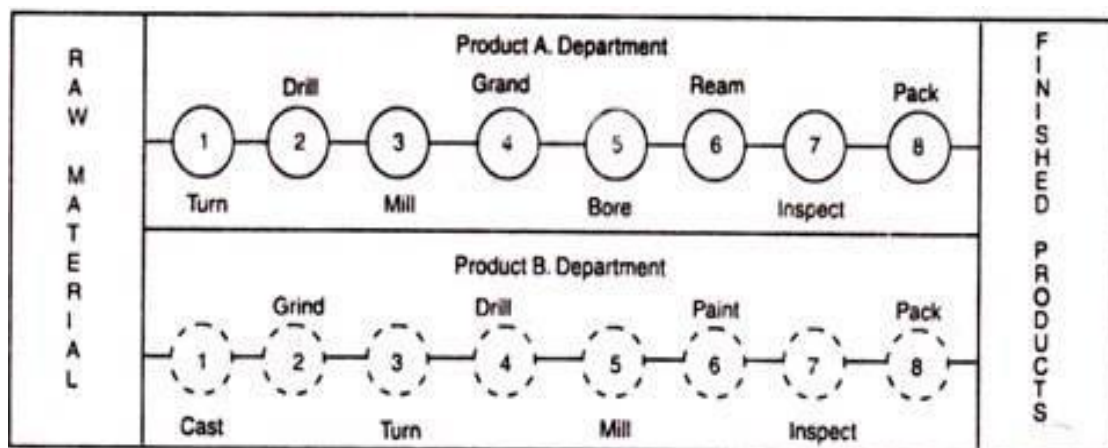


Fig. 8.3.

- Under this, machines and equipments are arranged in one line depending upon the sequence of operations required for the product.
- The raw material moves very fast from one workstation to other stations with a minimum work in progress storage and material handling.
- The grouping of machines should be done keeping in mind the following general principles.
- All the machine tools or other items of equipments must be placed at the point demanded by the sequence of operations
- There should no points where one line crossed another line.
- Materials may be fed where they are required for assembly but not necessarily at one point.

- All the operations including assembly, testing packing must be included in the line

Process or functional layout

- In this type of layout machines of a similar type are arranged together at one place. E.g. Machines performing drilling operations are arranged in the drilling department, machines performing casting operations be grouped in the casting department. Therefore the machines are installed in the plants, which follow the process layout.
- Hence, such layouts typically have drilling department, milling department, welding department, heating department and painting department etc. The process or functional layout is followed from historical period.
- The work, which has to be done, is allocated to the machines according to loading schedules with the object of ensuring that each machine is fully loaded. Process layout is shown in the following diagram.

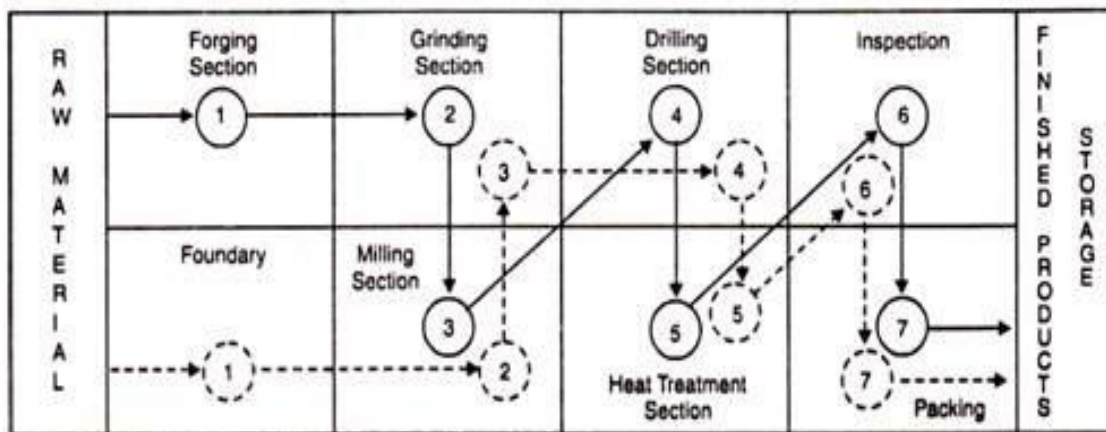
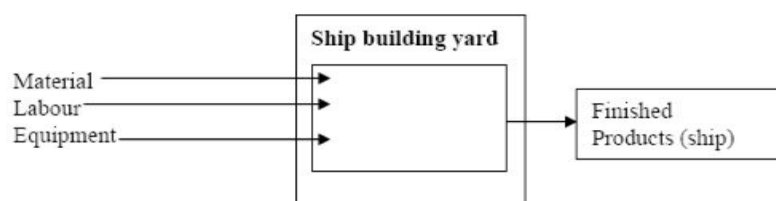


Fig. 8.4.

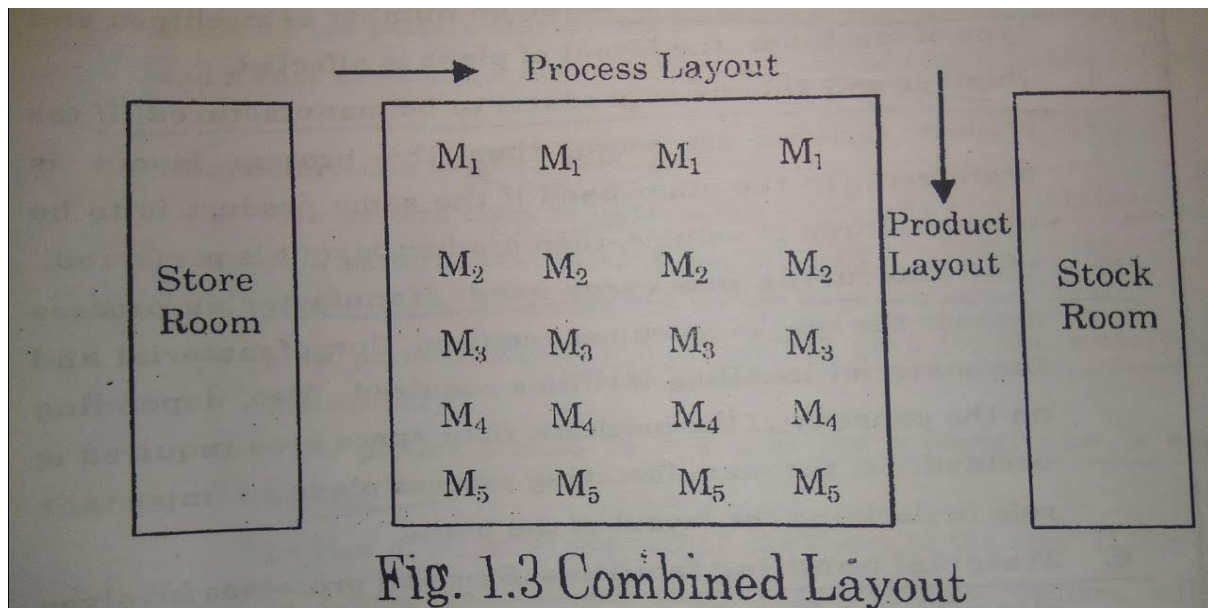
Fixed position or location layout

- In this type of layout, the major product being produced is fixed at one location.
- Equipment labour and components are moved to that location.
- All facilities are brought and arranged around one work center.
- This type of layout is not relevant for small scale entrepreneur.
- The following figure shows a fixed position layout regarding shipbuilding.



Combined or group layout

- Certain manufacturing units may require all three processes namely intermittent process (job shops), the continuous process (mass production shops) and the representative process combined process [i.e. miscellaneous shops].
- In most of industries, only a product layout or process layout or fixed location layout does not exist. Thus, in manufacturing concerns where several products are produced in repeated numbers with no likelihood of continuous production, combined layout is followed.



Advantages of a Good Plant Layout:

To the Worker:

- (i) Reduces the effort of the worker.
- (ii) Reduces the number of handlings.
- (iii) Extends the process of specialization.
- (iv) Permits working at optimum conditions by eliminating congestions.
- (v) Produces better working conditions by eliminating congestions.
- (vi) Reduces the number of accidents.
- (vii) Provides better employee service facilities/conditions.
- (viii) Provides basis for higher earning for employees.

In Labour Cost:

- (i) Increases the output per man-hour.
- (ii) Reduces set up time involved.
- (iii) Reduces the number of operations or some operations may be combined.
- (iv) Reduces the number of handlers. Thus reducing labour cost.
- (v) Reduces the length of hauls.
- (vi) Reduces lost motions between operations.
- (vii) Converts operator into a producer instead of a handler by eliminating the various unnecessary movements.

In Other Manufacturing Costs:

- (i) Reduces the cost of expensive supplies.
- (ii) Decreases maintenance costs.
- (iii) Decreases tool replacement costs.
- (iv) Effects a saving in power loads.
- (v) Decreases spoilage and scrap. Thus waste is minimized
- (v) Eliminates some of the waste in raw material consumption.
- (vii) Improves the quality of the product by decreasing handling.
- (viii) Provides better cost control.

In the Manufacturing Cycle:

- (i) Shortens the moves between work-stations.
- (ii) Reduces the manufacturing cycle in each department.
- (iii) Reduces the length of the travel by the product for completion.
- (iv) Reduces the overall time of manufacturing the product.

In Production Control:

- (i) Facilitates receipts, shipments and delivery of inputs and finished goods.
- (ii) Provides adequate and convenient storage facilities.
- (iii) Permits the maximum possible output with same input.

- (iv) Paces production & determines production flow.
- (v) Makes production time predictable.
- (vi) Makes scheduling and dispatching automatic.
- (vii) Sets up production centre & permits straight line layout by products for mass production.
- (viii) Permits layout by process for job order manufacturing.
- (ix) Moves work in process by most direct lines.
- (x) Reduces the number of lost or mishandled parts leading to waste minimization.
- (xi) Reduces the paper work for production control & reduces the number of stock chasers. Thus reduces production control expenses.

In Supervision:

- (i) Tends to ease the burden of supervision.
- (ii) Determines the supervisory control.
- (iii) Reduces the cost of supervision process.
- (iv) Reduces cost of piece counts.
- (v) Decreases the amount of inspection involved.

In Capital Investment:

- (i) Holds permanent investment at its minimum level.
- (ii) Keeps the plant from becoming obsolete before it is worn out.
- (iii) Reduces the investment in machinery and equipment by
 - (a) Increasing the production per machine.
 - (b) Utilizing idle machine time.
 - (c) Reducing the number of operations per machine.
- (iv) Maintains a proper balance of departments.
- (v) Eliminates wasted aisle space.
- (vi) Reduces the capital investment by proper space utilization of material handling equipment required.
- (vii) Reduces the inventory level of work in process and of finished product.

METHODS OF PRODUCTION

Job, batch and flow production

Production is about creating goods and services. Managers have to decide on the most efficient way of organizing production for their particular product.

There are three main types of production to choose from:



Business managers decide the best method of production for running their business. Depending upon their business this could be job production, batch production or flow production

Job production, where items are made individually and each item is finished before the next one is started. Designer dresses are made using the job production method.

Batch production, where groups of items are made together. Each batch is finished before starting the next block of goods. For example, a baker first produces a batch of 50 white loaves. Only after they are completed will he or she start baking 50 loaves of brown bread.

Flow production, where identical, standardised items are produced on an assembly line. Most cars are mass-produced in large factories using conveyor belts and expensive machinery such as robot arms. Workers have specialised jobs, for instance, fitting wheels.

WORK STUDY

Meaning of Work Study:

According to ILO — International Labour Organisation — work study is “**a term used to embrace the techniques of method study and work measurement which are employed to**

ensure the best possible use of human and material resources in carrying out a specified activity.”

In other words, “work study is a tool or technique of management involving the analytical study of a job or operation.” Work study helps to increase productivity.

Objectives of Work Study:

- (i) Work study brings higher productivity;
- (ii) Work study improves existing method of work for which cost becomes lower;
- (iii) It eliminates wasteful elements;
- (iv) It sets standard of performance;
- (v) It helps to use plant and human more effectively;
- (vi) It improves by saving in time and loss of material also.

Steps Involved in Work Study:

The steps of work study are:

- (i) It selects the jobs which are to be studied;
- (ii) It examines critically the recorded facts which are already done;
- (iii) It records from direct observations all the matters which are happened;
- (iv) It defines new method;
- (v) It also installs the new method;
- (vi) It also maintains the new standard;
- (vii) It develops most economic and appropriate methods;
- (viii) It measures the work content in the method, that is selected and compute a standard time.

Types of Work Study:

1. Method Study:

According to ILO, method study is “**the systematic recording, analysis and critical examination of existing and proposed ways of doing work and the development and application of easier and more effective method**”. In short, it is a systematic procedure to analyse the work to eliminate unnecessary operations.

Objectives:

The objectives of method study are:

- (i) It improves the proper utilisation of manpower, machine and materials;
- (ii) It also improves the factory layout, work place, etc.;
- (iii) It also improves the process and procedure;
- (iv) It develops better physical working environment;
- (v) It reduces undesirable fatigue.

Steps:

The steps of method study are:

- (i) At first select the proper work which are to be studied;
- (ii) Record all the facts of existing method;
- (iii) Examine the facts very critically;
- (iv) Develop the most practical, economic, and effective method;
- (v) Install the method and the same should be maintained.

2. Time And Motion Study:

According to ILO, Time Study means **“a technique for determining as accurately as possible from a limited number of observations the time necessary to carry out a given activity at a different standard of performance”**. In other words, “time study is the art of observing and recording time required to do each detailed element of an individual operation.” Practically, it studies the time taken on each element of a job.

Motion study, on the other hand, is the study of the body motion used in performing an operation, with the thought of improving the operation by eliminating unnecessary motion and simplifying necessary motion and thus establishing the most favourable motion sequence for maximum efficiency.

So, in short, ‘Time Study’ means the determination of standard time that is taken by a worker of average ability under normal working conditions for performing a job. But ‘Motion Study’ determines the correct method of doing a job to avoid wasteful movements, for which the workers are unnecessarily tired.

Steps:

1. Time and Motion studies eliminate wasteful movements;

2. They examine the proposed method critically and determine the most effective one;
3. They determine for each element having a stop-watch;
4. They record all the parts of a job which are done by the existing method;
5. They install the method as standard one;
6. They critically observe the workers who are engaged with the work;
7. They assess the proper speed of the operator who is working.

STATISTICAL QUALITY CONTROL THROUGH CONTROL CHARTS

Statistical Quality Control (SQC) is the use of statistical methods in the monitoring and maintaining of the quality of products and services.

Statistical Quality Control (SQC) is the term used to describe the set of statistical tools used by quality professionals. SQC is used to analyze the quality problems and solve them.

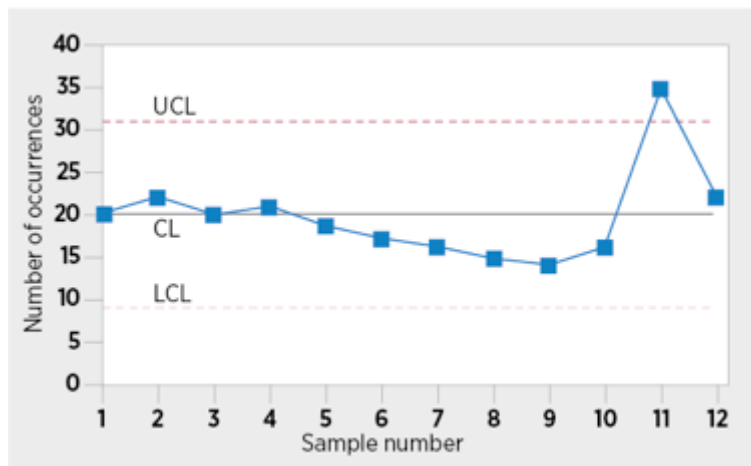
Statistical quality control refers to the use of statistical methods in the monitoring and maintaining of the quality of products and services.

All the tools of SQC are helpful in evaluating the quality of services. SQC uses different tools to analyze quality problem.

- 1) Descriptive Statistics: Descriptive Statistics involves describing quality characteristics and relationships.
- 2) Statistical Process Control (SPC): SPC uses graphical displays known as control charts to determine whether a process should be continued or should be adjusted to achieve the desired quality.
- 3) Acceptance Sampling: can be used when a decision must be made to accept or reject a group of parts or items based on the quality found in a sample

CONTROL CHARTS

The control chart is a graph used to study how a process changes over time. Data are plotted in time order. A control chart always has a central line for the average, an upper line for the upper control limit, and a lower line for the lower control limit. These lines are determined from historical data. By comparing current data to these lines, you can draw conclusions about whether the process variation is consistent (in control) or is unpredictable (out of control, affected by special causes of variation).



TYPES OF QUALITY CONTROL CHARTS

There are two types of control charts:

Variable control chart and attribute control chart.

1. Variable control charts: It is one by which it is possible to measure the quality characteristics of a product. The variable control charts are X-BAR chart, R-BAR chart.

2. Attribute control chart: It is one in which it is not possible to measure the quality characteristics of a product, i.e., it is based on visual inspection only like good or bad, success or failure, accepted or rejected.

The attribute control charts are p-charts, np-charts, c-charts, u-charts. It requires only a count of observations on characteristics e.g., the number of nonconforming items in a sample.

VARIABLE CONTROL CHARTS

X BAR CHART

In industrial statistics, the **X-bar chart** is a type of Shewhart control chart that is used to monitor the arithmetic means of successive samples of constant size, n . This type of control chart is used for characteristics that can be measured on a continuous scale, such as weight, temperature, thickness etc. For example, one might take a sample of 5 shafts from production every hour, measure the diameter of each, and then plot, for each sample, the average of the five diameter values on the chart.

For the purposes of control limit calculation, the sample means are assumed to be normally distributed, an assumption justified by the Central Limit Theorem.

The X-bar chart is always used in conjunction with a variation chart such as the R

chart or and s chart. The R-chart shows sample ranges (difference between the largest and the smallest values in the sample), while the s-chart shows the samples' standard deviation. The R-chart was preferred in times when calculations were performed manually, as the range is far

easier to calculate than the standard deviation; with the advent of computers, ease of calculation ceased to be an issue, and the s-chart is preferred these days, as it is statistically more meaningful and efficient. Depending on the type of variation chart used, the average sample range or the average sample standard deviation is used to derive the X-bar chart's control limits.

R CHART

An X-bar and **R** (range) **chart** is a pair of control **charts** used with processes that have a subgroup size of two or more. The standard **chart** for variables data, X-bar and **R chart** help determine if a process is stable and predictable.

ATTRIBUTE CONTROL CHART

P CHART

In statistical quality control, the **p-chart** is a type of control **chart** used to monitor the proportion of nonconforming units in a sample, where the sample proportion nonconforming is defined as the ratio of the number of nonconforming units to the sample size, n .

C CHART

A **c-chart** is an attributes **control chart** used with data collected in subgroups that are the same size. **C-charts** show how the process, measured by the number of nonconformities per item or group of items, changes over time. Nonconformities are defects or occurrences found in the sampled subgroup.

U CHART

A **u-chart** is an attributes control **chart** used with data collected in subgroups of varying sizes. **U-charts** show how the process, measured by the number of nonconformities per item or group of items, changes over time. Nonconformities are defects or occurrences found in the sampled subgroup.

CHARACTERISTICS OF CONTROL CHARTS

A control chart is a time-ordered diagram to monitor a quality characteristic, consisting of:

1. A nominal value, or centre line, the average of several past samples.
2. Two control limits used to judge whether action is required, an upper control limit (UCL) and a lower control limit (LCL).
3. Data points, each consisting of the average measurement calculated from a sample taken from the process, ordered overtime. By the Central Limit Theorem, regardless of the distribution of the underlying individual measurements, the distribution of the sample means will follow a normal distribution. The control limits are set based on the sampling distribution of the quality measurement.

BENEFITS OF USING CONTROL CHARTS

A control chart indicates when something may be wrong, so that corrective action can be taken.

1. The patterns of the plot on a control chart diagnosis possible cause and hence indicate possible remedial actions.
2. It can estimate the process capability of process.
3. It provides useful information regarding actions to take for quality improvement.

OBJECTIVES OF CONTROL CHARTS

To secure information to be used in establishing or changing specifications or in determining whether the process can meet specifications or not.

1. To secure information to be used on establishing or changing production procedures.
2. To secure information to be used on establishing or changing inspection procedures or acceptance procedures or both.
3. To provide a basis for current decision during production.
4. To provide a basis for current decisions on acceptance for rejection of manufacturing or purchased product.
5. To familiarize personnel with the use of control chart.

INVENTORY MANAGEMENT

Inventory: It defined as a comprehensive list of movable items which are required for manufacturing the products and to maintain the plant facilities in working conditions

Inventory Management: “Inventory management refers to the process of ordering, storing and using a company's inventory. This includes the management of raw materials, components and finished products, as well as warehousing and processing such items”.

OBJECTIVES OF INVENTORY MANAGEMENT

1. **To supply the required materials continuously:** there should be a continuous available of materials in the factory or finished goods for trade. The main objective of inventory management in to maintain required inventory so that production and sales process run smoothly.
2. **To minimize the risk of under and over stocking of material:** if a company keeps inventory without proper analysis, there will be a chance of overstocking, which will increase the cost of carrying the inventory or under stocking of inventory that create problem in smooth operation of a business. So one of the main objectives of the inventory management is to minimize the risk caused due to under and over stocking of inventory.

3. To maintain systematic record of inventory: management needs different information regarding inventory for planning and decision-making. A systematic recodes of inventory helps provides such information to the management. It also assists to evaluate the current inventory management policy.

4. To reduce losses, damages and misappropriation of materials: inventory management aims to reduce or remove the losses and misappropriate of materials. This is done by maintaining the proper stocks of materials with utmost care.

5. To minimize the cost associated with inventory: the proper maintenance of the information regarding inventory helps to make decisions like whether to take discounts or not, the size of order to be placed, when to order etc. the total cost associated with inventory may be minimized by analyzing the lot size to be acquired, the offer of discount on variable lot size and the timing of order. Such analysis helps to reduce the unnecessary inventory in inventories.

6. To make stability in price: an effective inventory management system minimizes the effects of regular price fluctuation. This in turn helps to gain the stability is selling price.

INVENTORY CONTROL:

Meaning of Inventory Control:

Inventory control is the technique of maintaining stock-keeping items at the desired level, whether they be raw-materials, goods in process or finished products.

Inventory control, also known as stock control, is regulating and maximizing your company's warehouse inventory.

NEED FOR INVENTORY CONTROL

A business firm maintains inventory for the following purposes:

(i) To carry on the day-to-day operations of business:

The inflow and outflow of inventory cannot be synchronised completely. There is a time lag between production and sales and inventories get built up during this time lag. This is the transactional motive of inventory.

(ii) To ensure regular production:

In the absence of adequate stock of raw materials production would be hampered. Both men and machines would become idle. By holding stocks of raw material a firm can keep its output at a constant level.

(iii) To maintain a safety margin:

It is not possible to predict demand exactly. Therefore, business firms maintain stock of finished goods to safeguard against fluctuations in demand. This is known as the precautionary motive of inventory.

(iv) To take advantage of price:

When a business firm expects rise in prices of inventory it may hold stock of goods when the price is low until the price becomes high. This is called the speculative motive of holding inventories. It is fruitful to hold inventory when expected price rise exceeds the cost of holding inventory.

Objectives of Inventory Control

1. To maintain the overall investment at the lowest level, consistent with operating requirements.
2. To supply the product, raw material, sub-assemblies, semi-finished goods etc. to its users as per their requirements at right time and at right price.
3. To keep inactive, waste, surplus, scrap and obsolete items at the minimum level.
4. To minimize holding, replacement and shortage costs of inventories and maximize the efficiency in production and distribution.
5. To treat inventory as investment which is risky. For some items, investment may lead to higher returns and for others less returns.
6. To protect against inflation since the prices of materials are constantly increasing. Thus it is important to invest in the inventories and save whenever the price of the materials goes up. However this benefit can only be availed if cost of holding inventory is taken care off.
7. To avail quantity discounts on bulk purchases.

Functions of inventory control:

- * To develop policies, plans and standards essential to achieve the objectives To build up a logical and workable plan of organization for doing the job satisfactory
- * To develop procedure and methods that will produce the desired results economically
- * To provide the necessary physical facilities
- * To maintain overall control by checking results and taking corrective actions.

ECONOMIC ORDER QUANTITY (EOQ)

Economic order quantity (EOQ) is the ideal order quantity a company should purchase to minimize inventory costs such as holding costs, shortage costs, and order costs.

Formula and Calculation of Economic Order Quantity

The formula for EOQ is:

$$Q = \sqrt{\frac{2DS}{H}}$$

where:

Q = EOQ units

D = Demand in units (typically on an annual basis)

S = Order cost (per purchase order)

H = Holding costs (per unit, per year)

Determine EOQ:

Step1:

Total Ordering cost per year = No. of orders placed per year x ordering cost per Order

= $(A/S) \times O$ (A = Annual demand, S = Size of each order (units per order), O = Ordering cost per order)

Step2:

Total Carrying cost per year = Average inventory level x Carrying cost per year = $(S/2) \times C$

(A = Annual demand, S = Size of each order (units per order), C = Carrying cost per unit)

Step3:

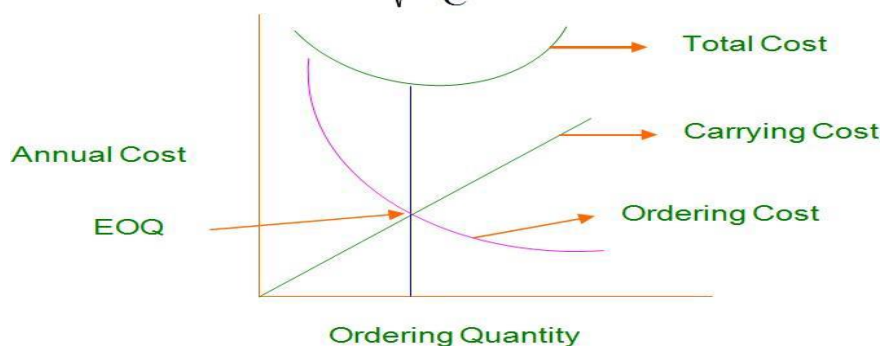
EOQ is one where the total ordering is equal to total carrying cost

$$\frac{A}{S} \times O = \frac{S}{2} \times C$$

$$2AO = S^2 \times C$$

$$S^2 = \frac{2AO}{C}$$

$$S = \sqrt{\frac{2AO}{C}}$$



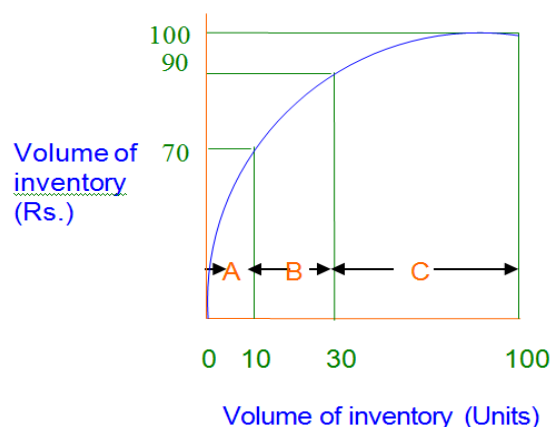
Where S is the Economic order quantity, A is the annual demand in units, O is the ordering cost per order and C is the carrying cost per unit.

ABC ANALYSIS

ABC method of inventory control involves a system that controls inventory and is used for materials and throughout the distribution management. It is also known as selective inventory control or SIC.

ABC analysis is a method in which inventory is divided into three categories, i.e. A, B, and C in descending value. The items in the A category have the highest value, B category items are of lower value than A, and C category items have the lowest value.

Inventory control and management are critical for a business. They help to keep their costs under control. The ABC analysis helps the business to control inventory by letting the management focus on the highest value goods (the A-items) and not on the many low-value goods (the C-items).



A - Refers to high value item

B - Refers to medium value item

C - Refers to low value item

A category comprises of inventory, which is very costly and valuable. Normally 70% of the funds are tied up in such costly stocks, which would be around 10% of the total volume of stocks. Because the stocks in this category are very costly, these require strict monitoring on a day-to-day basis.

B category comprises of inventory, which is less costly. Twenty percent of the funds are tied up in such stocks and these accounts for over 20% of the volume of stocks. These items require monitoring on a weekly or fortnightly basis.

C category consists of such stocks, which are of least cost. Volume-wise, they form 70% of the total stocks but value-wise, they do not cost more than 10% of the investment in the stocks. This category of stocks can be monitored on a monthly or bi-monthly basis.

The following table summarizes the concept of ABC analysis;

Category	Value (%)	Volume (%)	Desired Degree of Control
A	70	10	STRICT
B	20	20	MODERATE
C	10	70	LOW

2.2 MARKETING MANAGEMENT

What Is Marketing?

Marketing refers to activities a company undertakes to promote the buying or selling of a product or service. Marketing includes advertising, selling, and delivering products to consumers or other businesses.

Marketing is a social process by which individuals and groups obtain what they need and want through creating and exchanging products and value with others.

MARKETING MANAGEMENT

Marketing management is the process of planning and executing the conception, pricing, promotion, and distribution of ideas, goods, and services to create exchanges that satisfy individual and organizational goals.

“Marketing management may be defined as the process of management of marketing programmes for accomplishing organisational goals and objectives. It involves planning, implementation and control of marketing programmes or campaigns”.

NATURE OF MARKETING MANAGEMENT

The following points highlights the nature of marketing management

Customer Focus

Marketing is a customer-centered function of the business. It aims at finding out what customers want and fulfilling their needs by delivering them the right products.

Creates Market Offering

Marketing provides offers of various goods and services to potential customers. It is the one that communicates all information regarding products like its prices, uses, quality, and technology to customers.

Exchange Oriented

It is a process which aims at exchanging products among buyer and seller. Marketing attracts and influences people to buy the products of the company.

Continuous Activity

Marketing is a regular and continuous activity of business for selling their products. Businesses always need to monitor the marketing environment and should accordingly plan, implement, and control all marketing programs.

Goal-Oriented

Marketing is a goal-oriented business activity that aims at achieving the desired sales and profitability. It focuses on approaching more and more customers and thereby satisfying their needs by delivering them the required goods or services.

Manages 4 P's

It is a combination of four elements that are product, place, price, and promotion. The whole marketing system is made up of these variable factors which are influenced by customer behavior, competition, trade factors, etc.

Creates Utilities

Marketing creates various utilities such as form utility, time utility, and place utility. It creates form utility by manufacturing the right product using inputs, time utility by storing goods in warehouses, and place utility by delivering goods properly to end customers.

Economic Process

It is a process that involves exchanges of goods in monetary terms. Marketing is one by means of which monetary transactions as per the exchange value of goods take place for transferring goods among buyers and sellers.

Guiding element of business

Modern Marketing is the heart of industrial activity that tells what, when, how to produce. It is capable of guiding and controlling business.

System of Interacting Business Activities

Marketing is the system through which a business enterprise, institution or organization interacts with the customers with the objective to earn profit, satisfy customers and manage relationship. It is the performance of business activities that direct the flow of goods and services from producer to consumer or user.

FUNCTIONS OF MARKETING MANAGEMENT

1. Selling:

It is core of marketing. It is concerned with the prospective buyers to actually complete the purchase of an article. It involves transfer of ownership of goods to the buyer. Selling plays an important part in realising the ultimate aim of earning profit.

2. Buying and Assembling:

It involves what to buy, of what quality, how much from whom, when and at what price. People in business buy to increase sales or to decrease costs.

A manufacturer buys raw materials, spare parts, machinery, equipment's, etc. for carrying out his production process and other related activities. A wholesaler buys products to resell them to the retailers.

Assembling means to purchase necessary component parts and to fit them together to make a product.

3. Transportation:

Transportation is the physical means by which goods are moved from the places where they are produced to those places where they are needed for consumption. It creates place, utility.

Transportation is essential from the procurement of raw material to the delivery of finished products to the customer's places.

4. Storage:

It involves holding of goods in proper (i.e., usable or saleable) condition from the time they are produced until they are needed by customers (in case of finished products) or by the production department (in case of raw materials and stores); storing protects the goods from deterioration and helps in carrying over surplus for future consumption or use in production.

5. Standardization and Grading:

The other activities that facilitate marketing are standardisation and grading. Standardisation means establishment of certain standards or specifications for products based on intrinsic physical qualities of any commodity. A standard conveys a uniformity of the products.

Grading means classification of standardised products into certain well defined classes or groups. It involves the division of products into classes made of units possessing similar characteristics of size and quality.

6. Financing:

It involves the use of capital to meet financial requirements of agencies dealing with various activities of marketing. The services to provide the credit and money needed, the costs of getting merchandise into the hands of the final user is commonly referred to as finance function in marketing.

7. Risk Taking:

Risk means loss due to some unforeseen circumstances in future. Risk bearing in marketing refers to the financial risk interest in the ownership of goods held for an anticipated demand including the possible losses due to a fall in prices and the losses from spoilage, depreciation, obsolescence, fire and floods or any other loss that may occur with the passage of time.

8. Market Information:

The importance of this facilitating function of marketing has been recognised only recently. The only sound foundation on which marketing decisions may be based is correct and timely market information. Right facts and information reduce the aforesaid risks and thereby result in cost reduction.

MARKETING MIX

The term marketing mix was first coined by James Culliton and was described as 4P`s by Professor Jerome McCarthy. Since then the terms ‘marketing mix’ and ‘4P`s’ have been used synonymously.

Marketing Mix refers to the set of controllable factors that a firm can use to influence the buyer’s response within a given marketing environment. It is used to describe the combination of four inputs (product, price, place, promotion) which constitute the core of a company’s marketing system.

Marketing Mix is the combination of different marketing decision variables used by a firm to market its goods and services. These four marketing mix elements form the core of all marketing efforts.

Elements of Marketing Mix



Product – A product is anything that satisfies a consumer need. It may be a good, service, event, experience etc. The first step in marketing is to identify and develop a product that consumers’ demand.

It involves decisions regarding:

- Physical Product
- Product mix
- Product variables
- Product Quality
- Branding
- Packaging etc.

Price – Price is the amount of money that a buyer has to pay in order to own a product. A marketer makes price alterations in order to attract the customers through discounts, allowances, coupons etc. The element of price is concerned with determining:

- Level of prices (low, high)
- Range of prices
- Price–Quality relationship
- Competitors pricing
- Government control over prices
- Psychological effect of pricing

Place – It is concerned with the delivery of a product from the place of production to the final consumer. A producer may sell a product directly or through intermediaries. It involves decisions regarding:

- Channel levels – Manufacturer to Wholesaler to Retailer to Consumer and creation of:
 - ✓ Time utility – Product must be available when needed
 - ✓ Place utility – Locate products where they can be easily acquired.
 - ✓ Possession – Minimize the number of obstacles between the manufacturer and consumer.
- Inventory management and control
- Physical distribution of goods
- Order processing and Transportation

Promotion – Promotion refers to communication of information about a product and its features to potential customers through various forms of media. It is the process of educating customers through various forms of media about

- Product Quality
- Product Utility
- Product Price

It involves selection of a combination of tools of promotion like advertising, sales promotion, personal selling, publicity and public relations to persuade potential customers to buy a product or service.

In case of services, the producer-oriented model of marketing mix is consists of 7P's. Including the above 4P's there are additional 3P's - **Physical Evidence**, **People**, and **Process**. Physical evidence refers to elements like uniform of employees, signboards, and etc. People refers to the employees of the organization comes in contact with the customers in the process of marketing. Process refers to the systems and processes followed within organization.

CHANNELS OF DISTRIBUTION

Distribution refers to all the activities undertaken to transfer the product from the manufacturer to the consumer.

The distribution channel is made up of various kinds of intermediaries such as retailers, distributors, wholesalers, and agents.

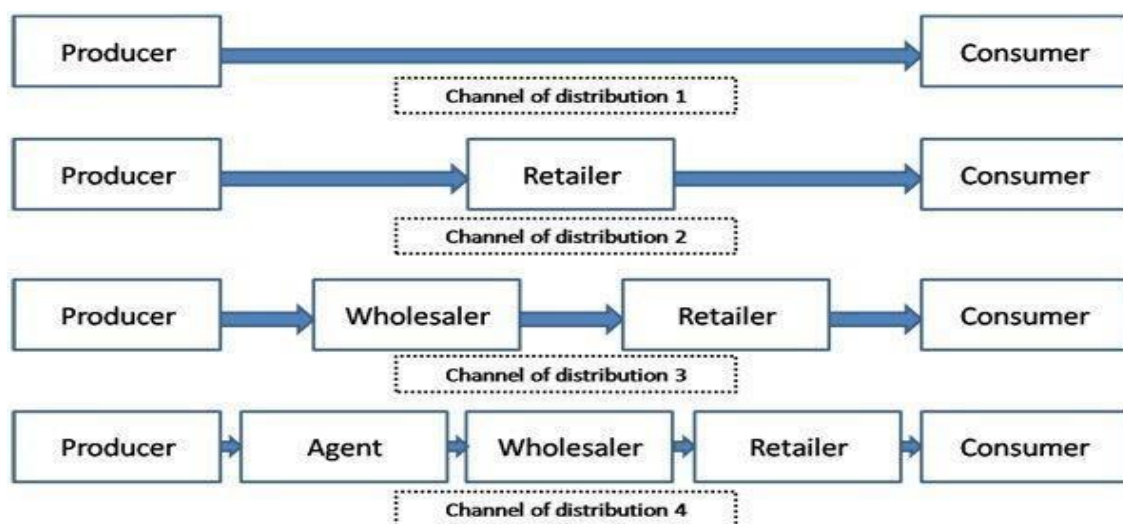
Distribution channel can be defined as the passage of goods or services from the producers or providers to the target customers through a plethora of modes or channels.

In other words, flow of goods from producers to consumers is called the channels of distribution.

Different types of channel of distribution are as follows:

Manufacturers and consumers are two major components of the market. Intermediaries perform the duty of eliminating the distance between the two. There is no standardised level which proves that the distance between the two is eliminated.

Based on necessity the help of one or more intermediaries could be taken and even this is possible that there happens to be no intermediary. Their description is as follows:



(A) Direct Channel or Zero Level Channels CD1:

When the manufacturer instead of selling the goods to the intermediary sells it directly to the consumer then this is known as Zero Level Channel. Retail outlets, mail order selling, internet selling and selling

(B) Indirect Channels:

When a manufacturer gets the help of one or more middlemen to move goods from the production place to the place of consumption, the distribution channel is called indirect channel. Following are the main types of it:

1. One Level Channel CD2:

In this method an intermediary is used. Here a manufacturer sells the goods directly to the retailer instead of selling it to agents or wholesalers. This method is used for expensive watches and other like products. This method is also useful for selling FMCG (Fast Moving Consumer Goods).

2. Two Level Channel CD3:

In this method a manufacturer sells the material to a wholesaler, the wholesaler to the retailer and then the retailer to the consumer. Here, the wholesaler after purchasing the material in large quantity from the manufacturer sells it in small quantity to the retailer.

Then the retailers make the products available to the consumers. This medium is mainly used to sell soap, tea, salt, cigarette, sugar, ghee etc.

3. Three Level Channel CD4:

Under this one more level is added to Two Level Channel in the form of agent. An agent facilitates to reduce the distance between the manufacturer and the wholesaler. Some big companies who cannot directly contact the wholesaler, they take the help of agents. Such companies appoint their agents in every region and sell the material to them.

Then the agents sell the material to the wholesalers, the wholesaler to the retailer and in the end the retailer sells the material to the consumers.

ADVERTISEMENT AND SALES PROMOTION

Advertising is a means of communication with the users of a product or service. Advertisements are messages paid for by those who send them and are intended to inform or influence people who receive them, as defined by the Advertising Association of the UK.

Advertisement (ad) is an efficient and effective technique to promote goods, services, and ideas. It is a paid form of non-personal communication wherein business information is made available for potential customers.

Objectives of Advertisements

objectives of advertisement, significant of them are –

- To promote newly launched products among the potential customers.
- To promote personal selling program.
- To create awareness among maximum people about your business in a short period of time.
- To enter national or even international market and motivate new group of customers.
- To enhance the goodwill and build credibility among the customers by promising to provide better quality of products and services.

TYPES OF ADVERTISING

Online Advertising

Online **advertisings** or **digital advertisings** as a form in which the message is conveyed via the internet.

SMS advertising

SMS marketing is the major source of mobile advertising. Users are informed about the product or service in 160 or fewer characters. This was when the internet was not available on mobile phones.

Television Ads

About a decade ago television was the most popular form of advertising. Events like the super bowl, international cricket games, Olympics where the top attractions for advertisers to advertise about their products.

Ads in Theatres

The advertisements in movie theatres before all the movies start or during the intimation are called movie ads.

Product Placement

Product placement is called covert advertising wherein a product is quietly embedded in the entertainment media.

Radio

Radio advertisements are the ones that are broadcast it through radio waves and heard on radios all over the place. These mostly consist of audible advertisements or jingles. While some

consider this to be an ineffective form of advertising there are still many followers listen to the radio every morning.

Print

Printing is the slowly decreasing form of advertising. There were days before the evolution of television when printing was a major source of advertising and considered to be one of the most effective media. But since the explosion of television usage, print advertisements have taken a backseat.

- **Magazine advertising**

These are also known as periodical advertisements in which a weekly fortnightly or monthly magazine are used for advertising.

- **Brochures or handouts**

Brochures are specific advertising materials used to promote a particular product usually given at a point of sale are handed out at different locations.

- **Newspaper**

Newspapers display a huge number of ads in them, right from matrimonial services to job hunt, to the notifications and circulars from the Governments.

Outdoor

Outdoor adv. consists of displaying large posters banners or hoardings with the advertisement. These are displayed on the side of the road, on the glass of large buildings, or on specifically targeted places that have huge inflow from the public.

Global

Google advertising is more of a strategy than the type of advertising but some have classified these as a separate type of advertising. In Global Advertising, a single Ad runs throughout all the countries where the product is present, irrespective of the audience and their language.

Outdoor Blank Space Advertising

It is a newer form of advertising which ensures a large reach of audience. Occupying the empty spaces for advertisements is known as space advertising. Examples include the spaces of metros, buses, cabs, flight seats & movie theatre seats (where advertising is done on the removable seat covers) etc.

Online E-mailers

While some have classified this under internet advertising E-mailers have gained a separate category owing to the fact of personalization. The emails sent from companies have a personalized message including a personalized greeting makes E-mailers more effective than most of the forms of advertising.

Importance of Advertising

Advertising is an integral part of our economic and social life. As a powerful technique of promoting sales, it has been doing wonders in the area of distribution. The role of advertising can be analyzed from five different angles namely, manufacturers; middleman; sales force; customer and society.

1. Importance of Advertising to Manufacturers:

Every manufacturer and producer, who wants to make available his goods to people at profit, do take full advantage of advertising to popularize their products and services.

The major benefits that are available to manufacturer are:

- (a) Increasing Sales – Even the best product cannot be sold on its own, unless people know about the product. In today's highly sensitive and competitive market, a firm cannot maximize its profit, unless it multiplies its sales turnover. A regular and frequent advertising helps the producer to obtain this objective.
- (b) Helps in maintain existing market and explore new market – Every forward looking company eyes on future prospects without losing its current position. A company's success is reflected by how it maintains its current position and future expansion. Advertising helps the manufacturer, in this regard, to face competition effectively.
- (c) Helps to control price of product – Through advertising, it is possible to control price of the product especially in retail market. Very often greedy retailer charge higher price from the customer. The manufacturer can help them by printing the price on the packages.

2. Importance of Advertising for Middlemen:

Middlemen are essential link between producer and consumer.

The benefits which advertising offers to middlemen are:

- (a) Quick Sale – Every retailer holds stock of different producer. Advertising, by making the range of products known to customer, helps the retailer to quickly sell its stock.
- (b) Act as Salesmen – Advertising has been rightly described as salesmanship in print. Advertising perform task of travelling salesmen at least cost. That is why; most of the retail organizations do not employ large number of travelling salesman. Instead, they spend on advertising, which attracts customers to the shops, where counter salesmen cater to their need.

3. Importance of Advertising for Consumer:

Ultimate aim of all marketing efforts is to satisfy the needs of the customer by providing the goods and services.

Advertising is essential for consumer due to following reasons:

(a) Quick decision making and saves time – In today's competitive world market is full of different types of product which satisfy needs and want. Every producer claims his product to be superior one. In such situation, advertising helps the consumer in comparing features; price; utility; quality etc. of the product, and select the best.

(b) Better Quality Product at Reasonable Price – Advertising promotes good quality product by printing their image in the minds of consumer. Due to this, bad product goes out of the market. Moreover, it increases competition in the market, which helps consumer in getting the product at reasonable price.

4. Importance of Advertising for Society:

Advertising is not only beneficial for manufacturer; retailer and customer but also for the society.

Various benefits offered by advertising to the society are:

(a) Uplift Standard of Living – Advertising has made it possible for general public to use those products which were luxuries of yesterday. Advertising increases the consumption pattern, it results in more production at least cost, which in turn increases earning of the society and thus, standard of living of the society.

(b) Generates Employment – Advertising generates gainful employment opportunities both directly and indirectly. It directly generates employment painters; artist; photographers; technician etc. Indirectly, it gives employment by supporting all those industries like – paper; colour; electronic etc. which supply inputs for the advertising activities.

SALES PROMOTION

Sales promotion is a set of marketing technologies aimed to stimulate the demand in particular products and increase brand awareness.

Sales promotion is a part of the promotional mix where the business uses many short-term customer-oriented strategies to stimulate the demand for its product by making it look more attractive and/or worthy.

Sales Promotion Definition

Sales promotion is a marketing strategy where the product is promoted using short-term attractive initiatives to stimulate its demand and increase its sales.

This strategy is usually brought to use in the following cases –

- To introduce new products,
- To sell out existing inventories,

- To attract more customers, and
- To lift sales temporarily.

American Marketing Association defines sales promotion as –

Media and nonmedia marketing pressure applied for a predetermined, limited period of time in order to stimulate trial, increase consumer demand, or improve product availability.

IMPORTANCE OF SALES PROMOTION

Sales promotion is a handy technique to fulfil the short term sales goals by persuading potential customers to buy the product. It is an important promotional strategy to –

- **Spread information about the brand** to new customers or new market
- **Stabilize sales volume** and fulfil short-term sales goals
- **Stimulate demand** for a short term by making the product look like a great deal.

OBJECTIVES OF SALES PROMOTION

The answer to the question *what is sales promotion?* also gives a hint to sales promotion objectives, the main objective being lifting the sales temporarily.

Other objectives include but are not limited to –

To Create Market for New Products

It is sometimes hard to establish demand for a new product in a market of similar products. In such cases, the company opt for increasing some sales by using sales promotion strategies like penetration pricing, offers, discounts, and scarcity principle.

To Remain Competitive

Companies use temporary sales promotion techniques to compete with competitor's short term marketing strategies.

To Gain Dealers Trust

Sales promotion techniques increase the sales of the products. This increases dealers' income and results in them preferring the brand more.

To Take Products to New Markets

New markets are often hard to enter. Sales promotion increases traction and makes more customers try the new product.

Increase Brand Awareness

It includes attractive incentives which help increase brand awareness, which eventually leads to more sales.

Woo Existing Customers

Sales promotion is also used to tackle the poaching strategies of competitors and keep existing customers with the brand.

SALES PROMOTION STRATEGIES

Sales promotion strategies can be divided into three broad types. These are –

- **Pull Strategy** – The pull strategy attempts to get the customers to ‘pull’ the products from the company. It involves making use of marketing communication and initiatives like seasonal discounts, financial schemes, etc.
- **Push Strategy** – The push strategy attempts to push the product away from the company to the customers. It involves convincing the intermediary channels to push the product from the [distribution channels](#) to the final consumers using promotional and personal selling efforts. This strategy involves making use of tactics developed especially for resellers, merchants, dealers, distributors, and agents.
- **Hybrid Strategy** – A hybrid sales promotion strategy makes use of both the pull and push strategy to sell the product with the least resistance possible. It involves attracting the customers using special coupons and also providing incentives to the merchants to sell the brand’s products.

TYPES OF SALES PROMOTIONS

Sales promotion can be broadly divided into two types according to whom the promotion is targeted to. These are –

Consumer Sales Promotion

When the sales promotion strategies are targeted to the end consumers, it is referred to as consumer sales promotion. An example would be offering 20% off on certain products to the customers. The main motive of consumer-oriented promotion is to increase sales directly by attracting new customers and wooing existing ones.

Sales Promotion Techniques Targeted To Consumers

Sales promotion tools used for consumer-oriented promotion are –

- **Free Samples:** Distributing free samples increases brand awareness and triggers the psychology of ownership where the person chooses the promoted product if he liked the sample.

- **Free Gifts** – Offering free gifts attract customers as they get more while paying for less.
- **Discounts/Discount Coupons** – Discount coupons are a great method of increasing sales for the short term. People go for discount coupons as they let them buy the products they couldn't afford otherwise.
- **Exchange Schemes** – Exchange schemes attract many customers as they get some value even for their old product.
- **Finance Schemes** – Finance schemes like no-cost EMI, low-interest EMI, etc. makes it easier for customers to purchase expensive products.
- **Shipping Schemes** – Sometimes huge shipping costs discourage the customers from buying products. Such short term shipping schemes remove friction.
- **Bundle Discounts** – These deals are a great way to reduce unsold inventory. It includes selling bundled products at a price lesser than when those number of products are bought separately.
- **Bulk Purchase Deals** – This is a great sales promotion tactic to reduce unsold inventory. It includes providing discount to customers who buy in bulk.

Trade Sales Promotion

When the promotion activities are strategized keeping in mind the dealers, distributors, or agents, it is called trade sales promotion. In this type of sales promotion, offers are provided within the trade channels with an aim to woo retailers, wholesalers, agents, or distributors. This is done to get more shelf space as compared to competitors, motivate the dealers to sell more of the brand's products and to increase the sales indirectly.

Sales Promotion Techniques Targeted To Traders

- **Point Of Purchase Displays** – This includes providing free point of purchase (POP) display units to the retailers to increase their sales.
- **Trade Shows** – Trade shows are a great sales promotion strategy where the business promotes its product to thousands of traders in the trade show. Trade shows also witness huge discounts as compared to when bought usually.
- **Push Money** – Also known as spiffs, this technique includes extra payments to traders to motivate them to meet specified goals. For example, giving them a \$50 bonus per unit for selling product A and \$30 for selling product B for a specified time period.
- **Deal Loaders** – These are the gifts provided to the traders (wholesalers and retailers) for ordering a certain quantity of product.

- **Trade Deals** – These are special concessions provided to the merchants to encourage them to promote a specific product and increase its sales for a limited time period.
- **Buying Allowances** – Special discounts provided to the sellers when they order a specified number of products.

Sales Promotion Examples

A product can be promoted for a limited time using innumerable tactics. Here are a few examples of sales promotion tactics that exist –

Black Friday Sale

Black Friday sale is a seasonal sale which occurs only once a year. It involves huge discounts and special offers which are limited to a day. As a result, it increases the sales manifold.

Buy One Get One

Buy One Get One (BOGO) is a popular type of sales promotion where two products are offered at a price of one. This works great to promote a new product or clear the inventory at the end of the season.

Referral Bonuses

Referral marketing is a great sales promotion strategy where the company pushes its own customers to bring in new customers. This is done by providing them with special discounts, offers, cashback, or actual monetary benefits.

MARKETING STRATEGY

A **marketing strategy** refers to a business's overall game plan for reaching prospective consumers and turning them into customers of their products or services.

A **marketing strategy** contains the company's value proposition, key brand messaging, data on target customer demographics, and other high-level elements.

A Marketing Strategy is the long term planning of business objectives that the company wants to achieve. For these to be achieved it is important to choose well the specific actions to consolidate the reputation of products and services or increase sales in the market.

Importance of having a Marketing Strategy in business

Choosing a suitable marketing strategies for the company will bring great advantages such as:

- Increased sales
- Creating sustainable growth for the company
- Understanding what your customers want

- You will meet the needs and exceed the expectations of the customers
- You will strengthen the relationship with the target market
- You will build the brand in the consumer's mind

TYPES OF MARKETING STRATEGIES

There are different types of marketing strategies available. Picking up a marketing strategy includes analyzing the needs of your business, your target audience and specifications of your products.

The two main types of marketing strategy are:

1. Business to business (B2B) marketing
2. Business to consumer (B2C) marketing

The most common form of marketing is business to consumer (B2C) marketing. Following are the different types of marketing strategies available.

1. Paid advertising

This includes multiple approaches for marketing. It includes traditional approaches like TVCs and print media advertising. Also, one of the most well-known marketing approach is internet marketing. It includes various methods like PPC (Pay per click) and paid advertising.

2. Cause marketing

Cause marketing links the services and products of a company to a social cause or issue. It is also well known as cause related marketing.

3. Relationship marketing

This type of marketing is basically focused on customer building. Enhancing existing relationships with customers and improving customer loyalty.

4. Undercover marketing

This type of marketing strategy focuses on marketing the product while customers remain unaware of the marketing strategy. It is also known as stealth marketing.

5. Word of mouth

It totally relies on what impression you leave on people. It is traditionally the most important type of marketing strategy. Being heard is important in business world. When you give quality services to customers, it is likely that they'd promote you.

6. Internet marketing

It is also known as cloud marketing. It usually happens over the internet. All the marketing items are shared on the internet and promoted on various platforms via multiple approaches.

7. Transactional marketing

Sales is particularly the most challenging work. Even for the largest retailers, selling is always tough especially when there are high volume targets. However with the new marketing strategies, selling isn't as difficult as it was. In transactional marketing the retailers encourage customers to buy with shopping coupons, discounts and huge events. It enhances the chances of sales and motivates the target audience to buy the promoted products.

8. Diversity marketing

It caters diverse audience by customizing and integrating different marketing strategies. It covers different aspects like cultural, beliefs, attitudes, views and other specific needs.

PRODUCT LIFE CYCLE

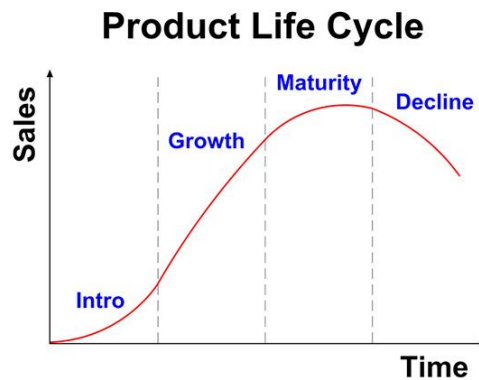
We have a life cycle, we are born, we grow, we mature, and finally we pass away. Similarly, products also have life cycle, from their introduction to decline they progresses through a sequence of stages. The major stages of the product life cycle are - introduction, growth, maturity, and decline. ***Product life cycle describes transition of a product from its development to decline.***

The time period of product life cycle and the length of each stage varies from product to product. Life cycle of one product can be over in few months, and of another product may last for many years. One product reach to maturity in years and another can reach it in few months. One product stay at the maturity for years and another just for few months. Hence, it is true to say that length of each stage varies from product to product.

Product life cycle is associated with variation in the marketing situation, level of competition, product demand, consumer understanding, etc., thus marketing managers have to change the marketing strategy and the marketing mix accordingly.

Product life cycle can be defined as ***"the change in sales volume of a specific product offered by an organisation, over the expected life of the product."***

Stages of the Product Life Cycle



The four major stages of the product life cycle are as follows :-

1. Introduction,
2. Growth,
3. Maturity, and
4. Decline.

Introduction Stage

At this stage the product is new to the market and few potential customers are aware with the existence of product. The price is generally high. The sales of the product is low or may be restricted to early adopters. Profits are often low or losses are being made, this is because of the high advertising cost and repayment of developmental cost. At the introductory stage :-

- The product is unknown,
- The price is generally high,
- The placement is selective, and
- The promotion is informative and personalised.

Growth Stage

At this stage the product is becoming more widely known and acceptable in the market. Marketing is done to strengthen brand and develop an image for the product. Prices may start to fall as competitors enters the market. With the increase in sales, profit may start to be earned, but advertising cost remains high. At the growth stage :-

- The product is more widely known and consumed,
- The sales volume increases,
- The price begin to decline with the entry of new players,

- The placement becomes more widely spread, and
- The promotion is focused on brand development and product image formation.

Maturity Stage

At this stage the product is competing with alternatives. Sales and profits are at their peak. Product range may be extended, by adding both width and depth. With the increases in competition the price reaches to its lowest point. Advertising is done to reinforce the product image in the consumer's minds to increase repeat purchases. At maturity stage :-

- The product is competing with alternatives,
- The sales are at their peak,
- The price reaches to its lowest point,
- The placement is intense, and
- The promotion is focused on repeat purchasing.

Decline Stage

At this stage sales start to fall fast as a result product range is reduced. The product faces reduced competition as many players have left the market and it is expected that no new competitor will enter the market. Advertising cost is also reduced. Concentration is on remaining market niches as some price stability is expected there. Each product sold could be profitable as developmental costs have been paid at earlier stage. With the reduction in sales volume overall profit will also reduce. At decline stage :-

- The product faces reduced competition,
- The sales volume reduces,
- The price is likely to fall,
- The placement is selective, and
- The promotion is focused on reminding.

3. HUMAN RESOURCE MANAGEMENT

What Is HRM?

Human resource management (HRM) is the process of employing people, training them, compensating them, developing policies relating to them, and developing strategies to retain them.

HRM is defined as the process of managing people in organizations in a structured and thorough manner.

In a line HRM encompasses effective management of human resources which includes certain distinct concepts such as recruitment, selection, training, T&D, promotions, compensation etc. which always deal with all employees in the organization.

NATURE OF HRM:

Here we can observe certain basic features of HRM which are taking place in the organization.

- By nature, it is pervasive because it can be seen or existed in all the organizations such as TATA, SBI.
- The fundamental nature of HRM is it moves around the employees.
- It directly deals with employee skills, creativity, talent, potentiality etc.
- By nature, it is result, dynamic – oriented. (Challenging, risk – oriented projects).
- Basically, HR department makes the employees in HR department more worthful or industry fit.
- By nature, it is multidisciplinary like selection, T&D, promotions, organization development, QWL (Quality work Life) etc.
- As HRM always deals with employees, HR department ensures high level employee job satisfaction, fair relations, healthy working conditions etc.
- HRM strongly believes that employees are biggest assets of the organization.
- By nature, it ensures employee attention, employee attachment (involvement) by following fair policies.
- It also deals with delights like free education, medical facilities etc.
- HRM deals with individual and team or group level.

FUNCTIONS OF HUMAN RESOURCE MANAGEMENT

Human Resource Management functions can be classified in following three categories.

- Managerial Functions,
- Operative Functions, and
- Advisory Functions

The **Managerial Functions** of Human Resource Management are as follows

- 1. Human Resource Planning** - In this function of HRM, the number and type of employees needed to accomplish organizational goals is determined. Research is an important part of this function, information is collected and analyzed to identify current and future human resource needs and to forecast changing values, attitude, and behaviour of employees and their impact on organization.
- 2. Organizing** - In an organization tasks are allocated among its members, relationships are identified, and activities are integrated towards a common objective. Relationships are established among the employees so that they can collectively contribute to the attainment of organization goal.
- 3. Directing** - Activating employees at different level and making them contribute maximum to the organization is possible through proper direction and motivation. Taping the maximum potentialities of the employees is possible through motivation and command.
- 4. Controlling** - After planning, organizing, and directing, the actual performance of employees is checked, verified, and compared with the plans. If the actual performance is found deviated from the plan, control measures are required to be taken.

The **Operative Functions** of Human Resource Management are as follows:

- 1. Recruitment and Selection** - Recruitment of candidates is the function preceding the selection, which brings the pool of prospective candidates for the organization so that the management can select the right candidate from this pool.
- 2. Job Analysis and Design** - Job analysis is the process of describing the nature of a job and specifying the human requirements like qualification, skills, and work experience to perform that job. Job design aims at outlining and organizing tasks, duties, and responsibilities into a single unit of work for the achievement of certain objectives.
- 3. Training and Development** - This function of human resource management helps the employees to acquire skills and knowledge to perform their jobs effectively. Training and development programs are organized for both new and existing employees. Employees are prepared for higher level responsibilities through training and development.
- 4. Performance Appraisal** - Human resource professionals are required to perform this function to ensure that the performance of employee is at acceptable level.
- 5. Wage and Salary Administration** - Human resource management determines what is to be paid for different type of jobs. Human resource management decides employees compensation which includes - wage administration, salary administration, incentives, bonuses, fringe benefits, and etc.,.

6. Employee Welfare - This function refers to various services, benefits, and facilities that are provided to employees for their well being.

7. Maintenance - Human resource is considered as asset for the organization. Employee turnover is not considered good for the organization. Human resource management always try to keep their best performing employees with the organization.

8. Labour Relations - This function refers to the interaction of human resource management with employees who are represented by a trade union. Employees comes together and forms an union to obtain more voice in decisions affecting wage, benefits, working condition, etc.,

9. Personnel Research - Personnel researches are done by human resource management to gather employees' opinions on wages and salaries, promotions, working conditions, welfare activities, leadership, etc,. Such researches helps in understanding employees satisfaction, employees turnover, employee termination, etc,.

10. Personnel Record - This function involves recording, maintaining, and retrieving employee related information like - application forms, employment history, working hours, earnings, employee absents and presents, employee turnover and other other data related to employees.

The **Advisory Functions** of Human Resource Management are as follows:

Human Resource Management is expert in managing human resources and so can give advice on matters related to human resources of the organization. Human Resource Management can offer advice to:

1. Advised to Top Management

Personnel manager advises the top management in formulation and evaluation of personnel programs, policies, and procedures.

2. Advised to Departmental Heads

Personnel manager advises the the heads of various departments on matters such as manpower planning, job analysis, job design, recruitment, selection, placement, training, performance appraisal, etc

IMPORTANCE/SIGNIFICANCE OF HRM

To maintain quality of work life

Quality of work life is the perception of employees regarding the relationship between working condition with management. Simply, it is the relationship between employees and total work environment.

Quality of work life is concerned with the employee's perception of physical as well as psychological wellbeing at workplace and it can be obtained by maintaining work autonomy, work freedom, job recognition, belongingness, rewards, etc.

To increase productivity and profit

Human resource management ensures right quality and quantity of personnel in workplace, apart from which, it creates opportunities to facilitate and motivate individual and group of employees to grow and advance their career. Motivated employees work hard to meet their personal career goal which directly influences productivity of organization. Minimizing cost and maximizing profit is the essence of productivity.

To produce employees who are easily adaptable to change

Human resource management performs various functions, among which training and development of employees is one of the vital ones. Training and development programs keep employees updated with skills necessary to adjust with alteration of organization's environment, structure and technology.

To match demand and supply of human resource

Human resource management probes existing human resource in certain interval of time to identify if the company has adequate number of workers or not. In case when company has inadequate human resource, human resource management performs activities like recruitment and selection to balance the need. In the same way, it lay-off inefficient employee in case when the company has excessive employees.

To retain employees and motivate them to accomplish company's goal

Utilizing human resource to their fullest and maintaining them in the company for a long term is a major function of human resource management. Under this function, HRM performs tasks like providing fringe benefits, compensation and rewards to the deserving employees. These activities do not only help in retaining employees in the company but also in achieving organization's goal effectively.

To recognize merit and contribution of employee

Human resource management performs timely appraisal of employee's performance in order to recognize excellent and poor workers. The excellent workers are then rewarded as a return for their contribution.

To create a feeling of belongingness and team spirit in the employee

Human resource management designs job on the basis of teamwork. As teamwork demands contribution from every team member, HRM focuses on making each and every employee feel valued so that employee contributes best from their side.

To sustain business in the market

Human resource management makes sure that the company has the best employee because promising employees are capable of taking their company ahead in the race by delivering excellent results. This guarantees organization's success and stability in future.

To resolve conflicts

In any company, either big or small, conflicts may arise between any parties/group. Conflicts are inevitable and they should not be ignored. Human resource management acts as a consultant to sort out such conflicts timely and conduct other organizational activities smoothly.

To develop corporate image

Every company should maintain good public image in order to sustain in the market. Any individual would like to work for companies which are known for moral and social behavior. On the other hand, companies that do not treat their employees in a good manner can run through employee deficiency and even collapse.

HUMAN RESOURCE PLANNING (HRP)

Human resource planning (HRP) is the continuous process of systematic **planning** ahead to achieve optimum use of an organization's most valuable asset—quality employees.

Human resources planning ensures the best fit between employees and jobs while avoiding manpower shortages or surpluses.

Human Resource Planning (HRP) is the process of forecasting the future human resource requirements of the organization and determining as to how the existing human resource capacity of the organization can be utilized to fulfill these requirements.

Need for Human Resource Planning

The need of HRP may arise because of the following reasons:

1. In India, unemployment is a grave concern. Scarcity of manpower and that too with the required skill sets and competence, has given rise for need of Human Resource Planning.
2. It comes handy for smooth and continuous supply of workers when a huge number of employees is retiring, or leaving the company or maybe they are incapable of working due to psychological or physical ailments.
3. There is a need for Human Resource Planning when there is an increase in employee turnover, which is obvious. Some examples of this turnover are promotions, marriages, end of contract, etc.
4. Technological changes lead to a chain of changes in the organization, right from skill sets product methods and administration techniques. These changes lead to an overall change in

the number of employees required and with entirely different skill set. It is here that the Human Resource Planning helps the organization deal with the necessary changes.

5. Human Resource Planning is required to meet the requirements of diversification and growth of a company.
6. There is a need for Human Resource Planning in downsizing the resources when there is a shortage of manpower. Similarly, in case of excess resources, it helps in redeploying them in other projects of the company.

Importance of Human Resource Planning

After the need for HRP, it is apt to discuss the importance of it. A few are mentioned below.

- It gives the company the right kind of workforce at the right time frame and in right figures.
- In striking a balance between demand-for and supply-of resources, HRP helps in the optimum usage of resources and also in reducing the labor cost.
- Cautiously forecasting the future helps to supervise manpower in a better way, thus pitfalls can be avoided.
- It helps the organization to develop a succession plan for all its employees. In this way, it creates a way for internal promotions.
- It compels the organization to evaluate the weaknesses and strengths of personnel thereby making the management to take remedial measures.
- The organization as a whole is benefited when it comes to increase in productivity, profit, skills, etc., thus giving an edge over its competitors

Human Resource Planning Process

Human Resource Planning Process



- 1. Determining the Objectives of Human Resource Planning:** The foremost step in every process is the determination of the objectives for which the process is to be carried on. The objective for which the manpower planning is to be done should be defined precisely, so as to ensure that a right number of people for the right kind of job are selected. The objectives can vary across the several departments in the organization such as the personnel demand may differ in marketing, finance, production, HR department, based on their roles or functions.
- 2. Analyzing Current Manpower Inventory:** The next step is to analyze the current manpower supply in the organization through the stored information about the employees in terms of their experience, proficiency, skills, etc. required to perform a particular job. Also, the future vacancies can be estimated, so as to plan for the manpower from both the internal (within the current employees) and the external (hiring candidates from outside) sources. Thus, it is to be ensured that reservoir of talent is maintained to meet any vacancy arising in the near future.
- 3. Forecasting Demand and Supply of Human Resources:** Once the inventory of talented manpower is maintained; the next step is to match the demand for the manpower arising in the future with the supply or available resources with the organization. Here, the required skills of personnel for a particular job are matched with the job description and specification.
- 4. Analyzing the Manpower Gaps:** After forecasting the demand and supply, the manpower gaps can be easily evaluated. In case the demand is more than the supply of human resources, that means there is a deficit, and thus, new candidates are to be hired. Whereas, if the Demand is less than supply, there arises a surplus in the human resources, and hence, the employees have to be removed either in the form of termination, retirement, layoff, transfer, etc.
- 5. Employment Plan/Action Plan:** Once the manpower gaps are evaluated, the action plan is to be formulated accordingly. In a case of a deficit, the firm may go either for recruitment, training, interdepartmental transfer plans whereas in the case of a surplus, the voluntary retirement schemes, redeployment, transfer, layoff, could be followed.
- 6. Training and Development:** The training is not only for the new joiners but also for the existing employees who are required to update their skills from time to time. After the employment plan, the training programmes are conducted to equip the new employees as well as the old ones with the requisite skills to be performed on a particular job.
- 7. Appraisal of Manpower Planning:** Finally, the effectiveness of the manpower planning process is to be evaluated. Here the human resource plan is compared with its actual implementation to ensure the availability of a number of employees for several jobs. At this stage, the firm has to decide the success of the plan and control the deficiencies, if any.

Thus, human resource planning is a continuous process that begins with the objectives of Human Resource planning and ends with the appraisal or feedback and control of the planning process.

JOB EVALUATION

Definition:

The **Job Evaluation** is the process of assessing the relative worth of the jobs in an organization. The jobs are evaluated on the basis of its content and the complexity involved in its operations and thus, positioned according to its importance.

Process of Job Evaluation

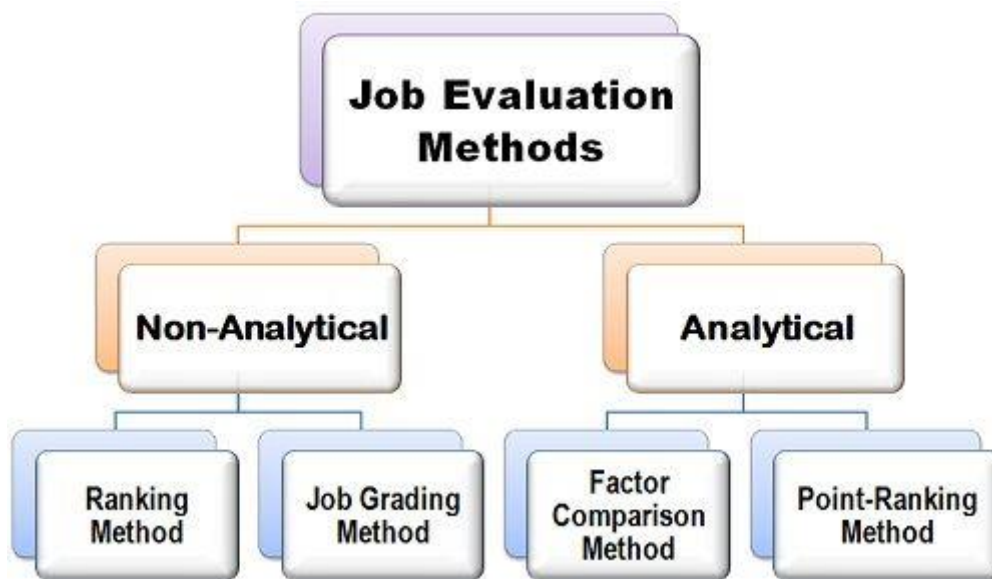
The process of job evaluation involves following steps:

- Securing acceptance from employees after explaining the purpose and use of job evaluation programme.
- Creating job evaluation committee consisting of experienced employees, union representatives and HR experts.
- Deciding the job to be evaluated, which may represent the type of work performed in the organisation.
- Analysing and preparing job description.
- Selecting method of evaluation, according to the job factors and organisational demand.
- Classifying the jobs on the basis of weightage and monetary values.
- Installing the programme in the whole organisation after explaining it to employees.
- Conducting periodical review in the light of changes in environment from time to time.



Job Evaluation Methods

There are non-analytical and analytical job evaluation methods that are employed by the organizations to realize the worth of a set of jobs.



Non-analytical Job Evaluation Methods

1. **Ranking Method:** This is the simplest and an inexpensive job evaluation method, wherein the jobs are ranked from the highest to the lowest on the basis of their importance in the organization. In this method, the overall job is compared with the other set of jobs and then is given a rank on the basis of its content and complexity in performing it.

Here the job is not broken into the factors, an overall analysis of the job is done. The main advantage of the ranking method is, it is very easy to understand and is least expensive. But however it is not free from the limitations, it is subjective in nature due to which employees may feel offended, and also, it may not be fruitful in the case of big organizations.

2. **Job Grading Method:** Also known as **Job-Classification Method**. Under this method the job grades or classes are predetermined and then each job is assigned to these and is evaluated accordingly.

For Example Class, I, comprise of the managerial level people under which sub-classification is done on the basis of the job roles such as office manager, department managers, departmental supervisor, etc.

The advantage of this method is that it is less subjective as compared to the ranking method and is acceptable to the employees. And also, the entire job is compared against the other jobs and is not broken into factors. The major limitation of this method is that the jobs may differ with respect to their content and the complexity and by placing all under one category the results may be overestimated or underestimated.

Analytical Job Evaluation Methods

1. **Factor-Comparison Method:** Under this method, the job is evaluated, and the ranks are given on the basis of a series of factors Viz. Mental effort, physical effort, skills required supervisory responsibilities, working conditions, and other relevant factors. These factors are assumed to be constant for each set of jobs. Thus, each job is compared against each other on this basis and is ranked accordingly. The advantage of this method is that it is consistent and less subjective, thus appreciable by all. But however it is the most complex and an expensive method.
2. **Point-Ranking Method:** Under this method, each job's key factor is identified and then the subfactors are determined. These sub-factors are then assigned the points by its importance.

For example, the key factor to perform a job is skills, and then it can be further classified into sub-factors such as training required, communication skills, social skills, persuasion skills, etc.

The point ranking method is less subjective and is an error free as the rater sees the job from all the perspectives. But however it is a complex method and is time-consuming since the points and wage scale has to be decided for each factor and the sub factors.

The important thing to note is, the job evaluation is considered only with the analysis of a job and not with the job holders.

To summarize, Job evaluation is associated with a process of establishing worth of different jobs. Job evaluation offers a basis for ranking or grading different jobs and developing a pay structure for them. It is the process of explaining the duties, authority relationships, skills, condition of work and other relevant information related to jobs. It provides valuable data and information to develop job description and specification documents. Job evaluation is a significant task of human resource manager which is done in order to determine the value or worth of each job within the organization.

RECRUITMENT

Meaning:

Recruitment is a positive process of searching for prospective employees and stimulating them to apply for the jobs in the organisation. When more persons apply for jobs then there will be a scope for recruiting better persons.

Definition:

According to Edwin B. Flippo, "It is a process of searching for prospective employees and stimulating and encouraging them to apply for jobs in an organisation." He further elaborates it, terming it both negative and positive.

He says, “It is often termed positive in that it stimulates people to apply for jobs, to increase the hiring ratio, i.e. the number of applicants for a job. Selection, on the other hand, tends to be negative because it rejects a good number of those who apply, leaving only the best to be hired.”

Process of Recruitment:

Recruitment Process Passes through the Following Stages:

- (i) Searching out the sources from where required persons will be available for recruitment. If young managers are to be recruited then institutions imparting instructions in business administration will be the best source.
- (ii) Developing the techniques to attract the suitable candidates. The goodwill and reputation of an organisation in the market may be one method. The publicity about the company being a professional employer may also assist in stimulating candidates to apply.
- (iii) Using of good techniques to attract prospective candidates. There may be offers of attractive salaries, proper facilities for development, etc.
- (iv) The next stage in this process is to stimulate as many candidates as possible to apply for jobs. In order to select a best person, there is a need to attract more candidates.

Factors Influencing Recruitment:

All enterprises, big or small, have to engage themselves in recruitment of persons. A number of factors influence this process.

Some Of The Main Factors Are Being Discussed Below:

1. Size of the Enterprise:

The number of persons to be recruited will depend upon the size of an enterprise. A big enterprise requires more persons at regular intervals while a small undertaking employs only a few employees. A big business house will always be in touch with sources of supply and shall try to attract more and more persons for making a proper selection. It can afford to spend more amounts in locating prospective candidates. So the size of an enterprise will affect the process of recruitment.

2. Employment Conditions:

The employment conditions in an economy greatly affect recruitment process. In under-developed economies, employment opportunities are limited and there is no dearth of prospective candidates. At the same time suitable candidates may not be available because of lack of educational and technical facilities. If the availability of persons is more, then selection from large number becomes easy. On the other hand, if there is a shortage of qualified technical persons, then it will be difficult to locate suitable persons.

3. Salary Structure and Working Conditions:

The wages offered and working conditions prevailing in an enterprise greatly influence the availability of personnel. If higher wages are paid as compared to similar concerns, the enterprise will not face any difficulty in making recruitments. An organisation offering low wages can face the problem of labour turnover.

The working conditions in an enterprise will determine job satisfaction of employees. An enterprise offering good working conditions like proper sanitation, lighting, ventilation, etc. would give more job satisfaction to employees and they may not leave their present job. On the other hand, if employees leave the jobs due to unsatisfactory working conditions, it will lead to fresh recruitment of new persons.

4. Rate of Growth:

The growth rate of an enterprise also affects recruitment process. An expanding concern will require regular employment of new employees. There will also be promotions of existing employees necessitating the filling up of those vacancies. A stagnant enterprise can recruit persons only when present incumbent vacates his position on retirement, etc.

SOURCES OF RECRUITMENT

There are two types of sources, they are:

1. Internal Sources of Recruitment
2. External Sources of Recruitment

Internal Sources of Recruitment:

(i) Transfer: it means shifting of employees from one job to another. There is no drastic change in the pay or status of the employees it generally remains the same.

(ii) Promotion: it means shifting the employee to higher position with the change in the pay or status.

(iii) Internal advertisement: it means appointing the employees from within the organization through advertisement.

(iv) Lent services: it means employing the person for newly started plant in the organization.

(v) Extension of services: it is extending the service of employees by recruiting them again.

(vi) Absorption of trainees: those who come in the organization for training employing them.

2. External sources of recruitment:

- (i) Advertisement:** it is a way to attract number of employees by giving advertisement in the newspaper.
- (ii) Casual callers:** the good organizations have a list of casual callers also who apply them and the organization instead of giving new advertisement give a call to these casual callers.
- (iii) Employment exchange:** it is a good source of recruitment where the candidates are called with the help of employment exchange.
- (iv) Educational institutions:** here the company's go to the educational institutions from where they select the suitable employees for the job.
- (v) Labor union:** the organization may select the candidate from the labor union of the company to satisfy the labors also.
- (vi) Recommendation of employees:** sometimes the employees working in the organization also suggest the name of suitable candidate for the appointment.
- (vii) Waiting lists:** generally the organization s prepares the waiting lists of the employees. Sometimes the selected candidates failed to join in that case the waiting list employees are called.

Thus the above said are the main sources of recruitment of the employees within the organization and outside the organization which are used by the organization to employ the employees.

SELECTION

Definition: The **Selection** is a process of picking the right candidate with prerequisite qualifications and capabilities to fill the jobs in the organization.

Selection is the process of picking or choosing the right candidate, who is most suitable for a vacant job position in an organization.

In others words, selection can also be explained as the process of interviewing the candidates and evaluating their qualities, which are required for a specific job and then choosing the suitable candidate for the position.

The selection of a right applicant for a vacant position will be an asset to the organization, which will be helping the organization in reaching its objectives.

Different authors define Selection in different ways. Here is a list of some of the definitions –

- Employee selection is a process of putting a right applicant on a right job.

- Selection of an employee is a process of choosing the applicants, who have the qualifications to fill the vacant job in an organization.
- Selection is a process of identifying and hiring the applicants for filling the vacancies in an organization.
- Employee selection is a process of matching organization's requirements with the skills and the qualifications of individuals.

Advantages of Selection

A good selection process offers the following advantages—

- It is cost-effective and reduces a lot of time and effort.
- It helps avoid any biasing while recruiting the right candidate.
- It helps eliminate the candidates who are lacking in knowledge, ability, and proficiency.
- It provides a guideline to evaluate the candidates further through strict verification and reference-checking.
- It helps in comparing the different candidates in terms of their capabilities, knowledge, skills, experience, work attitude, etc.

Selection Process

The selection procedure comprises of following systematic steps:



1. **Preliminary Interview:** The preliminary interview is also called as a screening interview wherein those candidates are eliminated from the further selection process who do not meet the minimum eligibility criteria as required by the organization.

Here, the individuals are checked for their academic qualifications, skill sets, family backgrounds and their interest in working with the firm. The preliminary interview is less formal and is basically done to weed out the unsuitable candidates very much before proceeding with a full-fledged selection process.

2. **Receiving Applications:** Once the individual qualifies the preliminary interview he is required to fill in the application form in the prescribed format. This application contains the candidate data such as age, qualification, experience, etc. This information helps the interviewer to get the fair idea about the candidate and formulate questions to get more information about him.
3. **Screening Applications:** Once the applications are received, these are screened by the screening committee, who then prepare a list of those applicants whom they find suitable for the interviews. The shortlisting criteria could be the age, sex, qualification, experience of an individual. Once the list is prepared, the qualified candidates are called for the interview either through a registered mail or e-mails.
4. **Employment Tests:** In order to check the mental ability and skill set of an individual, several tests are conducted. Such as intelligence tests, aptitude tests, interest tests, psychological tests, personality tests, etc. These tests are conducted to judge the suitability of the candidate for the job.
5. **Employment Interview:** The one on one session with the candidate is conducted to gain more insights about him. Here, the interviewer asks questions from the applicant to discover more about him and to give him the accurate picture of the kind of a job he is required to perform.

Also, the briefing of certain organizational policies is done, which is crucial in the performance of the job. Through an interview, it is easier for the employer to understand the candidate's expectations from the job and also his communication skills along with the confidence level can be checked at this stage.

6. **Checking References:** The firms usually ask for the references from the candidate to cross check the authenticity of the information provided by him. These references could be from the education institute from where the candidate has completed his studies or from his previous employment where he was formerly engaged. These references are checked to know the conduct and behavior of an individual and also his potential of learning new jobs.

7. **Medical Examination:** Here the physical and mental fitness of the candidate are checked to ensure that he is capable of performing the job. In some organizations, the medical examination is done at the very beginning of the selection process while in some cases it is done after the final selection.

Thus, this stage is not rigid and can take place anywhere in the process. The medical examination is an important step in the selection process as it helps in ascertaining the applicant's physical ability to fulfill the job requirements.

8. **Final Selection:** Finally, the candidate who qualifies all the rounds of a selection process is given the appointment letter to join the firm.

Thus, the selection is complex and a lengthy process as it involves several stages than an individual has to qualify before getting finally selected for the job.

PLACEMENT

Placement is the process of assigning a specific job to each one of the selected candidates.

In very simple words **placement** means sending the newly employed person to some department for work. It also implies assigning a specific rank and responsibility to an individual.

Importance

1. If the employees are properly placed, they will enjoy their work and organisation will not have to suffer the problem of employee turnover.
2. If employees don't like their work, they start making excuses from the job and remain absent. Effective placement will keep the absenteeism rate low.
3. Morale of workers increases because they get the work of their choice, if correctly placed.
4. Workers will work attentively and safety of workers will be ensured and lesser accidents will happen.
5. Workers will be satisfied with their jobs and there will be no reasons for disputes, so human relations will improve.
6. Through proper placement, misfit between the job and person can be avoided.
7. Efficient and effective performance of individual tasks will ensure the achievement of organisational goals.
8. Productivity i.e., ratio of output to input increases as wastage and abnormal losses decrease.

Placement should be done keeping into view the job and social, psychological & emotional needs of person.

Principles Followed while Making Placement of an Employee

The following principles are required to be observed while making placement of an employee:

- (i) Job requirement – Man should be placed on the job according to the requirement of the job rather than qualification and requirement of the man.
- (ii) Qualification – The job offered should match with the qualification possessed by an employee.
- (iii) Information – All the information relating to the job should be given to the employees along with the prevailing working conditions. They should also be made known that they have to pay penalty for wrong doing.
- (iv) Loyalty and Co-operation – Every effort should be made to develop a sense of loyalty and co-operation in employees to make them understand their responsibilities

INDUCTION

Induction means introduction of a new employee to the job and the organization.

It is the process of receiving and welcoming an employee when he first joins a company and giving him the basic information he needs to settle down quickly and happily and start work.

It is a welcoming process to make him feel at home and generate in him a feeling of belongingness to the organization.

In the process of induction the new comer is explained his duties and responsibilities, company rules, policies and regulations so as to make him familiar to the organization.

Induction covers the following:

History- The organization's traditions, customs, myths, background of founders and the present members in the top management.

Company Coals- Coals of the organization with rules, values or principles directing the organization.

Language- Jargons, slangs, and technical terms unique to the organization.

Politics- Ways and means to gain formal and informal information about jobs and relationships/power structures in the organization.

People- Nuances in initiating and maintaining relationships with peers.

Performance Proficiency- Effectiveness in acquiring and using knowledge, skills and abilities needed for the job.

In addition to the above, induction includes communication of HRM policies such as work hours, pay procedures, overtime requirements/company benefits, and duties and responsibilities of new members; In large organization the members are shown the various physical facilities available. Important aspect is introduction of the new employee to his supervisor and co-workers.

Induction is taken care of by the employee's supervisor/manager or the HRM personnel. Sometimes computer-based programmes are used.

Characteristics

The important characteristics of induction are:

- (i) Encouraging employees to ask questions.
- (ii) Including information on both technical and social aspects of the job.
- (iii) Making the new employee's manager responsible for the orientation.
- (iv) Avoiding embarrassment to the new members.
- (v) Arranging formal and informal interaction with managers and peers.
- (vi) Providing relocation assistance such as house hunting, information about the local society, and etc.
- (vii) Giving information about the company's products, services and customers.
- (viii) Familiarizing adequately with the culture of the organization such as how work is done, what matters in the organization, which work-related behaviours are acceptable or unacceptable and etc.

Need for Induction

- i. Employees have the right to a proper orientation to any new job position. Fowler (1983) stated that, new employees need to have realistic expectations of their jobs.
- ii. They need to be helped to fit rapidly into the organization and into the working group.
- iii. They need to know clearly what is expected of them- the standards and style of the work they are to perform.
- iv. They need to understand their organization's rules and regulations.
- v. They must know who's who at work, and who can make decisions about which issues.
- vi. They should be helped to develop their aptitudes and talents.

vii. They need to feel that their employer has a respect for them as individuals, as potentially resourceful humans rather than as a mechanistic human resource.

Once the objectives of induction are clear, it is wise to plan the content and duration of sessions carried out during the induction programme in order to get the best from the employees.

Objectives of Induction

The main object of induction is to make the new employee feel at home and develop a sense of pride in the organization and commitment to the job.

The following are the some of the other important objectives:

- (i) To help the new employee to develop a close and cordial relation with the existing employees.
- (ii) To give the new employee necessary information such as – location of the different building, company rules, leave rules, rest periods etc.
- (iii) To help the new employee overcome his natural shyness and nervousness in meeting the new people in the organization.
- (iv) To develop a sense of belongingness and loyalty among new employees.
- (v) To help the employees know the different facilities and opportunities available in the organization.
- (vi) To help the new employees to minimize the “reality shock” that they may undergo after reporting to duty.

WAGE AND SALARY ADMINISTRATION

Wage and salary administration is defined as the process by which **wage and salary** levels and structures are determined in organisational settings.

Wages are payments for labour services rendered frequency, expressed in hourly rates, while a **salary** is a similar payment, expressed in weekly, monthly or annual rates.

Administration of employee compensation is called **wage and salary administration**.

According to D.S. Beach “**Wage and Salary Administration** refers to the establishment and implementation of sound policies and practices of employee compensation. It includes such areas as job evaluation, surveys of wage and salaries, analysis of relevant organizational problems, development and maintenance of wage structure, establishing rules for administering wages, wage payment incentives, profit sharing, wage changes and adjustments, supplementary payments, control of compensation costs and other related items.”

Objectives of Wage and Salary Administration :

A sound plan of wage and salary administration seeks to achieve the following objectives:

- To establish a fair and equitable compensation offering similar pay for similar work.
- To attract competent and qualified personnel.
- To retain the present employees by keeping wage levels in tune with competitive units.
- To keep labor and administrative costs in line with the ability of the organization to pay.
- To improve motivation and morale of employees and to improve union management relations.
- To project a good image of the company and to comply with legal needs relating to wages and salaries.
- To establish job sequences and lines of promotion wherever applicable.
- To minimize the chances of favoritism while assigning the wage rates.

Principles of Wage and Salary Administration :

The following principles should be followed for an effective wage and salary administration ;

- Wage policy should be developed keeping in view the interests of all concerned parties viz., employer, employees, the consumers and the society.
- Wage and salary plans should be sufficiently flexible or responsive to changes in internal and external conditions of the organization.
- Efforts should be made to ensure that differences in pay for jobs are based on variations in job requirements such as skill, responsibility, efforts and mental and physical requirements.
- Wage and salary administration plans must always be consistent with overall organizational plans and programmes.
- Wage and Salary administration plans must always be in conformity with the social and economic objectives of the country like attainment of equality in income distribution and controlling inflation, etc.
- These plans and programmes should be responsive to the changing local and national conditions.
- Wage and salary plans should expedite and simplify administrative process.

- Workers should be associated, as far as possible, in formulation and implementation of wage policy.
- An adequate data base and a proper organizational set up should be developed for compensation determination and administration.
- The general level of wages and salaries should be reasonably in line with that prevailing in the labor market.
- There should be a clearly established procedure for hearing and adjusting wage complaints. This may be integrated with the regular grievance procedure, if it exists.
- The workers should receive a guaranteed minimum wage to protect them against conditions beyond their control.
- Prompt and correct payments to the employees should be ensured and arrears of payment should not accumulate.
- The wage and salary payments must fulfill a wide variety of human needs including the need for self actualization.
- Wage policy and programme should be reviewed and revised periodically in conformity with changing needs. For revision of wages, a wage committee should also be preferred to the individual judgement however unbiased of a manager.

Factors Affecting Wage Rate Decisions:

The wage payment is an important factor affecting the labour management relations. Workers are very much concerned with the rates of wages as their standard of living is linked to the amount of remuneration they get. Managements, however, do not come forward to pay higher wages because cost of production goes up and profits decrease to that extent. A number of factors, thus, Influence the remuneration payable to the employees. These factors can be categorised into (i) External Factors and (ii) Internal Factors.

(A) External Factors:

1. Demand and Supply:

The labour market conditions or demand and supply forces operate at the national and local levels and determine organisational wage structure. When the demand for a particular type of labour is more and supply is less than the wages will be more.

On the other hand, if supply of labour is more and demand on the other hand, is less then persons will be available at lower wage rates also. In the words of Mescon, 'the supply and demand compensation criterion is very closely related to the prevailing pay, comparable wage and ongoing wage concepts since, in essence, all these remuneration standards are determined by immediate market forces and factors.

2. Cost of Living:

The wage rates are directly influenced by cost of living of a place. The workers will accept a wage which may ensure them a minimum standard of living. Wages will also be adjusted according to price index number. The increase in price index will erode the purchasing power of workers and they will demand higher wages. When the prices are stable then frequent wage increases may not be undertaken.

3. Trade Unions' Bargaining Power:

The wage rates are also influenced by the bargaining power of trade unions. Stronger the trade union higher will be the wage rates. The strength of a trade union is judged by its membership, financial position and type of leadership. Union's last weapon is strike which may also be used for getting wage increases. If the workers are disorganised and disunited then employers will be successful in offering low wages.

4. Government Legislation:

To improve the working conditions of workers, government may pass a legislation for fixing minimum wages of workers. This may ensure them a minimum level of living. In underdeveloped countries bargaining power of labour is weak and employers try to exploit workers by paying them low wages. In India, Minimum Wages Act, 1948 was passed to empower government to fix minimum wages of workers.

5. Psychological and Social Factors:

Psychologically the level of compensation is perceived as a measure of success in life.

Management should take into consideration the psychological needs of the employees while fixing the wage rates so that the employees take pride in their work. Sociologically and ethically, the employees want that the wage system should be equitable, just and fair. These factors should also be taken into consideration while devising a wage programme.

6. Economy:

Economy also has its impact on wage and salary fixation. While it may be possible for some organisations to thrive in a recession, there is no doubt that economy affects remuneration decisions. A depressed economy will probably increase the labour supply. This, in turn, should lower the going wage rate.

7. Technological Development:

With the rapid growth of industries, there is a shortage of skilled resources. The technological developments have been affecting skill levels at faster rates. Thus, the wage rates of skilled employees constantly change and an organisation has to keep its level upto the mark to suit the market needs.

8. Prevailing Market Rates:

No enterprise can ignore prevailing or comparative wage rates. The wage rates paid in the industry or other concerns at the same place will form a base for fixing wage rates. If a concern pays low rates then workers leave their jobs whenever they get a job somewhere else. It will not be possible to retain good workers for long.

(B) Internal Factors:

1. Ability to Pay:

The ability to pay of an enterprise will influence wage rates to be paid. If the concern is running into losses then it may not be able to pay higher wage rate. A profitable concern may pay more to attract good workers. During the period of prosperity, workers are paid higher wages because management wants to share the profits with labour.

2. Job Requirements:

Basic wages depend largely on the difficulty level, and physical and mental effort required in a particular job. The relative worth of a job can be estimated through job evaluation. Simple, routine tasks that can be done by many people with minimum skills receive relatively low pay. On the other hand, complex, challenging tasks that can be done by few people with high skill levels generally receive high pay.

3. Management Strategy:

The overall strategy which a company pursues should determine the remuneration to its employees. Where the strategy of the organisation is to achieve rapid growth, remuneration should be higher than what competitors pay. Where the strategy is to maintain and protect current earnings, because of the declining fortunes of the company, remuneration level needs to be average or even below average.

4. Employee:

Several employees related factors interact to determine his remuneration.

These are:

(i) Performance or productivity is always rewarded with a pay increase. Rewarding performance motivates the employees to do better in future.

(ii) Seniority:

Unions view seniority as the most objective criteria for pay increases whereas management prefers performance to effect pay increases.

(iii) Experience:

Makes an employee gain valuable insights and is generally rewarded.

(iv) Potential:

Organisations do pay some employees based on their potential. Young managers are paid more because of their potential to perform even if they are short of experience.

(v) Luck:

Some people are rewarded because of their sheer luck. They have the luck to be at the right place at the right time.

TRAINING AND DEVELOPMENT

Training and development refers to educational activities within a company created to enhance the knowledge and skills of employees while providing information and instruction on how to better perform specific tasks.

Employee training and development implies a program in which specific knowledge, skills and abilities are imparted to the employees, with the aim of raising their performance level, in their existing roles, as well as providing them learning opportunities, to further their growth.

Training implies a process of upgrading an individual's knowledge, skills and competencies. When an employee joins the organization, he/she is given job-related training to ensure satisfactory performance of the tasks and duties assigned, so that the employee can contribute more to the organization.

On the other hand, **development has a long-term view.** It is all about preparing the employee for the current as well as future jobs, by providing them with learning opportunities to increase their capacities, to undertake more challenging and complex tasks.

Importance of Training and Development

1. For organization:

- Training and Development programme facilitates **better utilization of an organization's resources**, i.e. men, machine, material and money.
- It ensures **less wastage, with respect to resources and time.** When a worker is skilled and trained enough, the amount of wastage will be less, as well as the learning time will be reduced, and the work will be performed in very less time.
- It **improves the performance** of the employee which results in increased profitability.
- The employees learn **new and improved methods of performing an activity**, which helps the organization to survive, compete and grow in the market.

- It results in a **reduction in employee turnover, idle time and absenteeism.**
- It develops a **positive attitude** in employees towards the work and organization, as well as boosts their morale, which helps them adapt themselves to the changing environment.

2. For Employees:

- An individual focuses more on career than a job. And so, the **skills and knowledge acquired from the training add to their resume.**
- Improved performance is a sort of **motivation** for an individual which encourages them to perform much better than before and contribute to the organization's success, which ultimately increases their incentives.
- When an employee is trained enough, he/she will **handle any situation or problem and deal with the challenges** in a better way.
- Training makes workers machine or equipment-friendly, which **reduces the number of accidents** during the process.
- It helps the employee to **stay up-to-date with the technological changes.**
- It results in **job satisfaction and boosts employees morale.**

Training is often imparted to middle or low-level employees. In contrast, the development programmes are specifically designed for the employees working in higher positions in the organisational hierarchy.

Inputs in Training and Development

- **Skills:** Basic skills should be imparted to the workers, to help them operate the machinery and other equipment correctly, with least wastage and damage.
- **Education:** It aims at teaching theoretical concepts along with providing a hands-on experience to the employee. It also improves their judgement and reasoning power.
- **Development:** Training involves development, wherein more stress is given on increasing knowledge, rather than the skills. Here, knowledge means an understanding of the business environment, human relations, management, etc.
- **Ethics:** Ethics are the principles that regulate the conduct of an employee, at the time of undertaking any activity. So, the training and development programme must provide knowledge regarding ethical conduct.
- **Change in attitude:** One of the important benefits of training is the attitudinal changes in employees. Attitude indicates the feelings, outlook, reaction and beliefs of the employee

towards others. It has a great impact on an employee's morale, motivation, loyalty, commitment and satisfaction.

- **Extraordinary Skills:** Apart from basic skills, training and development scheme should impart some other skills to higher level employees such as problem-solving, decision making, industrial analysis, etc.
- **Literacy:** Here literacy is not related to education, but proficiency in the corporate language and he/she must be fully aware of the corporate culture, to deal with international clients and customers.

In a nutshell, any activity that intends to **enhance the present or future performance of an employee**, by improving his/her ability through the process of learning, by changing approach, sharpening skills, strengthening competencies and increasing the knowledge base, comes under the ambit of training and development.

METHODS OF TRAINING

There are two methods of training

A. On-the job training

B. Off-the job training

a. On-the-Job Training

Definition: The **On-the-Job Training** is a technique wherein the workers, i.e., operative staff, is given the direct instructions to perform their jobs on the actual work floor.

The workers can learn the skills that are required to be performed in the actual work conditions and also gets accustomed to the working environment. Also, the organizations need not to bear any additional cost of setting up a classroom or a simulated setup for imparting training to the workers, away from the actual work floor, as in the case of Off-the -Job training.

On-the-Job Training Methods

1. **Coaching:** Under this method, the superior or an experienced staff gives instructions to the workers to perform a job. It is one-to-one training designed for the workers where they can find answers to their queries through the instructions and demonstrations given by the superior.
2. **Mentoring:** This training is given to the managerial level people, wherein the senior or the manager gives instructions to the immediate subordinate to carry out the day to day functioning.

It is again a one-to-one training method, where the manager is considered as a mentor to the subordinate and guides him in the situations of difficulty.

3. **Job Rotation:** Under the job rotation, an employee is often shifted to the other related jobs, with the intention to make him well versed with other job backgrounds.

This helps him to escape the boredom caused by performing the same kind of work again and again and also helps in developing a rapport with other people in the organization.

4. **Job Instructional Training:** Under this training, a trainer designs a step by step training program, wherein the worker is given the instructions to perform the job as required. Firstly, the overview of the job along with the desired results is explained to the trainee, and then the skills required for the job is demonstrated by the trainer.

Then a worker is allowed to perform the job as per his acquired skills or expertise, and then finally the workers are asked to give their feedback and ask for any query arising out of the training program.

5. **Understudy:** Here, the superior gives training to the subordinate as an understudy or an assistant who is likely to perform a superior's job in case of the vacancy arising out of superior's retirement, transfer, promotion or death.
6. **Apprenticeship:** This type of training is generally given to the people in crafts, trade and technical fields that require a long-term learning before they actually gain the proficiency in their respective disciplines.

This training is a blend of classroom and on-the-job training and is conducted under the close supervision. This can be extended up to 3 to 4 years as apprentices need to go through the learning process till they become an expert in their fields.

E.g. The craftsmen job, mechanic, electrician, plumber, tool maker, etc. have to undergo this type of training.

On-the-Job training is based on the principle of “**learning by doing**”, i.e. the workers learn the job while performing it within the actual work environment. This type of training is beneficial for both the workers and the organization

Off-the-Job Training

Definition: The **Off-the-Job Training** is the training method wherein the workers/employees learn their job roles away from the actual work floor.

Simply, off-the-job training comprises of a place specifically allotted for the training purpose that may be near to the actual workplace, where the workers are required to learn the skills and get well equipped with the tools and techniques that are to be used at the actual work floor.

Off-the-Job Training Methods

1. **Special lectures:** This is also called as classroom training wherein the employees are given lectures about the job requirements and the necessary skills required for implementing the job.

There is generally a classroom or a workshop wherein the complete job knowledge is given to the workers by the experts or specialists from the professional institutes. The main purpose of this training is to make the employees well informed about their job roles and discussing their queries arising out of the lectures.

2. **Simulation:** Under this training, the trainee is required to learn the operations of machines and equipment, that are reasonably designed to look similar to those installed at the actual work floor.

This is one of the most common method of training wherein the worker learns to operate tools and machinery that look alike to those, they would be using in the actual work environment.

3. **Vestibule Training:** This type of training is specifically given to the technical staff, office staff and the employees who learn the operations of tools and equipment assembled at a place away from the actual work floor.

This type of training is conducted to give the real feel to the trainees, that they would be experiencing at the actual plant.

4. **Case Studies:** Under this method, the trainees are given the situation or a problem in the form of a case study, and are required to solve it as per their learning from the training program.
5. **Role playing:** This type of training is essential in case of customer services. Under this, the trainees assume roles and enact as per the given situations. It is also called as *socio-drama or psycho-drama*, wherein the employees act as if, they are facing the situation and have to solve it spontaneously without any guidance.
6. **Management Games:** Under this method, the trainees are divided into groups and then they are presented with the simulated marketplace or the situations, wherein they are required to apply their learning and solve the problems accordingly.

Under off-the-job training, the worker concentrates only on the learning of a job and is not accountable for the production.

Training Process

Definition: The **Training Process** comprises of a series of steps that needs to be followed systematically to have an efficient training programme.

1. **Needs assessment:** The first step in the training process is to assess the need for training the employees. The need for training could be identified through a diagnosis of present and future challenges and through a gap between the employee's actual performance and the standard performance.

The needs assessment can be studied from two perspectives: Individual and group. The individual training is designed to enhance the individual's efficiency when not performing adequately. And whereas the group training is intended to inculcate the new changes in the employees due to a change in the organization's strategy.

2. **Deriving Instructional Objectives:** Once the needs are identified, the objectives for which the training is to be conducted are established. The objectives could be based on the gaps seen in the training programmes conducted earlier and the skill sets developed by the employees.
3. **Designing Training Programme:** The next step is to design the training programme in line with the set objectives. Every training programme encompasses certain issues such as: Who are the trainees? Who are the trainers? What methods are to be used for the training? What will be the level of training? etc. Also, the comprehensive action plan is designed that includes the training content, material, learning theories, instructional design, and the other training requisites.
4. **Implementation of the Training Programme:** Once the designing of the training programme is completed, the next step is to put it into the action. The foremost decision that needs to be made is where the training will be conducted either in-house or outside the organization.

Once it is decided, the time for the training is set along with the trainer who will be conducting the training session. Also, the trainees are monitored continuously throughout the training programme to see if it's effective and is able to retain the employee's interest.

5. **Evaluation of the Training Programme:** After the training is done, the employees are asked to give their feedback on the training session and whether they felt useful or not. Through feedback, an organization can determine the weak spots if any, and can rectify it in the next session. The evaluation of the training programme is a must because companies invest huge amounts in these sessions and must know it's effectiveness in terms of money.

Thus, every firm follows the series of steps to design an effective training programme that serves the purpose for which it was intended.

PERFORMANCE APPRAISAL

Definition: Performance Appraisal is defined as a systematic process, in which the personality and performance of an employee is assessed by the supervisor or manager, against predefined standards, such as knowledge of the job, quality and quantity of output, leadership abilities, attitude towards work, attendance, cooperation, judgment, versatility, health, initiative and so forth.

It is also known as performance rating, performance evaluation, employee assessment, performance review, merit rating, etc.

Performance Appraisal is carried out to identify the abilities and competencies of an employee for future growth and development. It is aimed at ascertaining the worth of the employee to the organization, in which he/she works.

Objectives of Performance Appraisal

- To promote the employees, on the basis of performance and competence.
- To identify the requirement for training and development of employees.
- To provide confirmation to those employees who are hired as probationary employees, upon completion of the term.
- To take a decision regarding the hike in employees pay, incentives etc.
- To facilitate communication between superior and subordinate.
- To help employees in understanding where they stand in terms of performance.

Process of Performance Appraisal

1. **Establish Performance Standards or Goals** – The appraisal process begins with the establishment of performance standards. The evaluators must determine what outputs, accomplishments and skills will be evaluated. These standards should have evolved out of job analysis and job description. These performance standards should be clear and objectives are to be understood and measured.
2. **Communicate Performance Expectations to Employees** – Once the performance standards are established, these need to be communicated to the respective employees so that they come to know what expected of them. Past experience indicates that not communicating standards to the employees compounds the appraisal problems. Here, it must be noted that mere transference of information from the evaluator to the staff is not communication. It become communication only when the transference of information has taken place and has been received and understood by the staff. The feedback from the staff on the standards communicated to them must be obtained. If required, the

standards may be modified or revised in the light of feedback obtained from the employees.

3. **Measure Actual Performance** – This is the third step involved in the appraisal process. In this stage, the actual performance of the employees is measured on the basis of information available from various sources such as personal observation, statistical reports, oral reports, and written reports. The evaluator's feeling should not influence the performance measurements of the staff. Measurement must be objective based on facts and findings. This is because what we measure is more critical and important to the evaluation process than how we measure.
4. **Compare Actual Performance with Standards** – In this stage, the actual performance is compared with the predetermined standards. Such a comparison may reveal the deviation between standard performance and actual performance and will enable the evaluator to proceed to the fifth step on the process i.e. Review and discussion of the appraisal with the concerned staff.
5. **Discuss and Review the Appraisal with the Staff** – The fifth step in the appraisal process is to communicate to and discuss with the employees the results of the appraisal. This is one of the most challenging tasks the evaluators face to present an accurate appraisal to the staff and then make them accept the appraisal in a constructive manner. A discussion on appraisal enables employees to know their strengths and weaknesses. This has, in turn, impact on their future performance. The impact may be positive or negative depending upon how the appraisal is presented and discussed with the employees.
6. **Initiate Corrective Action** – The final step in the appraisal process is the initiation of corrective action when it is necessary. The areas needing improvement are identified and then, the measures to correct or improve the performance are identified and initiated.

Method of Performance Appraisal

These methods are broadly classified into two categories:

a. Traditional Methods of Performance Appraisal

1. **Rating Scales:** The numerical scale is prepared that represents the job evaluation criterion such as the output, initiative, attendance, attitude, dependability, etc. and ranges from excellent to poor. Thus, each employee is given the rating based on his performance against each set criteria and then the consolidated score is computed accordingly. The rating scales are one of the easiest traditional methods of performance appraisal and can be applied to any job.
2. **Checklist:** A checklist of employee traits in the form of statement is prepared where the rater put a tick mark in "Yes" or "No" column against the trait checked for each

employee. Once the checklist gets completed the rater forwards the list to the HR department for the final evaluation of the employee.

3. **Forced Choice Method:** Under this method, the rater is forced to answer the ready-made statements as given in the blocks of two or more, about the employees in terms of true or false. Once he is done with the list, it is forwarded to the HR department for the final assessment of the employee.
4. **Forced Distribution Method:** This method of performance appraisal is based on the assumption that employee's job performance conforms to the normal distribution curve i.e. a bell shaped curve. Hence, the rater is compelled to put employees on each point on the scale. It is seen that cluster of employees is placed at the highest point on a rating scale.
5. **Critical Incidents Method:** Under this method, the critical behavior of each employee that make a difference in the effective or non-effective performance is recorded by the supervisor and is taken into consideration while evaluating his performance.
6. **Behaviorally Anchored Rating Scale:** The descriptive statements about employees' behavior, both effective and ineffective are put on the scale points, and the rater is asked to indicate which behavioral point describes the employee behavior the best.
7. **Field Review Method:** Under this method, the performance evaluation of an employee is done by someone who does not belong to his department or is usually from the corporate office or the HR department. The outsider reviews the performance of an employee through his records and holds interviews with the assessee and his superiors.
8. **Performance Tests and Observations:** The test either written or oral is conducted to test the knowledge and the skills of employees. Sometimes the employee is asked to demonstrate his skills in the given situation and then he is evaluated on that basis.
9. **Confidential Reports:** The confidential reports are mostly prepared by the government departments, wherein the employee is evaluated on some of the following parameters:
 - Attendance
 - Leadership
 - Self-expression
 - Ability to work with others
 - Initiative
 - Technical ability
 - Integrity

- Responsibility, etc.

- 10. Essay Method:** Under this method, the detailed description of the employee performance is written by the rater. The performance of an employee, his relations with other Co-workers, requirements of training and development programs, strengths and weaknesses of the employee, etc. are some of the points that are included in the essay. The efficiency of this traditional method of performance appraisal depends on the writing skills of the rater.
- 11. Cost Accounting Method:** In this method, the employee's performance is evaluated in monetary terms, i.e. how much cost company is incurring on keeping the employee and how much he is contributing to the firm in terms of his output.
- 12. Comparative Evaluation Approaches:** Under this method, several comparative analysis are done, wherein the performance of one employee is compared with that of another Co-worker, and the rating is determined accordingly. The Ranking Method and the Paired Comparison Method are the usual comparative forms used in this approach.
 - **Ranking Method:** In the Ranking Method, superiors give the rank from best to worst, to their subordinates on the basis of their merits. Here, the detailed description of why best or why the worst is not given.
 - **Paired Comparison Method:** Under this method, the performance of each employee is compared with the other employee and then the decision on whose performance is better is made.

Thus, the performance of an individual can be evaluated using either of the methods that best measures the behavior of an employee and gives the exact evaluation of worker's doings.

b. Modern Methods of Performance Appraisal

- 1. Management by Objectives:** This concept was given by Peter.F.Drucker, according to him, the performance of an employee can be assessed on the basis of the targets achieved by him as set by the management of an organization. Firstly, the management sets the goals and communicate the same to the employees, and then the performance of an employee is compared against these set goals and is evaluated on this basis.

In case the employee is not able to achieve the pre-established goals, then management decides on a new strategy or policy that should be undertaken for the accomplishment of unattainable goals.
- 2. Psychological Appraisals:** This is one of the most frequently used modern methods of performance appraisal, wherein the psychologist assesses the employee's potential for the future performance rather than the past one. Under this method, the psychologist conducts

the in-depth interviews, psychological tests, discussions with the supervisors and the reviews of other evaluations.

This assessment is done to determine the intellectual, emotional, motivational and other related characteristics of an employee that is necessary to predict his potential for the future performance.

3. **Assessment Centres:** The assessment center is the central location where the managers come and perform the job exercises. Here the assessee is requested to participate in in-basket exercises, role playing, discussions, computer simulations, etc. Where they are evaluated in term of their persuasive ability, communication skills, confidence, sensitivity to the feelings of others, mental alertness, administrative ability, etc.

This entire exercise is done under the trainer who observes the employee behavior and then discusses it with the rater who finally evaluates the employee's performance.

4. **360-Degree Feedback:** It is a feedback method wherein the details of an employee's performance is gathered from other stakeholders such as superiors, peers, team members and self. In this method, the employee is asked about himself, i.e. what he feels about his performance, and then accordingly he can realize his strengths and weaknesses.

Also, the peers or team members are asked about assessee's performance through which the employee knows about what others feel about him and can overcome his disbeliefs if any. Thus, this method is used to have a detailed evaluation of an employee from all the perspectives.

5. **720-Degree Feedback:** This is an another feedback method, wherein the assessment is done not only by the stakeholders within the company but also from the groups outside the organization. These external groups who assesses the employee's performance are customers, investors, suppliers and other financial institutions.

It is one of the most crucial modern methods of performance appraisal because this is the only group that determines the success of the organization as a whole.

Nowadays, companies use the modern methods of performance appraisal which have a broader scope than the traditional methods and provides a more accurate and comprehensive evaluation of an individual.

EMPLOYEES GRIEVANCES

Employee grievance refers to the dissatisfaction of an employee with what he expects from the company and its management.

A company or employer is expected to provide an employee with a safe working environment, realistic job preview, adequate compensation, respect etc.

However, employee grievance is caused when there is a gap between what the employee expects and what he receives from the employer.

Steps in Employee Grievance Handling

1. Employee grievance should be submitted in a proper channel.
2. The supervisor of the employee should be informed and spoken to.
3. A review committee should examine the grievance for its validity and against the company's policy.
4. Resolution should be provided if the grievance is valid.
5. If the employee grievance is not resolved there should be a further body where it can be appealed.

Causes of Grievances:

Grievances may occur due to a number of reasons:

- 1. Economic:** Employees may demand for individual wage adjustments. They may feel that they are paid less when compared to others. For example, late bonus, payments, adjustments to overtime pay, perceived inequalities in treatment, claims for equal pay, and appeals against performance- related pay awards.
- 2. Work environment:** It may be undesirable or unsatisfactory conditions of work. For example, light, space, heat, or poor physical conditions of workplace, defective tools and equipment, poor quality of material, unfair rules, and lack of recognition.
- 3. Supervision:** It may be objections to the general methods of supervision related to the attitudes of the supervisor towards the employee such as perceived notions of bias, favouritism, nepotism, caste affiliations and regional feelings.
- 4. Organizational change:** Any change in the organizational policies can result in grievances. For example, the implementation of revised company policies or new working practices.
- 5. Employee relations:** Employees are unable to adjust with their colleagues, suffer from feelings of neglect and victimization and become an object of ridicule and humiliation, or other inter- employee disputes.
- 6. Miscellaneous:** These may be issues relating to certain violations in respect of promotions, safety methods, transfer, disciplinary rules, fines, granting leaves, medical facilities, etc.

Effects of Grievance:

Grievances, if not identified and redressed, may adversely affect workers, managers, and the organization.

The effects are the following:

1. On the production:

- a. Low quality of production
- b. Low productivity
- c. Increase in the wastage of material, spoilage/leakage of machinery
- d. Increase in the cost of production per unit

2. On the employees:

- a. Increase in the rate of absenteeism and turnover
- b. Reduction in the level of commitment, sincerity and punctuality
- c. Increase in the incidence of accidents
- d. Reduction in the level of employee morale.

3. On the managers:

- a. Strained superior-subordinate relations.
- b. Increase in the degree of supervision and control.
- c. Increase in indiscipline cases
- d. Increase in unrest and thereby machinery to maintain industrial peace

Effective Grievance Management:

The Manager should immediately identify all grievances and must take appropriate steps to eliminate the causes of such grievances so that the employees remain loyal and committed to their work. Effective grievance management is an essential part of personnel management. The managers should adopt the following approach to **manage grievance effectively**:

1. **Quick action-** As soon as the grievance arises, it should be identified and resolved. Training must be given to the managers to effectively and timely manage a grievance. This will lower the detrimental effects of grievance on the employees and their performance.
2. **Acknowledging grievance-** The manager must acknowledge the grievance put forward by the employee as manifestation of true and real feelings of the employees. Acknowledgement by

the manager implies that the manager is eager to look into the complaint impartially and without any bias. This will create a conducive work environment with instances of grievance reduced.

3. **Gathering facts-** The manager should gather appropriate and sufficient facts explaining the grievance's nature. A record of such facts must be maintained so that these can be used in later stage of grievance redressal.
4. **Examining the causes of grievance-** The actual cause of grievance should be identified. Accordingly remedial actions should be taken to prevent repetition of the grievance.
5. **Taking Decision-** After identifying the causes of grievance, alternative course of actions should be thought of to manage the grievance. The effect of each course of action on the existing and future management policies and procedure should be analyzed and accordingly decision should be taken by the manager.
6. **Execution and review-** The manager should execute the decision quickly, ignoring the fact, that it may or may not hurt the employees concerned. After implementing the decision, a follow-up must be there to ensure that the grievance has been resolved completely and adequately.

Need for a Formal Procedure to Handle Grievances:

A grievance handling system serves as an outlet for employee frustrations, discontents, and gripes like a pressure release valve on a steam boiler. Employees do not have to keep their frustrations bottled up until eventually discontent causes explosion.

The existence of an effective grievance procedure reduces the need of arbitrary action by supervisors because supervisors know that the employees are able to protect such behavior and make protests to be heard by higher management. The very fact that employees have a right to be heard and are actually heard helps to improve morale. In view of all these, every organization should have a clear-cut procedure for grievance handling.

10 Techniques to Better Handle Staff Complaints and Grievances

1. **Be a good listener:** Never interrupt when team members are talking, even if you disagree with the opinions expressed. Complaints often dissolve and resolve themselves when people simply have a chance to talk about them.
2. **Ask questions:** Your questions indicate interest and a desire for more information. When you ask open ended questions such as, "Why do you think that happened?" you may uncover underlying causes or related problems. When you ask good questions, you communicate that you are not unfairly pre-judging people or situations.
3. **Do not argue:** Present any information you have in a persuasive manner rather than an argumentative one. Arguing builds resistance and can make employees more determined to get their way regardless of the facts. Asking questions can be an effective tool for

disarming a potential argument. Your point of view is more persuasive when you refuse to be drawn into an argument.

4. **Make sure you understand:** Some people have difficulty expressing themselves and this can be even more difficult if they are stressed or emotional. Use all of your questioning and listening skills to make sure that you fully understand their position. Restate, summarise and ask additional questions to make sure you understand their point of view.
5. **Treat all employees with respect:** Ridicule or comments that minimise a person's concerns are powerful and devastating. They have no place in the management and leadership roles of today. If you attempt to make someone else feel foolish, you destroy the lines of communication and trust. Let others save face and retreat gracefully. Criticising and belittling people in front of others should also be avoided as this also destroys communication, trust and respect.
6. **Let the person know when to expect a response from you:** Many times, the problem can be settled on the spot. However, if it will take time to resolve, establish a time-frame for your action and response.
7. **Gather the facts:** If you are unable to make a decision during the meeting, investigate what the team member has said, check the situation, refer to employment agreements or other relevant documents and if appropriate, consult with higher management before making a final decision.
8. **Make a decision:** Once you make a decision (even if it is unpopular), stick to it firmly unless new evidence that deserves consideration is presented.
9. **Explain your decision:** If your decision is distasteful to the team member, take the time to explain it and answer any questions. Team members may not agree and may appeal your decision, but they will respect you for your stand.
10. **Thank the team member:** Express your appreciation for the person's willingness to communicate openly about problems. This encourages even more open communication in the future.

A legitimate complaint can signal not only discontent in the workplace, but it can also help to shine the light on serious infractions at your office. Psychologically, listening to your employees and addressing their concerns can be great for your company culture. It also shows them respect and fosters a sense of pride and accountability in their work. These benefits easily justify the implementation of a useful and effective system for addressing employee complaints.

4.1 STRATEGIC MANAGEMENT

Strategic Management

Strategic management is the process of setting goals, procedures, and objectives in order to make a company or organization more competitive.

Definition: The term 'strategic management' is used to denote a branch of management that is concerned with the development of strategic vision, setting out objectives, formulating and implementing strategies and introducing corrective measures for the deviations (if any) to reach the organization's strategic intent. It has two-fold objectives:

- To gain competitive advantage, with an aim of outperforming the competitors, to achieve dominance over the market.
- To act as a guide to the organization to help in surviving the changes in the business environment.

Here, changes refer to changes in the internal environment, i.e. within the organization, introduced by the managers such as the change in business policies, procedures etc. and changes in the external environment as in changes in the government rules that can affect business, competitors move, change in customer's tastes and preferences and so forth.

Strategic Management Process



1. Defining the levels of strategic intent of the business:
 - Establishing vision
 - Designing mission
 - Setting objectives
2. Formulation of strategy
 - Performing environmental and organizational appraisal
 - Considering strategies

- Carrying out strategic analysis
- Making strategies
- Preparing strategic plan

3. Implementation of strategy

- Putting strategies into practice
- Developing structures and systems
- Managing behavioural and functional implementation

4. Strategic Evaluation and Control

- Performing evaluation
- Exercising control
- Recreating strategies

Strategic Management is all about specifying organization's vision, mission and objectives, environment scanning, crafting strategies, evaluation and control.

Importance of Strategic Management

- It guides the company to move in a specific direction. It defines organization's goals and fixes realistic objectives, which are in alignment with the company's vision.
- It assists the firm in becoming proactive, rather than reactive, to make it analyse the actions of the competitors and take necessary steps to compete in the market, instead of becoming spectators.
- It acts as a foundation for all key decisions of the firm.
- It attempts to prepare the organization for future challenges and play the role of pioneer in exploring opportunities and also helps in identifying ways to reach those opportunities.
- It ensures the long-term survival of the firm while coping with competition and surviving the dynamic environment.
- It assists in the development of core competencies and competitive advantage, that helps in the business survival and growth.

VISION

A **vision** states what the organization aspires to become in the future.

According to Kotler “It is a description of something (an org, corporate culture, business, technology, and act) in the future.

What is a Vision Statement?

A vision statement describes what a company desires to achieve in the long-run, generally in a time frame of five to ten years, or sometimes even longer. It depicts a vision of what the company will look like in the future and sets a defined direction for the planning and execution of corporate-level strategies.

Vision **is long-term statement and typically generic & grand.** Therefore a vision statement does not change unless the company is getting into a totally different kind of business.

There are two components of vision

Core ideology – It rests on core values and core purpose. It defines the enduring character of an organization that remains unchanged.

Envisioned future – It is a long term audacious goal and a vivid description of achievement i.e. what an org will be in the future

Advantages of a good vision –

1. A good vision is inspiring and exhilarating
2. It helps in creation of common identity and shared sense of purpose
3. Vision is competitive, original and unique
4. It fosters risk taking and experimentation
5. It fosters long term thinking
6. It represents integrity Vision is used to convey what they stand for and what principles guide them to make strategic decisions.

Key Elements of a Good Vision Statement

While companies should not be too ambitious in defining their long-term goals, it is critical to set a bigger and further target in a vision statement that communicates a company's aspirations and motivates the audience. Below are the main elements of an effective vision statement:

- Forward-looking
- Motivating and inspirational

- Reflective of a company's culture and core values
- Aimed at bringing benefits and improvements to the organization in the future
- Defines a company's reason for existence and where it is heading

MISSION

According to David F. Harvey "A mission provides the basis of awareness of a sense of purpose, the competitive environment, degree to which the firm's mission fits its capabilities and the opportunity which the government offers.

According to Hynger "It is the purpose or reason for organizations existence" A mission states the role an organization plays in the society.

Nature of a Mission –

1. It gives social reasoning i.e. specifies the role an org plays in the society
2. It is philosophical and visionary i.e. it relates to top management values and has a long term perspective.
3. It legitimizes social existence
4. It is stylistic objectives i.e. it reflects corporate philosophy, identity, character and image of an organization

Characteristics of a Mission –

- It should be feasible
- It should be precise
- It should be clear
- It should be motivating
- It should be distinctive
- It should have a societal linkage.
- It should indicate major components of strategy
- It should indicate how objectives are to be accomplished
- It must be dynamic

Example of a good mission statement

Eg. India Today "the complete new magazine"

Ranbaxy Industries "to become a research based international pharmaceuticals company"

Formulation and communication of mission statements –

- It is formulated from the following sources:
- National priorities projected in plan documents and industrial policy statements

- Corporate philosophy as developed over the years
- Major strategists have a vision to develop mission statements
- The services of a consultant may be hired
- It is communicated through annual reports, posters, employee manuals, company information kits, word of mouth publicity, seminars and workshops, newsletters and advertisements.

Importance of vision and mission statement in an organization

Both the vision and mission statements play an important role in the organization. Below is a look at these roles:

1. The vision and mission statements **define the purpose** of the organization and instill a sense of belonging and identity to the employees. This motivates them to work harder in order to achieve success.
2. The mission statement acts as a “**North Star**”, where it provides the direction that is to be followed by the organization while the vision statement provides the goal (or the destination) to be reached by following this direction.
3. The vision and mission statements help to properly **align** the resources of an organization towards achieving a successful future.
4. The mission statement provides the organization with **a clear and effective guide for making decisions**, while the vision statement ensures that all the decision made are properly aligned with what the organization hopes to achieve.
5. The vision and mission statements provide a focal point that helps to align everyone with the organization, thus ensuring that everyone is working towards a single purpose. This helps to **increase efficiency and productivity** in the organization.

The vision and mission statements are important **tools of strategic planning**, and thus they help to shape the strategy that will be used by an organization to achieve the desired future.

GOALS

Goal is defined as an “intermediate result to be achieved by a certain time as part of the grand plan . A plan can, there for have many goals.”

Goal is a specific target that a firm intend to reach in long term.

- A Goal describes clearly the activities and task to be completed by an individual, a department or an organization.
- Goals should be measurable, quantitative, challenging, realistic, consistent and prioritized.

- Provide basics for measuring companies performance and the process it is making towards the vision.
- Strategic goals help managers to establish end result of activities in general without getting bogged down in details, such as issues of measurement and timing

Two types of Goals

- 1) Financial goals :** These goals focus on achieving a certain level of financial performance, measured in terms of return in investment or growth of revenues
- 2) Strategic goals :** The goals focus on achieving strategic or competitive advantages within the industry, like technology leadership, creativity and innovation and superior customer service

Characteristics of goal:

According to M.D.Richards :

- 1) Specific :** so that they are precise and measurable, would assist management in monitoring the progress towards achievement of goals at each specific point of time

2) Issues of goal:

- ☐ Short-term goals and objectives should be left to lower level managers to identify, plan and achieve
- ☐ Issues like lowering of cost and improving quality should be included in goals of middle level managers

3) Should be well constructed, realistic and challenging:

- ☐ Challenging goals motivates managers to be innovative, creative and ambitious in improving operations, marketing, sales, etc.

4) Specification of time period

Examples of goals

Strategic goals of 'El paso Electrical Company' in various operations are :

Customer Service :

- ☐ Provide quality service to customer at least equal to the highest standard in the industry
- ☐ Maintain reliability of service to customer at a level above 99%
- ☐ Ensure that customer are educated about the safety aspects of using electricity.

Community service :

- ☐ Promote economic growth and increased development of the company's total service area
- ☐ Provide job opportunities and investment in the service area which promotes a higher standards of living for all citizens.
- ☐ Cooperate with and serve the educational institutions located in the service area in a manner consistent with other leaders in the industry.

STRATEGY

What is Strategy?

Strategy comes from a Greek word "strategos" which means generalship i.e. art of the general.

A strategy could be

- A plan or course of action or a set of decisions making a pattern or creating a common thread.
- A pattern or a common thread derived from the policies goals and objectives of an organization.
- The activities that move an organization from its current state to desired future state.
- The resources necessary for implementing a plan or following a course of action.

Prof. Henry Mintzberg distinguished between intended and emergent strategies.

Intended strategies refer to the plans that managers develop while emergent strategies are the actions that take place over a period of time. It is simply the means to achieve objectives.

A strategy has the following components –

- It includes clear set of long term goals.
- It defines the scope of the firm(type of products)
- It is a statement of competitive advantage.
- It represents a firm's internal activities which will allow it to achieve competitive advantage in the industry.

Therefore strategy is a large scale future oriented plan used to interact with competitive environment to achieve objectives. It provides a framework for managerial decisions. it reflects company's awareness of the main elements of competition.

DIMENSIONS OF STRATEGY/STRATEGIC DECISIONS

1. Strategic issues require top management decisions.
2. Strategic issues involve the allocation of large amount of company's resources.

3. Strategic issues are likely to have a significant impact on the long term prosperity of the firm.
4. Strategic issues are future oriented.
5. Strategic issues usually have major multi-functional or multi-business consequences.
6. Strategic issues necessitate consideration of factors in the firm's external environment.

LEVELS OF STRATEGY

A. CORPORATE LEVEL– Board of Director, CEO, Administration and management.

B. BUSINESS LEVEL – Business and corporate managers, SBU*, SBA*

C. FUNCTIONAL LEVEL – Product managers, geographic and functional level managers

* SBU (Strategic business unit) – It is any part of the organization which is operated separately for strategic management purposes.

* SBA (Strategic business area) – A distinctive segment of the external environment in which the firm operates.

CORPORATE PLANNING PROCESS

We can define corporate planning as follows:

Corporate planning is a total system of planning which involves the determination of the objectives for the company as a whole and for each department of the it; formulation of strategies for the attainment of these objectives (all this being done against the background of SWOT analysis); conversion of strategies into tactical plans (or operational plans); implementation of tactical plans and a review of the progress of tactical plans against the corporate planning objectives.

Analysis of the above definition:

On the basis of the above definition, we can state the following main features of corporate planning:

- (i) Corporate planning is a total system of planning, under which concept objectives are determined for the company as a whole and for each department of it. This means that under the concept of corporate planning, no department of company is allowed to have its own independent planning. All departmental plans are a part of corporate planning, in a unified structure.
- (ii) To realize the objectives of corporate planning, strategy formulation is done. Strategy formulation is the core aspect of corporate planning. Success of corporate planning depends on the success of strategy formulation

(iii) Determination of objectives of corporate planning and strategy formulation – both are done against the background of SWOT analysis.

(iv) Strategies are translated (or converted) into tactical plans (or operational plans), which are detailed in nature.

(v) Tactical plans are put to action at the right time, as decided by management. This is the practical aspect of corporate planning.

(vi) Performance of tactical plans is judged in the light of the objectives of corporate planning; so that necessary modifications might be made in the corporate planning process and better corporate planning might be done in future.

(vii) Corporate planning has a long-term perspective; while operational plans have a short-term prospective.

David Hussey (Corporate Planning: Theory and Practice) defines corporate planning as follows:

“Corporate planning includes the setting of objectives, organizing the work, people and systems to enable those objectives to be achieved, motivating through the planning process and through the plans, measuring performance and so controlling progress of the plans and developing people through better decision-making, clearer objectives, more involvement, and awareness of progress.”

Process of Corporate Planning:

Major steps involved in corporate planning are as follows:

(i) Environmental Analysis and Diagnosis:

The first steps (which is, in fact, the background step), involved in corporate planning is environmental analysis and diagnosis. (A detailed account of this step is attempted subsequently, in the discussion about corporate planning).

(ii) Determination of Objectives:

All planning starts with a determination of the objectives for the plan; and corporate planning is no exception to this generality. In corporate planning, after environmental analysis and diagnosis, the planners determine objectives for the company as a whole and for each department of it; which become the beginning point of corporate planning.

All objectives of corporate planning must represent an integrated or coordinated system of objectives. In order to make corporate planning a realistic approach to attaining objectives; objective setting for corporate planning is done in the light of environmental analysis and diagnosis.

(iii) Strategy Formulation:

Strategy formulation is the core aspect of corporate planning. Strategy is, in fact, the weapon of the planner devised for attaining objectives of corporate planning. It is easier to set objectives; it is difficult to realize them. Strategies facilitate the attainment of objectives.

There is no doubt about it that success of strategies is the success of corporate planning; and vice-versa. Strategy formulation is also done in the light of environmental analysis and diagnosis.

(iv) Development of Tactical Plans:

Strategies are translated into action plans called tactical plans or operational plans. Tactical plans are necessary for implementation of strategies leading to the attainment of corporate planning objectives. For example, if the strategy of a company is to develop the skills and talents of manpower for realizing objectives; then designing of suitable training programmes would amount to making tactical plans.

Corporate planning and strategy formulation have a long-term perspective; while tactical plans have a short-term perspective, as the latter are to be implemented immediately, in the usual course of organisational life.

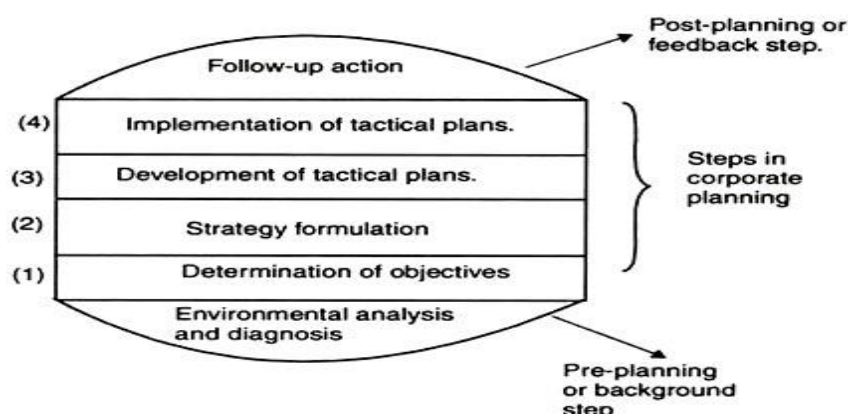
(v) Implementation of Tactical Plans:

Mere paper planning is no planning; unless and until it is put into practice. As such, tactical plans are put into a process of implementation, just at the right time, as decided by management. For implementation purposes, necessary communications are made to the operating staffing; who are also provided with necessary facilities to implement the tactical plans.

(vi) Follow-Up-Action:

After the tactical plans have been put into practice; a review of progress is done i.e. an examination of what results are following from the implementation of the plan and what feedback action is necessary, for the betterment of the corporate planning process.

The following chart depicts the corporate planning process:



ENVIRONMENTAL SCANNING

Environment is anything and everything surrounding us. It consists of the living and nonliving things around us.

Characteristics of environment

- It is complex – It consists of many interrelated factors
- It is dynamic – Due to its shape and character
- It is multifaceted – The environment developed is perceived differently by different people
- It has a far reaching compact – The growth and profitability of the organization critically depends upon its environment.

Components of environment

Internal or micro environment

Customers
Suppliers
Organization
Competition
Market – cost structure ,price sensitivity,
distribution system, technology, market
stability
Intermediaries

External or macro environment

Economic environment
Demographic environment
Technological environment
Legal and political environment
Socio-cultural environment

Global/International environment

ENVIRONMENTAL SCANNING

Environmental Scanning is the process by which an organization monitors the relevant environment that identify opportunities and threats affecting the business for the purpose of making strategic decisions.

Factors to be considered while conducting an environmental appraisal –

Events – Important and specific occurring taking place in different environment sectors

Trends – General tendencies or course of action around which events and trends take place.

Issues – Current concern that arise or respond to events and trends

Expectations – Demand which are made by interested group in the light of current issue

Dimensions of Environmental Scanning:

SPECTACLES – Social, Political, Economic, Cultural, Technological, Aesthetic, Customer, Legal, Environmental and Sectoral

Approach's to environmental scanning

1. Systematic approach – In this approach information for environmental scanning is collected systematically. Information related to markets and customers, change in legislation and regulations etc. have a direct impact on the business therefore it is continually monitored for relevant facts. This approach is beneficial for strategic management and operational activities.

2. Ad-hock approach -An organization may conduct a special survey related to specific environmental issues from time to time... such studies may be undertaken when an organization has to take up new projects or update existing strategies. Changes and unforeseen developments may be investigated which affect the organization

3. Processed form approach-The organization under this approach uses information in a processed form available from both internal and external environment.eg. Information supplied by government agencies, private institutions, it uses secondary data method.

Sources of information for environment scanning

Secondary sources of information- newspapers,magazines,journals,documentaries

Mass media- radio, TV. , internet

Internal sources- Internal files , documents, reports ,records , database

External agencies-Customers, market intermediaries, suppliers, retailers etc.

Spying and surveillance- through ex-employees of competitors, industrial espionage, planting a mole in rival company.

Factors affecting Environmental scanning / Environmental Appraisal

1. Strategist related factors – Since strategist play a central role in formulation of strategies ,there characteristics such as age, education, experience, motivation, cognitive styles ,ability to withstand time pressure and responsibility have an impact to the extent to which they are able to appraise the environment.

2. Organization related factors – Characteristics of the organization which affect the environment are nature of the business, its age, size, complexity, nature of it market, product or service that it provides.

3. Environment related factors – The nature of environment faced by the organization determines how its appraisal could be done. The nature of the environment depends upon its complexity, volatility, hostility and diversity.

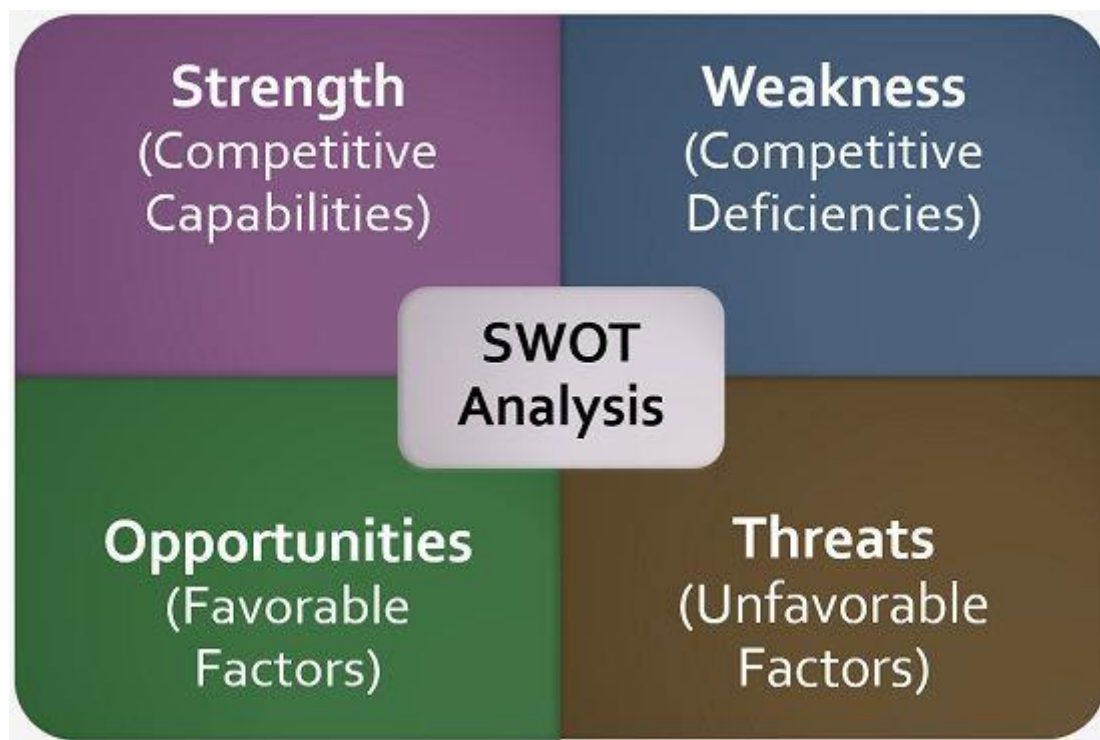
Techniques of Environmental Scanning / Environmental Appraisal

SWOT Analysis, ETOP Analysis, ANSOFF Matrix, BCG Matrix.

SWOT Analysis

Definition: SWOT Analysis is a strategic management tool that assists an enterprise in discerning their internal Strengths, and Weaknesses, and external Opportunities, and Threats, to determine its competitive position in the market.

The SWOT Analysis helps in ascertaining the factors that influences the efficiency and effectiveness of any product, project, or business entity. These are explained as under:



1. **Strengths:** The strengths of a company are the core competencies, in which the business has an edge over its competitors. It covers aspects such as:
 - Strong financial condition
 - A large customer base.
 - Strong brand name or a unique product
 - Latest technology or patents
 - Influential advertising and promotion.
 - Cost Advantage
 - Quality in product and customer service.
2. **Weaknesses:** Weaknesses can be described as the areas of limitations of the business, that hinders the growth of the company and even leads to a strategic disadvantage.

These are the areas which need improvement to perform competitively. It encompasses:

- Obsolete facilities and outdated technology.
- The unit cost of a product is higher than the competitors.
- No or less internal control.
- Less quality in products and services offered.
- Weak brand image.
- Financial condition is not very sound.
- Underutilization of plant capacity.
- Lack of major skills or competencies, and intellectual capital.

3. **Opportunities:** Opportunities can be understood as the condition, which is favourable or beneficial to the organization in the business environment, that the business could exploit to gain an advantage. These are:

- Looking for areas of development, by utilizing skills and technology to enter new markets
- Adding new products to the existing product line to increase customer base.
- Forward and backward integration.
- Acquiring rivals businesses.
- Joint ventures, mergers and alliances to increase market coverage.

4. **Threats:** Threat implies an adverse condition which can lead the business enterprise to losses, and can also harm the overall position and reputation of the enterprise. It entails:

- A downtrend in market growth.
- A new entrant to the market.
- Substitute products that can decrease sales.
- Increasing the bargaining power of customers and suppliers.
- New regulatory requirements

- Changes in a demographic environment that will decrease demand for firm's product.

Importance of SWOT Analysis

- **Logical framework of analysis:** SWOT Analysis equips the management with an insightful framework for eliminating issues in a systematic manner, that can influence the condition of business, formulation of various strategies and their selection.
- **Presents a comparative report:** The analysis facilitates in presenting systematic information about the internal and external environment. This helps in making a comparison of external opportunities and threats with internal strengths and weaknesses, as well as reconciling the internal and external business environment, to help the managers in choosing the best strategy, by considering various patterns.
- **Strategy Identification:** Every organization has its strengths weakness, opportunities and threats. So, the SWOT Analysis acts as a guide to the strategist to reckon the exact position, i.e. where the business stands, so as to identify the primary objective of the strategy under consideration.

SWOT Analysis helps the company's management in designing a business model specific to the firm. The model perfectly suits or aligns the company's resources or competencies, as per the needs of the business environment, wherein the organization operates and helps in gaining a competitive advantage over the rivals. This will increase the profitability, market share and the chances to survive in the dynamic competitive business environment.

There are four sequential steps of conducting a SWOT Analysis –

- Setting up objectives for the concerned organization
- Identifying the strengths, weaknesses, opportunities and threats of the business
- Answering the 4 major questions –
 - ✓ How to maximize the internal strengths of the organization?
 - ✓ How to minimize the internal weaknesses of the organization?
 - ✓ How to capitalize on opportunities present in the external environment of the organization?
 - ✓ How to protect the organization from threats in the external environment?
- Based upon the results of the above analysis a SWOT matrix is prepared which consists of Strengths, Weaknesses, Opportunities and Threats present in the Organization's internal and external environment and its impact on the business is studied.

A firm must direct its strengths towards exploitation of opportunities and blocking threats which minimizing the exposure of weaknesses at the same time.

SWOT Analysis is an important tool for auditing the overall strategic position of a business and its environment. The basic objective of SWOT Analysis is to provide a frame work to reflect a firm's ability to overcome barriers (threats) and avail opportunities emerging in the environment.

Advantages of SWOT Analysis –

- It is simple to use
- Low cost is involved
- It is flexible and can be adapted to varying situations
- It leads to clarification of issues
- It helps in development of goal oriented objectives
- It is useful as a starting point for strategic analysis

Disadvantages of SWOT Analysis

- Realities may be more complex than represented by swot matrix due to simplicity of its use
- It may result in compiling of lists rather than focusing on organizational objectives
- Strengths may sometimes be confused with opportunities
- The Person conducting the analysis may be biased towards a view-point and may misinterpret the situation

STRATEGY FORMULATION

Definition: Strategy Formulation is an **analytical process of selection of the best suitable course of action to meet the organizational objectives and vision**. It is one of the steps of the strategic management process. The strategic plan allows an organization to examine its resources, provides a financial plan and establishes the most appropriate action plan for increasing profits.

It is examined through **SWOT** analysis. SWOT is an acronym for strength, weakness, opportunity and threat. The strategic plan should be informed to all the employees so that they know the company's objectives, mission and vision. It provides direction and focus to the employees.

Steps of Strategy Formulation

The steps of strategy formulation include the following:



1. **Establishing Organizational Objectives:** This involves establishing long-term goals of an organization. Strategic decisions can be taken once the organizational objectives are determined.
2. **Analysis of Organizational Environment:** This involves SWOT analysis, meaning identifying the company's strengths and weaknesses and keeping vigilance over competitors' actions to understand opportunities and threats.

Strengths and weaknesses are internal factors which the company has control over. Opportunities and threats, on the other hand, are external factors over which the company has no control. A successful organization builds on its strengths, overcomes its weakness, identifies new opportunities and protects against external threats.

3. **Forming quantitative goals:** Defining targets so as to meet the company's short-term and long-term objectives. Example, 30% increase in revenue this year of a company.
4. **Objectives in context with divisional plans:** This involves setting up targets for every department so that they work in coherence with the organization as a whole.
5. **Performance Analysis:** This is done to estimate the degree of variation between the actual and the standard performance of an organization.

6. **Selection of Strategy:** This is the final step of strategy formulation. It involves evaluation of the alternatives and selection of the best strategy amongst them to be the strategy of the organization.

Strategy formulation process is an integral part of strategic management, as it helps in framing effective strategies for the organization, to survive and grow in the dynamic business environment.

Levels of strategy formulation

There are three levels of strategy formulation used in an organization:



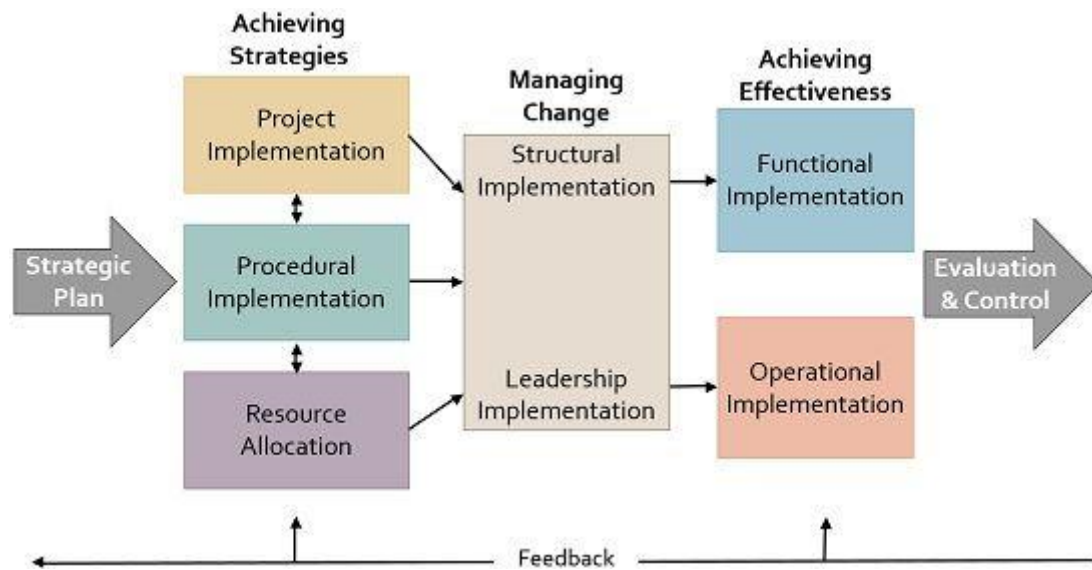
- **Corporate level strategy:** This level outlines what you want to achieve: growth, stability, acquisition or retrenchment. It focuses on what business you are going to enter the market.
- **Business level strategy:** This level answers the question of how you are going to compete. It plays a role in those organization which have smaller units of business and each is considered as the strategic business unit (SBU).
- **Functional level strategy:** This level concentrates on how an organization is going to grow. It defines daily actions including allocation of resources to deliver corporate and business level strategies.

Hence, all organisations have competitors, and it is the strategy that enables one business to become more successful and established than the other.

STRATEGY IMPLEMENTATION

Definition: Strategy Implementation refers to the **execution of the plans and strategies**, so as to accomplish the long-term goals of the organization. It converts the opted strategy into the moves and actions of the organisation to achieve the objectives.

Simply put, strategy implementation is the technique through which the firm develops, utilises and integrates its structure, culture, resources, people and control system to follow the strategies to have the edge over other competitors in the market.



Strategy Implementation is the **fourth stage of the [Strategic Management](#) process**, the other three being a determination of strategic mission, vision and objectives, environmental and organisational analysis, and [formulating the strategy](#). It is followed by Strategic Evaluation and Control.

Process of Strategy Implementation

1. Building an organization, that possess the capability to put the strategies into action successfully.
2. Supplying resources, in sufficient quantity, to strategy-essential activities.
3. Developing policies which encourage strategy.
4. Such policies and programs are employed which helps in continuous improvement.
5. Combining the reward structure, for achieving the results.
6. Using strategic leadership.

The process of strategy implementation has an important role to play in the company's success. The process takes places after environmental scanning, SWOT analyses and ascertaining the strategic issues.

Prerequisites of Strategy Implementation

- **Institutionalization of Strategy:** First of all the strategy is to be institutionalized, in the sense that the one who framed it should promote or defend it in front of the members, because it may be undermined.

- **Developing proper organizational climate:** Organizational climate implies the components of the internal environment, that includes the cooperation, development of personnel, the degree of commitment and determination, efficiency, etc., which converts the purpose into results.
- **Formulation of operating plans:** Operating plans refers to the action plans, decisions and the programs, that take place regularly, in different parts of the company. If they are framed to indicate the proposed strategic results, they assist in attaining the objectives of the organization by concentrating on the factors which are significant.
- **Developing proper organisational structure:** Organization structure implies the way in which different parts of the organisation are linked together. It highlights the relationships between various designations, positions and roles. To implement a strategy, the structure is to be designed as per the requirements of the strategy.
- **Periodic Review of Strategy:** Review of the strategy is to be taken at regular intervals so as to identify whether the strategy so implemented is relevant to the purpose of the organisation. As the organization operates in a dynamic environment, which may change anytime, so it is essential to take a review, to know if it can fulfil the needs of the organization.

Even the best-formulated strategies fail if they are not implemented in an appropriate manner. Further, it should be kept in mind that, if there is an alignment between strategy and other elements like resource allocation, organizational structure, work climate, culture, process and reward structure, then only the effective implementation is possible.

Aspects of Strategy Implementation

- Creating budgets which provide sufficient resources to those activities which are relevant to the strategic success of the business.
- Supplying the organization with skilled and experienced staff.
- Conforming that the policies and procedures of the organisation assist in the successful execution of the strategies.
- Leading practices are to be employed for carrying out key business functions.
- Setting up an information and communication system, that facilitate the workforce of the organisation, to perform their roles effectively.
- Developing a favourable work climate and culture, for proper implementation of the strategy.

Strategy implementation is the time-taking part of the overall process, as it puts the formulated plans into actions and desired results.

STRATEGY EVALUATION –

Meaning

Strategy evaluation is that phase of the strategic management process in which manager tries to assure that the strategic choice is properly implemented and is meeting the objectives of the enterprise. When one talks of evaluation one cannot forget control aspect.

‘Strategy evaluation’ is the process through which the strategists know the extent to which a strategy is able to achieve its objectives. In the words of Professor William F. Glueck and Lawrence R. Jauch, “Evaluation of strategy is that phase of strategic management process in which the top managers determine whether their strategic choice as implemented is meeting the objectives of the enterprise.”

Strategy Evaluation – Basic Requirements

Strategy evaluation system must meet several basic requirements to be effective.

1. Activities must be economical.
2. Should not provide too much information and provide required information at right time.
3. Should be done with average control and avoid too many controls.
4. Activities should specifically relate to a firm’s objectives.
5. Should be designed to provide a true picture of what is happening.
6. Process should not dominate strategic decisions, it should foster mutual understanding, trust and commonsense.

The process of strategy evaluation consists of following steps:

1. Fixing Benchmark of Performance:

While fixing the benchmark, strategists encounter questions such as – what benchmarks to set, how to set them and how to express them. In order to determine the benchmark performance to be set, it is essential to discover the special requirements for performing the main task.

The performance indicator that best identify and express the special requirements might then be determined to be used for evaluation. The organization can use both quantitative and qualitative criteria for comprehensive assessment of performance.

A quantitative criterion includes determination of net profit, ROI, earning per share, cost of production, rate of employee turnover etc. Among the Qualitative factors are subjective evaluation of factors such as – skills and competencies, risk taking potential, flexibility etc.

2. Measurement of Performance:

The standard performance is a bench mark with which the actual performance is to be compared. The reporting and communication system help in measuring the performance. If appropriate means are available for measuring the performance and if the standards are set in the right manner, strategy evaluation becomes easier.

But various factors such as managers' contribution are difficult to measure. Similarly divisional performance is sometimes difficult to measure as compared to individual performance. Thus, variable objectives must be created against which measurement of performance can be done.

The measurement must be done at right time else evaluation will not meet its purpose. For measuring the performance, financial statements like – balance sheet, profit and loss account must be prepared on an annual basis.

3. Analyzing Variance:

While measuring the actual performance and comparing it with standard performance there may be variances which must be analyzed. The strategists must mention the degree of tolerance limits between which the variance between actual and standard performance may be accepted.

The positive deviation indicates a better performance but it is quite unusual exceeding the target always. The negative deviation is an issue of concern because it indicates a shortfall in performance. Thus in this case the strategists must discover the causes of deviation and must take corrective action to overcome it.

4. Taking Corrective Action:

Once the deviation in performance is identified, it is essential to plan for a corrective action. If the performance is consistently less than the desired performance, the strategists must carry a detailed analysis of the factors responsible for such performance. If the strategists discover that the organizational potential does not match with the performance requirements, then the standards must be lowered.

Another rare and drastic corrective action is reformulating the strategy which requires going back to the process of strategic management, reframing of plans according to new resource allocation trend and consequent means going to the beginning point of strategic management process

After developing a number of strategic alternatives, they should be evaluated against the criteria, in order to select the best strategy. The process of evaluation is discussed below.

The steps in the process of strategic evaluation are:

(i) The first step is a strategic analysis in order to gain a clear understanding of the circumstances affecting the organisation's strategic situation.

- (ii) The second step is to produce a range of strategic options.
- (iii) The third step is to develop a basis of comparison. This may be available from the strategic analysis or may need to be specially developed.
- (iv) It is helpful to establish the underlying rationale for each strategy by explaining why the strategy might succeed. This is often done in qualitative terms and by using techniques like scenario building product portfolio analysis and the assessment of synergy.
- (v) At this stage, the large number of strategic alternatives may be narrowed down, before a more detailed analysis is undertaken. Strategic alternatives may be ranked, based on their relative merits and demerits.
- (vi) Suitability of each alternative should be tested. There are a number of techniques for testing. The specific choice of technique will depend upon the circumstances.
- (vii) The next stage is assessing the feasibility and acceptability of strategies which appear reasonably suitable based on the analysis. The choice of the technique should be based on the circumstances of the company.
- (viii) Finally, the company will need some system for selecting future strategies as a result of these evaluations.

Importance of strategic Evaluation

- Strategic evaluation helps to keep a check on the validity of a strategic choice.
- An ongoing process of evaluation would, in fact, provide feedback on the continued relevance of the strategic choice made during the formulation phase

Nature of Strategic Evaluation

- Nature of the strategic evaluation and control process is to test the effectiveness of strategy.
- During the two proceedings phases of the strategic management process, the strategists formulate the strategy to achieve a set of objectives and then implement the strategy.
- There has to be a way of finding out whether the strategy being implemented will guide the organization towards its intended objectives.

Participants in Strategic Evaluation

- Shareholders
- Board of Directors
- Chief Executives
- Profit-centre heads
- Financial controllers
- Company secretaries

- External and Internal Auditors
- Audit and executive committees
- Corporate planning staff or department
- Middle-level managers

Barriers in Evaluation

- Limits of control
- Difficulties in measurement
- Resistance to evaluation
- Rely on short-term implication of activities

Strategic Control

The types of strategic controls are:

- Premise control
- Implementation control
- Strategic surveillance

4.2 PROJECT MANAGEMENT

PROJECT

A project is a temporary venture that exists to produce a defined outcome. Each project will have agreed and unique objectives as well as its own **project plan, budget, timescale, deliverables** and **tasks**. A project may also involve people from different teams within an organization who are brought together to accomplish a specific goal.

PROJECT MANAGEMENT

Project management can be defined as the discipline of applying specific processes and principles to initiate, plan, execute and manage the way that new initiatives or changes are implemented within an organization.

Project management refers to the application of man, machinery, money and knowledge to achieve objectives of a project.

Project management is different to management of **business as usual** activity, which is an ongoing process, as it involves creating new work packages to achieve agreed ends or goals.

Project management involves the planning and organization of a company's resources to move a specific task, event, or duty towards completion.

Key components of project management are:

- **Time** – the intended duration of the work
- **Cost** – the budget allocated for the work
- **Scope** – what innovations or changes will be delivered by the project
- **Quality** – the standard of the outcome of the project.

Increasing or decreasing any one of these components will affect the others.

For example, reducing the **time** allocated to complete the project will also reduce the amount of work that can be done (**scope**), which may then affect the **quality** and the **cost** of the project.

Project management stages

Although there are different **project management methodologies** and approaches, most projects follow these stages:

- **Initiating the project** – the project manager defines what the project will achieve and realize, working with the project sponsor and stakeholders to agree deliverables.
- **Planning** – the project manager records all the tasks and assigns deadlines for each as well as stating the relationships and dependencies between each activity.

- **Execution** – the project manager builds the project team and also collects and allocates the resources and budget available to specific tasks.
- **Monitoring** – the project manager oversees the progress of project work and updates the project plans to reflect actual performance.
- **Closing** – the project manager ensures the outputs delivered by the project are accepted by the business and closes down the project team.

NETWORK ANALYSIS

Network Analysis refers to a number of techniques for the planning and control of complex projects. The two most frequently used forms of network planning are:

1. Programme Evaluation and Review Technique (PERT)
2. Critical Path Method (CPM)

WHAT IS A NETWORK?

A network is a graphical diagram consisting of certain configuration of “Arrows” (>) and “Nodes” (•) for showing the logical sequence of various tasks to be performed to achieve the project objective.

Importance of Network Analysis/Network technique:

A project has divided into many small activities and these activities can be analyzed with the help of network technique to achieve the objectives of the project.

- Network analysis helps management to minimize the total cost and total maintenance time. With the use of network analysis cost of production can be minimized through reducing the maintenance time.
- Network analysis ensures the effective utilization of limited resources. It also ensures the optimal use of resources and help to control the idle resources so that project can be effectively executed within the budgeted costs and scheduled time.
- Network analysis facilitates co-ordination among the activities as well the persons responsible for project.
- Time management plays a crucial role in every project. Sometimes available resources have to be arranged with a view to reduce the total time for the project rather than reducing the cost of the project.
- Network analysis helps the managers to manage activities without any delay.
- Network analysis is great tool which helps in planning, scheduling and controlling the activities of the project.
- Network analysis also creates inter-relationship as well as inter-dependence of various activities of project. It helps in integrating the project planning and this relationship assists in

bringing out the technological inter-dependence of the various activities.

- Network analysis provides the project formulation team an apparent picture of the work elements and also sequential relationship of the project.

Classification of Network Techniques:

There are number of network techniques which are used by the various people according to their purpose. The main techniques are given below:

- 1. CPM:** It is popularly known as Critical Path Method. Critical path method is a project management tool used to formulate a time frame for a project in order to determine where potential delays are most likely to take place.
- 2. PERT:** The Programme Evaluation and Review Technique is basically a scheduling technique. It helps project manager in planning, scheduling, monitoring, evaluating, and controlling large and complex projects. It is a probabilistic model and introduces uncertainties in project network.
- 3. GERT:** The Graphical Evaluation and Review Technique is a new technique and superior over the above mentioned techniques. In this analysis only simulation can be used.
- 4. LOB:** It is known as Line of Balance technique. Line of balance is a graphical technique to show the progress achieved during the project with the help of key events.
- 5. PERT/Cost:** It is an extension of the PERT technique to cover the cost of project. It is not only helpful to plan the completion of project within a specific time but also within a specific cost.
- 6. WASP:** It is known as Workshop Analysis Scheduling Programme. This method is propounded by The British Energy Authority. Though there are several techniques but the most commonly used network techniques are PERT and CPM which will be discussed in detail.

Programme Evaluation and Review Technique (PERT):

Project Programme Evaluation and Review Technique, which is generally popular as PERT, is a sophisticated and a fairly new tool used by the management for planning and control in the case of special project. It is primarily a scheduling technique. It shows any project or job as a set of processes of operations called activities which must take place in a certain sequence. It involves diagrammatic presentation of activities and events involved in a long term project.

The diagrammatic presentation is known as Network Drawing/technique and these techniques are most commonly used in project management. PERT was developed in 1958 as a result of collaboration between the operational research division of the United States Navy and a team of management consultants known as Boose Allen and Hamilton.

It was developed as a management tool to aid for completing Polaris Ballistic Missile Project which had 250 prime contractors and over 9000 sub contractors engaged in research, development, construction, testing and production of missiles, guidance system and maintenance system.

Since then, it has been adopted by many enterprises in different industries. So PERT schedules the sequence of activities to be completed in order to accomplish the project within a short period of time. Basic objective of PERT is to control time. The execution of project becomes very difficult where long times involved in the planning and scheduling of the project because it involved lot of complexities and inter related activities. So for the successful implementation of the project, project manager is to take some important decisions such as estimation of resource requirement, time for each activity, and maintaining inter-relationship amongst the activities.

Thus each project involves various decisions about:

1. What is the expected completion time of project?
2. What will be the effect of any delay on the activity/ project?
3. What type of additional resources needed if project has to be completed before scheduled time?
4. What is the probability of completing the project in time?

PERT is helpful to the project manager for taking decisions about these questions. It is a technique which helps project manager in planning, scheduling, monitoring, evaluating and controlling large and difficult projects.

In simple words we can say that projects whose time duration of activities is not exactly known, PERT is used. It depends upon three time estimates of activities.

In PERT below three estimates of time are used:

- The most optimistic time (t_o): The minimum time that would be required to perform the activity if everything goes extremely well, the chance of such an optimum activity actually takes place is one in hundred.
- The most likely time (t_m): The length of time that will, in all probability, be required to perform the job under the given circumstances or normal circumstances.
- The most pessimistic time (t_p): This is the longest or maximum probable time involved if everything that might logically go wrong does actually go wrong. It includes time for unusual days or unforeseen circumstances. The chance of its happening might also be one in hundred or very less. With the help of these above mentioned time estimates i.e. optimistic time, most likely time, and pessimistic time, average expected time for each activity would be determined.

Average expected time of the activity = $t_o + 4t_m + t_p / 6$

Procedure followed in PERT

- First of all, the network of activities is drawn to indicate what activity follows what.
- Then estimation of time to complete each activity is noted on the network.
- Estimation of minimum time taken to complete the project.
- Identification of critical activities and allocation of resources so that project can be completed in time.
- Calculation of project variability duration and profitability of the project in given period.
- In order to complete project in time closer watch on critical and other activities.

Advantages of PERT

1. It is very helpful in determine the schedule for a project within time limit.
2. It helps the management to optimum allocation of resources for the project.
3. It helps in taking right decision for the projects at a right time.
4. It is very helpful in determine the expected duration of activities.
5. It helps the management in handling the uncertainties involved in the project.
6. It helps the management to reduce the risk element in the project.
7. It suggests area of increasing efficiency, decreasing cost and maximizing profits.
8. It helps in co-coordinating the various activities involved in a project.
9. It enables the use of statistical analysis.
10. It enables a manager to know in advance, where the trouble may occur, where more supervision needed, and where resources may be transferred to keep the project on schedule.

Limitations of PERT

Although PERT have many advantages, but it also suffers from certain limitations. These are:

1. PERT emphasis only on time. It ignores the cost of a project.
2. It cannot be useful for programmes that are indefinite and vague.
3. Assumption of normal probability distribution is not true.
4. It does not consider the matter of resources required for various types of activities of a project.
5. It seems to be simple but in reality its application is too complex.
6. It is not practicable for routine planning of recurring events.

CPM (Critical Path Method): Another important method of networking is Critical Path Method. Critical path method was developed in 1956-1957 at the E.I. DuPont Nemours and Co., USA in connection with the periodic overhauling and maintenance of a chemical plant. Critical Path Method basically is a special application of analysis for planning and scheduling. Critical path method was basically developed with the objective of reducing duration and cost of the project.

Critical path method is a special application of network analysis. It uses network analysis for scheduling production, construction projects as well as research and development activities. It is also useful in situations which require estimates of time and performance.

Critical path method deals with repetitive type projects, such as overhaul of generating plant, which has to be carried repeatedly after set time intervals.

The critical path, is the overall time, it will take to complete the project. It is the longest path in time through the network. In other words, the longest path in the network is called critical path. Identifying the critical path is of great importance as it determines the duration of entire project. Critical path method differentiates between the planning and scheduling of the project.

A Critical path method is a very important project management tool used to formulate a time frame for a project in order to determine where potential delays are most likely to occur. The process includes a step by step process that provides the developer with a visual representation of potential bottleneck, throughout the course of the project.

Identification of the Critical Path: Project CP is the longest time path through the network. The path can be identified by determining the following parameters for each activity:

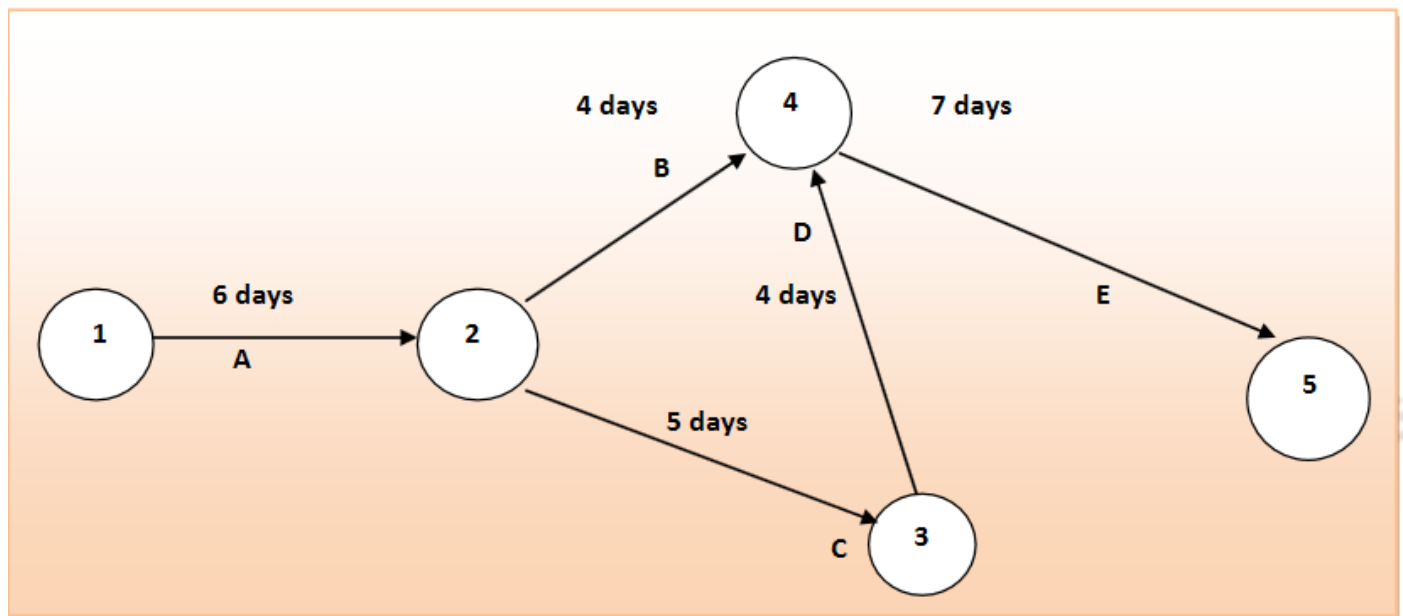
- **Earliest start time for activity (ES):** It is the earliest possible time at which the activity should start if only the ongoing activities are first completed.
- **Earliest finish time (EF):** It is the earliest possible time to finish the activity. It is equal to the earliest start time for activity plus the time required completing the activity.
- **Latest possible finish time for activity (LF):** It is the latest time at which the activity can be completing without any postpone or within the time framework.
- **Latest possible start time for activity (LS):** it is the latest start time for an activity and equal to the latest finish time minus the time required to complete the activity.
- **Slack time:** Slack time is the difference between earliest start time for activity and latest start time for activity, or between earliest finish time for activity and latest finish time for activity.

Now we will understand it with the help of following example.

Here the information regards the jobs involved in the project with their time duration. Now we have to find out the earliest possible time for completing the project.

Job	Activity	Time Duration
A	1-2	6 days
B	2-4	4 days
C	2-3	5 days
D	3-4	4 days
E	4-5	7 days
	Total	26

From the given data, we can draw the following network diagram:



In this diagram, the jobs have been shown in the form of arrows leading from one circle to another. In simple words the arrow connecting circles 1 and 2 represents job A and so on. All the activities have been reduced to a network diagram together with the information as regard to time required for the completion of different jobs. In

this diagram, there are two paths- the first connects circles 1, 2, 4, and 5 and takes 17 days. The second path connects circles 1, 2, 3, 4, and 5 and takes 22 days. The longest path, called the critical path takes 22 days and hence the minimum time required to complete the project will be 22 days.

So in order to complete the project within scheduled time, attention will have to be given to this path since any delay in time on any job or activity in this path will delay the completion of the project. In case the completion time is to be reduced, steps will have to be taken to reduce the time required for various activities of this path.

Advantages of Critical path method

The following are the advantages of CPM:

- It is very useful for scheduling and controlling of large projects.
- It is simple concept and not mathematically complex.
- It is very helpful in pinpoint activities that needed to be closely watched.
- In CPM, Project documentation and graphics point out who is responsible for various activities.
- It is applicable to a wide variety of projects.
- It is very useful in monitoring schedules and costs.
- It makes better and detailed planning possible.
- It is helpful at many stages of project management.
- It enables standard method for communicating project plans, schedules, time and cost performance
- With the help of CPM most critical activities are identified and thus more attention can be paid to these activities for the successful completion of project.

Limitations of CPM

- ☐ It ignores to incorporate statistical analysis in determining the time estimates.
- ☐ It is presumed in CPM that there is a precise known time that each activity in the project will take. But in reality it is not happen.
- ☐ Basically it developed as a static planning model and not as a dynamic controlling device. So it cannot be used as a dynamic controlling device.

TERMS USED IN A NETWORK

1. Activity: An effort that is required to complete a part of the project. It is represented by “>”.
2. Node: It represents the beginning or completion of an activity. It is represented by “.”

RULES OF NETWORK CONSTRUCTION

1. Each defined activity is represented by one and only one arrow in the network.
2. Before an activity can be undertaken, all activities preceding it must be completed.
3. The arrows depicting various activities are indicative of logical procedure only. The length and bearing of the arrows are of no significance.

4. The arrow direction indicates the general progression in time. Head events and Tail events.
5. When a number of activities terminate at one event, it indicates that no activity emanating from that event may start unless all activities terminating there have been completed.
6. Events are identified by numbers.
7. The activities are identified by the numbers of their starting and ending events or by alphabets.
8. A network should have only one initial and terminal node.
10. Parallel activities between two events, without intervening events, are prohibited. When two or more parallel activities in a project have the same head and tail events, dummy activities are needed in constructing the network. Dummy activities do not consume time or resources. An efficient network contains a minimum number of dummy activities required to portray the correct precedence relationships.
11. Looping is not permitted in a network.

PROJECT COST ANALYSIS

Cost analysis is an economic evaluation technique that involves the systematic collection, categorization, and analysis of different types of cost.

- It is useful to know the cost per unit of the product manufactured.
- If proper analysis of the cost spent on each and every aspect is estimated and calculated and that helpful in proper arrangement.

IMPORTANCE / NEED FOR COST ANALYSIS

- Deciding to accept / reject the project.
- Providing statement for operating policy.
- Provide data for fixing sales price.
- Submitting project for funding agency.
- Assessment of cost per unit.

STAGES OF COST ANALYSIS

Identification stage: To distinguish which types of capital expenditure projects are necessary to accomplish the objectives.

Search stage: To explore alternative capital investments that will achieve the objectives.

Information-acquisition stage: To consider the expected costs and the expected benefits of alternative capital investments

Selection stage: To choose projects for implementation.

Financing stage: To obtain project funding.

Implementation and control stage: To get projects underway and monitor their performance

COST ANALYSIS OF A PROJECT

- A well-planned project is essential for the project success.
- Important elements / issues to be considered for managing a project:
 - ✓ Size of the project.
 - ✓ Required resources.
 - ✓ Resources allocation.
 - ✓ Interface with satellite systems or projects.
 - ✓ Using managerial tools.
 - ✓ Periodic team meetings.

COSTS OF A PROJECT

- Cost incurred on Raw Material.
- Cost incurred on Procedure.
- Cost incurred on Instrumentation.
- Cost incurred on Clinical Trials.

COST ANALYSIS DEVELOPMENT

- Begins once the trial submitted for IRB review.
- Cost analysts works with a research team to identify all required labs and procedures within the study.
- Required information includes expected number of enrollment and estimated number of treatment cycles.

COST ANALYSIS METHODS (FOR HEALTH SERVICES RESEARCH AND PUBLIC HEALTH)

- Main types of cost analysis.
- Quadrants of cost-effectiveness.
- Key Attributes of cost analysis.
- Collecting cost data alongside clinical studies

PROJECT CRASHING

Project crashing is a method for shortening the project duration by reducing the time of the project duration by reducing the time of one (or more) of the critical project activities one (or more) of the critical project activities to less than its normal activity time. This reduction in the normal activity time is referred to as crashing.

Project Crashing is a schedule compression technique used to reduce or shorten the project schedule.

The PM can various measures to accomplish this goal. Some of the common methods used are

- **Adding additional resources to the critical path tasks**

This option has various constraints such as the securing of the budget to add the resources, and the availability of the resources.

- **Reduce the project requirements or scope**

This can be done only if the sponsor and major stakeholders agree to reduce the scope

After applying the crashing, the critical path might have changed and result in creating a different critical path. Always revisit the project schedule to ensure the schedule has been crashed.

5. CONTEMPORARY MANAGEMENT PRACTICES

MANAGEMENT INFORMATION SYSTEM

A **management information system (MIS)** is an **information system** used for decision-making, and for the coordination, control, analysis, and visualization of **information** in an organization. The study of the **management information systems** involves people, processes and technology in an organizational context.

The three components of MIS provide a more complete and focused definition, where **System** suggests integration and holistic view, **Information** stands for processed data, and **Management** is the ultimate user, the decision makers.

Management information system can thus be analyzed as follows –

Management

Management covers the planning, control, and administration of the operations of a concern. The top management handles planning; the middle management concentrates on controlling; and the lower management is concerned with actual administration.

Information

Information, in MIS, means the processed data that helps the management in planning, controlling and operations. Data means all the facts arising out of the operations of the concern. Data is processed i.e. recorded, summarized, compared and finally presented to the management in the form of MIS report.

System

Data is processed into information with the help of a system. A system is made up of inputs, processing, output and feedback or control.

Thus MIS means a system for processing data in order to give proper information to the management for performing its functions.

Definition

Management Information System or 'MIS' is a planned system of collecting, storing, and disseminating data in the form of information needed to carry out the functions of management.

Objectives of MIS

The goals of an MIS are to implement the organizational structure and dynamics of the enterprise for the purpose of managing the organization in a better way and capturing the potential of the information system for competitive advantage.

Following are the basic objectives of an MIS –

- **Capturing Data** – Capturing contextual data, or operational information that will contribute in decision making from various internal and external sources of organization.
- **Processing Data** – The captured data is processed into information needed for planning, organizing, coordinating, directing and controlling functionalities at strategic, tactical and operational level. Processing data means –
 - making calculations with the data
 - sorting data
 - classifying data and
 - summarizing data
- **Information Storage** – Information or processed data need to be stored for future use.
- **Information Retrieval** – The system should be able to retrieve this information from the storage as and when required by various users.
- **Information Propagation** – Information or the finished product of the MIS should be circulated to its users periodically using the organizational network.

Characteristics of MIS

Following are the characteristics of an MIS –

- It should be based on a long-term planning.
- It should provide a holistic view of the dynamics and the structure of the organization.
- It should work as a complete and comprehensive system covering all interconnecting sub-systems within the organization.
- It should be planned in a top-down way, as the decision makers or the management should actively take part and provide clear direction at the development stage of the MIS.
- It should be based on need of strategic, operational and tactical information of managers of an organization.
- It should also take care of exceptional situations by reporting such situations.
- It should be able to make forecasts and estimates, and generate advanced information, thus providing a competitive advantage. Decision makers can take actions on the basis of such predictions.

- It should create linkage between all sub-systems within the organization, so that the decision makers can take the right decision based on an integrated view.
- It should allow easy flow of information through various sub-systems, thus avoiding redundancy and duplicity of data. It should simplify the operations with as much practicability as possible.
- Although the MIS is an integrated, complete system, it should be made in such a flexible way that it could be easily split into smaller sub-systems as and when required.
- A central database is the backbone of a well-built MIS.

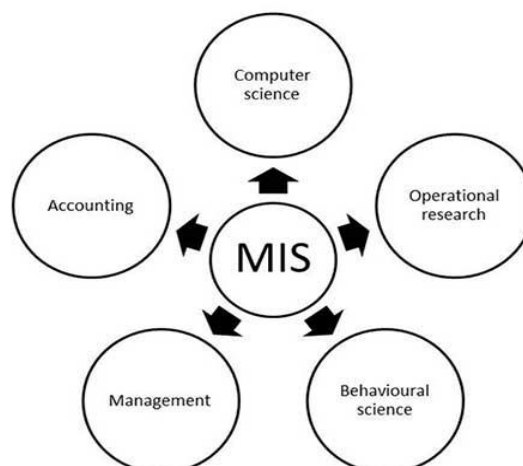
Characteristics of Computerized MIS

Following are the characteristics of a well-designed computerized MIS –

- It should be able to process data accurately and with high speed, using various techniques like operations research, simulation, heuristics, etc.
- It should be able to collect, organize, manipulate, and update large amount of raw data of both related and unrelated nature, coming from various internal and external sources at different periods of time.
- It should provide real time information on ongoing events without any delay.
- It should support various output formats and follow latest rules and regulations in practice.
- It should provide organized and relevant information for all levels of management: strategic, operational, and tactical.
- It should aim at extreme flexibility in data storage and retrieval.

Nature and Scope of MIS

The following diagram shows the nature and scope of MIS –



Important roles of the MIS:

- MIS satisfies the diverse needs through a variety of systems such as Query System, Analysis System, Modelling System and Decision Support System.
- It helps in strategic planning, management control, operational control and transaction processing.
- It helps in the clerical transaction processing.
- It answers the queries on the data pertaining to the transaction, the status of a particular record and reference on a variety of documents.
- MIS for Junior Management: Providing the operational data for planning, scheduling and control, and helps them further in decision-making at the operation level to correct an out of control situation.
- MIS for Middle Management: In short-term planning, target setting and controlling the business functions which is supported by the use of the management tools of planning and control.
- MIS for Top- Level Management: in goal setting, strategic planning and evolving the business plans and their implementation.
- It plays the role of information generation, communication, problem identification and helps in the process of decision-making.

BASIC CONCEPTS OF MIS

1. MATERIALS REQUIREMENT PLANNING
2. JUSI – IN – TIME
3. TOTAL QUALITY MANAGEMENT
4. SIX SIGMA
5. CAPABILITY MATURITY MODELS
6. SUPPLY CHAIN MANAGEMENT
7. ENTERPRISE RESOURCE PLANNING
8. PERFORMANCE MANAGEMENT
9. BUSINESS PROCESS OUTSOURCING
10. BUSINESS PROCESS RE-ENGINEERING
11. BENCH MARKING
12. BALANCE SCORE CARD

MATERIAL REQUIREMENTS PLANNING (MRP)

Material requirements planning (MRP) is a system for calculating the **materials** and components needed to manufacture a product. It consists of three primary steps: taking inventory of the **materials** and components on hand, identifying which additional ones are needed and then scheduling their production or purchase.

HISTORY

Prior to MRP, and before computers dominated industry, reorder point (ROP)/reorder-quantity (ROQ) type methods like EOQ (economic order quantity) had been used in manufacturing and inventory management.

- MRP was created initially to supply the Polaris program then, in 1964, as a response to the Toyota Manufacturing Program, Joseph Orlicky developed material requirements planning (MRP). The first company to use MRP was Black & Decker in 1964, with Dick Alban as project leader. Orlicky's 1975 book Material Requirements Planning has the subtitle The New Way of Life in Production and Inventory Management. By 1975, MRP was implemented in 700 companies. This number had grown to about 8,000 by 1981.
- In 1983, Oliver Wight developed MRP into manufacturing resource planning (MRP II). In the 1980s, Joe Orlicky's MRP evolved into Oliver Wight's manufacturing resource planning (MRP II) which brings master scheduling, rough-cut capacity planning, capacity requirements planning, S&OP in 1983 and other concepts to classical MRP. By 1989, about one third of the software industry was MRP II software sold to American industry (\$1.2 billion worth of software).

The scope of MRP in manufacturing

Dependent demand vs independent demand

Independent demand is demand originating outside the plant or production system, while dependent demand is demand for components. The bill of materials (BOM) specifies the relationship between the end product (independent demand) and the components (dependent demand). MRP takes as input the information contained in the BOM.

The basic functions of an MRP system include: inventory control, bill of material processing, and elementary scheduling. MRP helps organizations to maintain low inventory levels. It is used to plan manufacturing, purchasing and delivering activities.

- If a company purchases insufficient quantities of an item used in manufacturing (or the wrong item) it may be unable to meet contract obligations to supply products on time.
- If a company purchases excessive quantities of an item, money is wasted - the excess quantity ties up cash while it remains as stock that might never be used at all.

- Beginning production of an order at the wrong time can cause customer deadlines to be missed.

"Manufacturing organizations, whatever their products, face the same daily practical problem - that customers want products to be available in a shorter time than it takes to make them. This means that some level of planning is required."

Companies need to control the types and quantities of materials they purchase, plan which products are to be produced and in what quantities and ensure that they are able to meet current and future customer demand, all at the lowest possible cost. Making a bad decision in any of these areas will make the company lose money. A few examples are given below:

MRP is a tool to deal with these problems. It provides answers for several questions:

- What items are required?
- How many are required?
- When are they required?...

MRP can be applied both to items that are purchased from outside suppliers and to sub-assemblies, produced internally, that are components of more complex items.

Objectives of MRP:

The objectives of material requirement planning in operations management are:

- (a) It determines the quantity and timing of finished goods demanded.
- (b) It determines the time phased requirements of the demand for materials, components and sub-assemblies over a specified planning time horizon.
- (c) It computes the inventories, work-in-process batch sizes and manufacturing and packing lead times.
- (d) It controls inventory by ordering materials and components in relation to orders received rather than ordering them from stock level point of view.
- (e) It improves customer service by meeting delivery schedules promised and shortening the delivery lead times.
- (f) It reduces inventory cost by reducing inventory levels.
- (g) It improves plant operating efficiency by better use of productive resources.

Benefits of MRP:

MRP is a new way of managing manufacturing operations. It is not only to calculate how much material to order and when. It is a decision support system which provides timely and valuable information to operations managers.

The potential benefits of MRP system are:

a. Inventory:

MRP will substantially reduce inventory investment in dependent demand items while improving operational efficiency by removing the risk of shortages associated with the EOQ. The MRP system has numerous benefits over the fixed order size system (EOQ) for control of production items.

MRP system focuses on actual requirements. It is product oriented. Whereas EOQ focuses on replenishing supply and is part oriented. MRP is based on future production data while EOQ is on past demand data.

b. Purchases:

MRP helps in generating purchase order and in case of any changes, reschedule notices are also generated. These reports facilitate the purchase department in listing out the priority items and making them available in time.

c. Planning, Engineering and Schedule:

MRP plans orders for purchasing and shop scheduling for the quantity of items that must be available in each time period to produce the end items. The orders are planned for enough ahead to allow adequate time for scheduled completion of the final product without having material waiting unnecessarily for entry into a particular stage of the production process.

d. Production:

MRP can improve flow of work, thereby reducing intermittent delays and reducing the manufacturing cycle time for the jobs.

e. Sales:

As one of the functions of MRP is to report the changes in demand, it improves the company's ability to react to changes in customer orders, improves customer services by helping production meet assembly dates and helps to reduce delivery lead times.

Although MRP is an excellent tool for initial planning and scheduling, its greatest benefits may be its ability to re-plan and reschedule in view of unforeseen contingencies. The MRP system can predict shortages and overages soon enough so that something can be done to prevent them. It can help order priorities to help up-to-date by planning and re-planning order due dates.

MRP provides exception reporting whenever a mismatch of timing between demand and supply exists. It is a priority system: typical messages are to delay, expedite, or cancel an existing order, launch a new order, etc. It attempts to make the due date and need date coincide, so operations proceed as planned while inventory investment is minimised.

If a component to an assembly is not available when planned, MRP can reschedule all other components for the same assembly to a later date while rescheduling shop priority. MRP does not actually reschedule orders, but it prints messages specifying exactly where changes are appropriate. The decision to make changes remains with management personnel.

Limitations of MRP:

The effective operation and efficiency of an MRP system depends on the integrity of the files and records of relevant data. The quality is directly influenced by data accessibility, up-to-dateness, and accuracy. Lack of record integrity is the major reason for the failure of MRP system.

Computer-based MRP, even more than manual, will not perform satisfactorily with poor files and records. The outputs from computer-based MRP systems are dependent on its inputs.

Following are the weaknesses of MRP:

- a. It assumes that lead times are known constants that are independent of the lot size.
- b. It requires fixed routing for the items. It is not able to access and plan for the use of alternate routings.
- c. The sequencing logic priorities order only by period or date. It provides no priority for sequencing within a period based on similar set-ups, tooling, favoured customers, etc.
- d. It is time consuming process as it requires lot of processing and analytical times. This is achieved usually by lot of iterations.
- e. To make MRP system effective, an organisation must have effective communication system, motivated personnel, right leadership and an effective computer system. It should also have support and commitment of the top management.

Supplier lead times must be short and reliable. The MPS must be prepared before the start of actual production. What is produced must be known with certainty and quantity and timing of receipts of raw materials and components must be dependable.

JUST-IN-TIME (JIT) System

Just-in-time also known as JIT is an inventory management method whereby labour, material and goods (to be used in manufacturing) are re-filled or scheduled to arrive exactly when needed in the manufacturing process.

JIT approach has the capacity, when adequately applied to the organization, to improve the competitiveness of the organisation in the market significantly by minimizing wastes and improving production efficiency and product quality.

KEY TAKEAWAYS

- The just-in-time (JIT) inventory system is a management strategy that minimizes inventory and increases efficiency.
- Just-in-time manufacturing is also known as the Toyota Production System (TPS) because the car manufacturer Toyota adopted the system in the 1970s.
- Kanban is a scheduling system often used in conjunction with JIT to avoid overcapacity of work in process.
- The success of the JIT production process relies on steady production, high-quality workmanship, no machine breakdowns, and reliable suppliers.

Elements Involved in JIT

Continuous improvement

- Attacking fundamental problems and anything that does not add value to the product,
- Devising systems to identify production and allied problems,
- Simplicity: Simple systems are simple easy to understand, easily manageable and the chances of going wrong are very low.
- A product: oriented layout for less time spent on materials and parts movement,
- Quality control at source to ensure every worker is solely responsible for the quality of their own produced output
- Eliminating waste: There are seven types of waste:
 1. Waste from product defects
 2. Waste of time
 3. Transportation waste.
 4. Inventory waste.
 5. Waste from overproduction.
 6. Processing waste.

Waste minimization is one of the primary objectives of Just In Time system. This needs effective inventory management throughout the whole supply chain. Initially, a manufacturing

entity will seek to reduce inventory and enhance operations within its own organization. In an attempt to reduce waste attributed to ineffective inventory management, SIX principles in relation to JIT have been stated by Schniederjans and they are:

1. Reduce buffer inventory.
2. Try for zero inventory.
3. Search for reliable suppliers.
4. Reduce lot size and increase the frequency of orders.
5. Reduce purchasing cost.
6. Improve material handling.

How does it differ from traditional manufacturing?

- ▶ In traditional manufacturing we try to predict what the customer will want and we will create a forecast (or guess) against which we will produce our products. We will also try to produce those products in large batches as the belief is that will make machines and processes more efficient, especially if those machines require a long time to setup. This will typically result in long lead times through our processes, huge amounts of Work In Process (WIP) stocks and also large quantities of finished goods stocks that have not yet been ordered by our customers. This is what many now call "Just in case" manufacturing.
- ▶ If the customer does order something that is not in our current stocks they will either have to wait many weeks or even months for the product to be manufactured or work will be hurried through the system by progress chasers causing a huge amount of disruption to the production schedule.
- ▶ These systems are often run by Manufacturing Resource Planning (MRP2) programs that will try to schedule each and every process within the facility. These software packages will seek to control every step and everything requires careful and often complex planning.
- ▶ A Just in Time system on the other hand will seek to use simple visual tools known as Kanbans to pull production through the processes according to what the customer actually takes. It massively reduces the amount of stock held and will reduce lead times by a significant amount, often from weeks to just a few hours or days.

Requirements for implementing Just in Time

> Just in Time is just one of the pillars of a lean manufacturing system and as such it cannot be implemented in isolation and without a firm foundation which to build. Trying to reduce batch sizes without tackling setup times for instance cannot be done. The following are some of the things that must be implemented for JIT to be able to work.

- > Reliable Equipment and Machines; if your machinery is always breaking down or giving you quality problems then it will frequently manifest in big issues with any JIT flow. The implementation of Total Productive Maintenance is required to ensure that you can rely on your equipment and to minimize the impact that any failures have on your processes.
- > Well designed work cells; poor layout, unclear flow, and a host of other issues can all be cleared up by the implementation of 5s within your production. This simple and very easy to implement lean tool will make a significant improvement in your efficiencies all by itself.
- > Quality Improvements; an empowered workforce that is tasked with tackling their own quality problems with all of the support that they need is another vital part of any lean and JIT implementation. Setting up kaizen or quality improvement teams and using quality tools to identify and solve problems is vital,
- > Standardized Operations; only if you know how each operation is going to be performed can you be sure what the reliable outcome will be. Defining standard ways of working for all operations will help to ensure that your processes are reliable and predictable.
- > Pull Production; Just in time does not push raw materials in at the front end to create inventory (push production), it seeks to pull production through the process according to customer demand. It achieves this by setting up "supermarkets" between different processes from which products are taken or by the use of Kanbans which are signals (flags) to tell the previous process what needs to be made.
- > Single piece Flow; the ideal situation is one in which you will produce a single product as ordered by the customer. This for some industries is not immediately possible but should always be your end goal. To achieve this you will need to work on reducing batch sizes significantly through the use of Single Minute Exchange of Die (SMED) which seeks to significantly reduce the time taken for any setup. It will also often require the use of smaller dedicated machines and processes rather than all singing all dancing mega machines.
- > Flow at the beat of the customer; the demand of your customer is often referred to as your Takt time. You need to ensure that your cells and processes are organized, balanced and planned to achieve the pull of the customer.

The Advantages and disadvantages of a JIT system

The following are some of the many benefits that you could gain through the implementation of just in time

Order to Cash: The Order to Cash Timeline; as you reduce your lead time the quicker you get your cash. Reduction in the order to payment timeline: cash, as they say is king in business. Many businesses will suffer with cash flow problems as they will often have to purchase large amounts of raw materials prior to manufacturing and subsequent payment by the customer.

Often this gap is many months. Through implementing JIT you are able to considerably reduce that time period.

Reduction in Inventory costs; one of the main aims with any JIT implementation is to improve stock turns and the amount of stock being held. Personal experience has seen reductions of more than 90% stock in some industries. Along with the reduction in the stock come many other associated benefits.

Reduction in space required; by removing large amounts of stock from the system and moving processes closer together we will often see a significant reduction in the amount of floor space being used. Results from 100's of projects run within companies in the UK through the Manufacturing Advisory Service saw average reductions of 33% for simple 5 day implementation projects.

Reduction in handling equipment and other costs; if you don't have to move large batches there is less need for complex machinery to move them and all of the associated labor and training.

Lead time reductions; one of the most significantly impacted areas is that of the time it takes for products to flow through the process. Instead of weeks or months most JIT implementations result in lead times of hours or a few days.

Reduced planning complexity; the use of simple pull systems such as Kanban, even with your suppliers, can significantly reduce the need for any form of complex planning. With many implementations the only planning is the final shipping process.

Improved Quality; the removal of large batch manufacturing and reduction in handling often results in significant quality improvements; often in the region of 25% or more.

Productivity increases; to achieve JIT there are many hurdles that must be overcome with regards to how the process will flow. These will often result in productivity improvements of 25% upwards.

Problems are highlighted quicker; often this is cited as being a negative aspect of JIT in that any problems will often have an immediate impact on your whole production process. However this is the perfect way to ensure that problems are highlighted and solved immediately when they occur.

Employee empowerment; one requirement of JIT as with most other aspects of Lean manufacturing is that employees are heavily involved in the design and application of your system.

Inventory Hides Problems: Many definitions of Just in Time and Lean itself will state that waste reduction is the aim; waste reduction, however, is actually something that you achieve

through trying to implement the flow and other improvements required to achieve just in time and not an end in itself.

Disadvantages of Adopting JIT Systems

>> JIT approach states ZERO tolerance for mistakes, making re-work difficult in practice, as inventory is kept to a minimum level.

>> A successful application of JIT requires a high reliance on suppliers, whose performance is outside the purview of the manufacturer.

>> Due to no buffers in JIT, production line idling and downtime can occur which would have an unfavourable effect on the production and also on the finances.

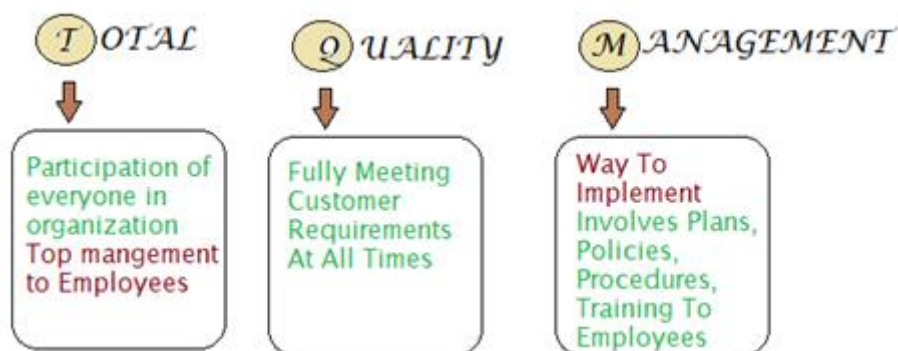
>> Chances are quite high of not meeting an unexpected increase in orders as there will be no excess inventory of finished goods.

TOTAL QUALITY MANAGEMENT

A core definition of total quality management (TQM) describes a management approach to long-term success through customer satisfaction.

Total quality management (TQM) is the continual process of detecting and reducing or eliminating errors in manufacturing, streamlining supply chain management, improving the customer experience, and ensuring that employees are up to speed with training.

Total quality management aims to hold all parties involved in the production process accountable for the overall quality of the final product or service.

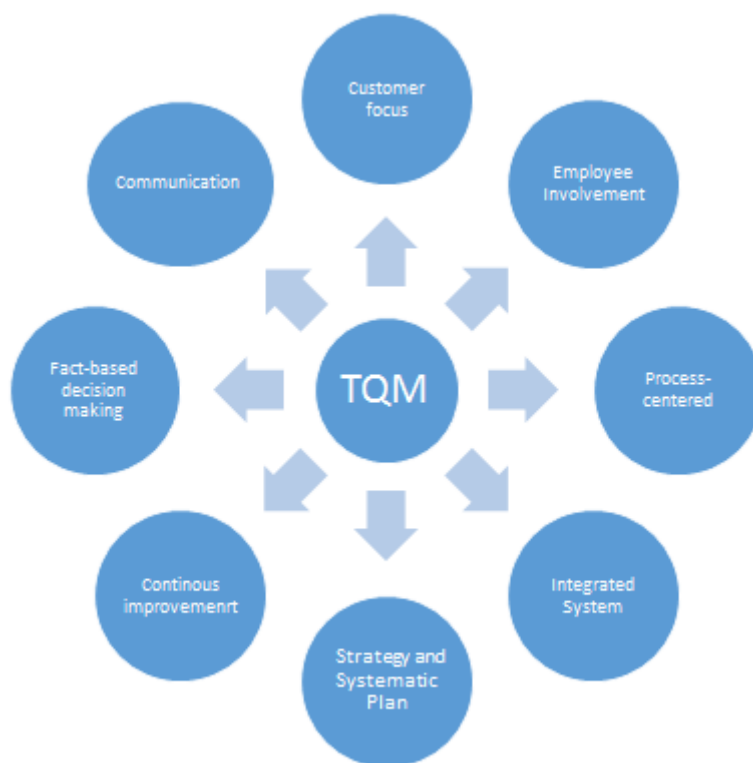


KEY TAKEAWAYS

- Total quality management (TQM) is an ongoing process of detecting and reducing or eliminating errors.
- It is used to streamline supply chain management, improve customer service, and ensure that employees are trained.

- The focus is to improve the quality of an organization's outputs, including goods and services, through continual improvement of internal practices.
- Total quality management aims to hold all parties involved in the production process accountable for the overall quality of the final product or service.

Principles of Total Quality Management



1. Customer Focus

The purpose of quality management is to provide customer satisfaction. With customer focus as an approach, the organization provides services or products meeting the customer expectations. From having several quality checks to friendly support centers, the ever-improving quality can be achieved. Organizations that are more customer focused are the one with larger customer share in the market.

2. Employee Involvement

As Total Quality Management Notes requires the involvement of every employee, it's the responsibility of the organization to foster the employee-friendly work culture. The work done by employees should be appreciated and their opinions and feedback are taken into consideration. A workplace that encourages employees to give their best achieves continuous improvement in quality.

3. Process-Centered

This process needs to be well defined and need to be monitored for any deviation during the execution. Best practices can be adopted in this approach.

4. Integrated System

Although the processes are divided vertically structured departments, it is horizontally that they interconnect cross-functionally. For example, manufacturing depends on the inventory of the procurement. These horizontal processes need to be modeled properly. Adopting standards ISO 9000 or any quality standard suitable to company requirements helps in integrating the systems.

5. A Strategic and Systematic Plan

A strategic and systematic approach is needed to achieve the organization's goals and visions. Incorporating quality as part of the strategic plan as the core element is based in TQM.

6. Continuous Improvement

One of the popular approaches in quality management is continuous improvement. It implies to finding creative ways to implement existing processes. This, in turn, helps in being more competitive and meeting the expectations of stakeholders.

7. Fact-Based Decision Making

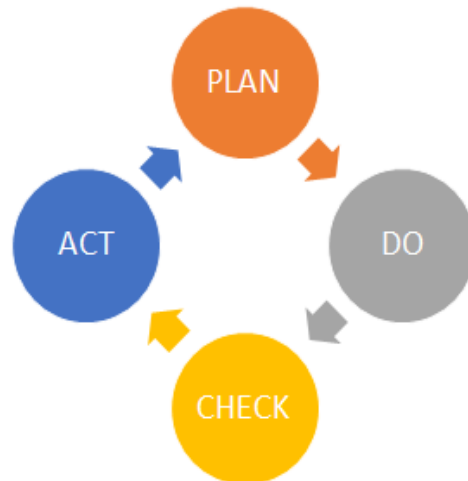
The performance of an organization can be found from the data collected and analyzed for various parameters. To improve the decision, this data needs to be considered. An accurate decision can be taken if a proper fact-based decision model is created.

8. Communication

This is the building block of a successful business. Communication plan and training employees on communication etiquettes goes a long way in fostering healthy work culture. Communication involves internal amongst the employees and external with the stakeholders. Effective communication makes opinions and feedbacks easier.

So this is the principles of TQM. To implement this principle, the PDCA cycle was coined by William Deming. It is also known as the Deming cycle/circle. DMAIC (Define, Measure, Analyze, Improve, Control) is the adoption of PDCA in six sigma.

PDCA Cycle



PDCA cycle falls under below four categories:

- Plan
- Do
- Check
- Act

PDCA is considered to cycle because it is based on continuous improvement. Any new idea to improve the existing processes goes through the PDCA cycle.

Plan

A thorough plan with smart goals for improvement is the first phase for Total Quality Management Notes. The plan is checked for the availability of the resources and also if fits the stakeholder's requirement.

Do

This phase involves the implementation and realization of the plan created for the improvement of processes. Outputs and other parameters are measured and results are documented.

Check

In this, results are monitored and analyzed to match with the plan. The difference known as planning-gap is measured. If the gap is significant, root-cause analysis is carried out for the same.

Act

Once the detected planning –the gap is significant, it's the time to act quickly. Adjustments are made in the plan and the cycle is initiated again.

Essential requirements for successful implementation of TQM

- **Commitment:** Quality improvement (in all aspects) must be everyone's job in the organization. An apparent commitment from the top management, breaking down the barriers for continuous quality improvement and steps required to provide an environment for changing attitudes must be provided. Training and support for this should be extended.
- **Culture:** There should be proper training to effect the changes in attitude and culture.
- **Continuous Improvement:** Recognize improvement as a continuous process, and not merely a one-off program.
- **Customer Focus:** Perfection in service with zero defects and full satisfaction to the end-user whether it's internal or external.
- **Control:** Ensure monitoring and control checks for any deviation from the intended course of implementation.
 - Plan
 - Do
 - Check
 - Act

This is also referred to as the PDCA cycle.

- **Planning Phase:** This phase is the most crucial phase of total quality management. Under this phase, employees have to come up with their respective queries and problems which need to be addressed. The employees apprise the management of different challenges which they are facing in their day to day operations and also analyze the root cause of the problem. They need to do the required research and collect significant data which would help them find solutions to all the problems.
- **Doing Phase:** In this phase, a solution for the identified problems in the planning phase is developed by the employees. Strategies are devised and implemented to crack down the challenges faced by employees. The efficiency and effectiveness of solutions and strategies are also evaluated in this stage.
- **Checking Phase:** Under this phase, a comparison analysis of before and after is done in order to assess the effectiveness of the processes and measure the results.
- **Acting Phase:** This is the last phase of the cycle, in this phase employees document their results and prepare themselves to address other problems

Beliefs about Total Quality Management

Following are the universal Total Quality Management beliefs:

- Satisfaction of the customer/owner is the measure of quality.
- Everyone is an owner.
- Continuous Quality improvement must be there.
- Analysis of the processes is the key to quality improvement.
- Constant TQM is not possible without consistent, active and enabling leadership by managers at all levels.
- It is important to incessantly improve the quality of the products and services which we are supposed to provide to our customers/owners.

Benefits of Total Quality Management

We have seen the concept of total quality management in previous sections and now let's move ahead to with benefits it provides to the organizations and business as a whole:

1. Since TQM is based on the intention to satisfy the customer needs, it focuses on market requirements. It is universal to be used and so helps in understanding the needs of the market.
2. It is suitable for changing markets or emerging products. Adaptable to the change.
3. Increase in productivity as redundant processes are eliminated.
4. Removal of waste and as it helps identify best practices in the processes. This also reduces the cost of repairs, support, warranty, and repair.
5. Increase in the skills of employees owing to the continuous improvement and high-quality standards.
6. Enhances the trust of stakeholders and customers in the organization because of the quality services provided.
7. Improvement in the communication amongst employees and stakeholders too.
8. Encourages a healthy work environment as the feedback of employees is taken into consideration.

There are numerous advantages of Total Quality Management and but now let's see some of the disadvantages.

Disadvantages of Total Quality Management

1. The major drawback of this method is that it requires organization-wide efforts. The entire leadership team needs to be on the same grounds when it comes to implementing. Commitment to high levels is expected.
2. The employees may be adamant to change the existing processes that they are following for years. Always looking for better ideas can be strenuous for some.
3. Methods like six sigma, lean, kaizen are gaining popularity as they don't require the involvement of the entire company. Departments or teams can implement these in their respective work.

Although disadvantages can be grave, with strenuous efforts these can be reduced. So this was about total quality management. It maybe a gigantic step for implementation but long-term results are beneficial to the companies, businesses or even small-scale enterprises.

SIX SIGMA

Six Sigma is a quality-control methodology developed in 1986 by Motorola, Inc.

The method uses a data-driven review to limit mistakes or defects in a corporate or business process.

“Six Sigma is a highly disciplined process that helps us focus on developing and delivering near-perfect products and services”.

Six Sigma emphasizes cycle-time improvement while at the same time reducing manufacturing defects to a level of no more than 3.4 occurrences per million units or events.

In other words, the system is a method to work faster with fewer mistakes.

Methodologies of Six Sigma

There are two major methodologies used within Six Sigma, both of which are composed of five sections, according to the 2005 book “JURAN Institute Six Sigma Breakthrough and Beyond” by Joseph A. De Feo and William Barnard.

DMAIC: The DMAIC method is used primarily for improving existing business processes. The letters stand for:

- **D**efine the problem and the project goals
- **M**easure in detail the various aspects of the current process
- **A**nalyze data to, among other things, find the root defects in a process
- **I**mprove the process

- Control how the process is done in the future

DMADV: The DMADV method is typically used to create new processes and new products or services. The letters stand for:

- **D**efine the project goals
- **M**easure critical components of the process and the product capabilities
- **A**nalyze the data and develop various designs for the process, eventually picking the best one
- **D**esign and test details of the process
- **V**erify the design by running simulations and a pilot program, and then handing over the process to the client

Features of Six Sigma

- Six Sigma's aim is to eliminate waste and inefficiency, thereby increasing customer satisfaction by delivering what the customer is expecting.
- Six Sigma follows a structured methodology, and has defined roles for the participants.
- Six Sigma is a data driven methodology, and requires accurate data collection for the processes being analyzed.
- Six Sigma is about putting results on Financial Statements.
- Six Sigma is a business-driven, multi-dimensional structured approach for –
 - Improving Processes
 - Lowering Defects
 - Reducing process variability
 - Reducing costs
 - Increasing customer satisfaction
 - Increased profits

The word *Sigma* is a statistical term that measures how far a given process deviates from perfection.

The central idea behind Six Sigma: If you can measure how many "defects" you have in a process, you can systematically figure out how to eliminate them and get as close to "zero defects" as possible and specifically it means a failure rate of 3.4 parts per million or 99.9997% perfect.

Key Concepts of Six Sigma

At its core, Six Sigma revolves around a few key concepts.

- **Critical to Quality** – Attributes most important to the customer.
- **Defect** – Failing to deliver what the customer wants.
- **Process Capability** – What your process can deliver.
- **Variation** – What the customer sees and feels.
- **Stable Operations** – Ensuring consistent, predictable processes to improve what the customer sees and feels.
- **Design for Six Sigma** – Designing to meet customer needs and process capability.

Our Customers Feel the Variance, Not the Mean. So Six Sigma focuses first on reducing process variation and then on improving the process capability.

Myths about Six Sigma

There are several myths and misunderstandings surrounding Six Sigma. Some of them few are given below –

- Six Sigma is only concerned with reducing defects.
- Six Sigma is a process for production or engineering.
- Six Sigma cannot be applied to engineering activities.
- Six Sigma uses difficult-to-understand statistics.
- Six Sigma is just training.

Levels of Six Sigma

	Six Sigma Level	% Accuracy	DPMO
Virtual Perfection	6	99.9997%	3.4
	5	99.98%	233
	4	99.4%	6210
Good	3.5	97.7%	22,700
	3	93.3%	66,807
Improvement Needed	2	69.1%	308,537

Benefits of Six Sigma

Six Sigma offers six major benefits that attract companies –

- Generates sustained success
- Sets a performance goal for everyone
- Enhances value to customers
- Accelerates the rate of improvement
- Promotes learning and cross-pollination
- Executes strategic change

Origin of Six Sigma

- Six Sigma originated at Motorola in the early 1980s, in response to achieving 10X reduction in product-failure levels in 5 years.
- Engineer Bill Smith invented Six Sigma, but died of a heart attack in the Motorola cafeteria in 1993, never knowing the scope of the craze and controversy he had touched off.
- Six Sigma is based on various quality management theories (e.g. Deming's 14 point for management, Juran's 10 steps on achieving quality).

Objectives of Six Sigma

1. To reduce variation.
2. To solving the problems in scientific manner.
3. Six Sigma places an emphasis on the DMAIC approach (define, measure, analyze, improve, and control) to problem solving.
4. To develop the bottom line responsibilities towards continuous improvement.
5. Organizations using Six Sigma often utilize teams that are assigned well-defined projects with a direct impact on the bottom line.

Three Key Characteristics of SIX SIGMA

⌘ **Leadership Commitment:** • Achieving Six Sigma is not easy – it requires serious commitment in the form of time, effort, and resources. For a company to be successful, such commitment must come first from the top executive leadership of the organization and must be practiced by everyone.

⌘ Managing Decisions with Data:

- ♣ It is not enough to run a business based on one's experience or "tribal knowledge." Decisions must be based on data versus the typical "I think", "I feel", or "In my opinion" practices that often exist today.
- ♣ With the maturation of the information economy, data is available to virtually everyone in the organization, along with the tools for analyzing that data.
- ♣ Properly using data to Measure, Analyze, Improve, and Control performance forms the foundation of the Six Sigma methodology.

⌘ Training and Cultural Change: • Improved performance does not and will not happen automatically. High-calibre training is required. Disciplined implementation must follow, and people at all levels have to change the way they go about doing their jobs.

• In short, new ways of thinking, communicating, and operating must pervade the entire organization. You also need a methodology. DMAIC and DFSS provide a structured problem solving roadmap and tools towards obtaining the results you expect.

CAPABILITY MATURITY MODEL

The Software Engineering Institute (SEI) Capability Maturity Model (CMM) specifies an increasing series of levels of a software development organization. The higher the level, the better the software development process, hence reaching each level is an expensive and time-consuming process.

CMM was developed by the Software Engineering Institute (SEI) at Carnegie Mellon University in 1987.

- It is not a software process model. It is a framework which is used to analyse the approach and techniques followed by any organization to develop software products.
- It also provides guidelines to further enhance the maturity of the process used to develop those software products.
- It is based on profound feedback and development practices adopted by the most successful organizations worldwide.
- This model describes a strategy for software process improvement that should be followed by moving through 5 different levels.
- Each level of maturity shows a process capability level. All the levels except level-1 are further described by Key Process Areas (KPA's).

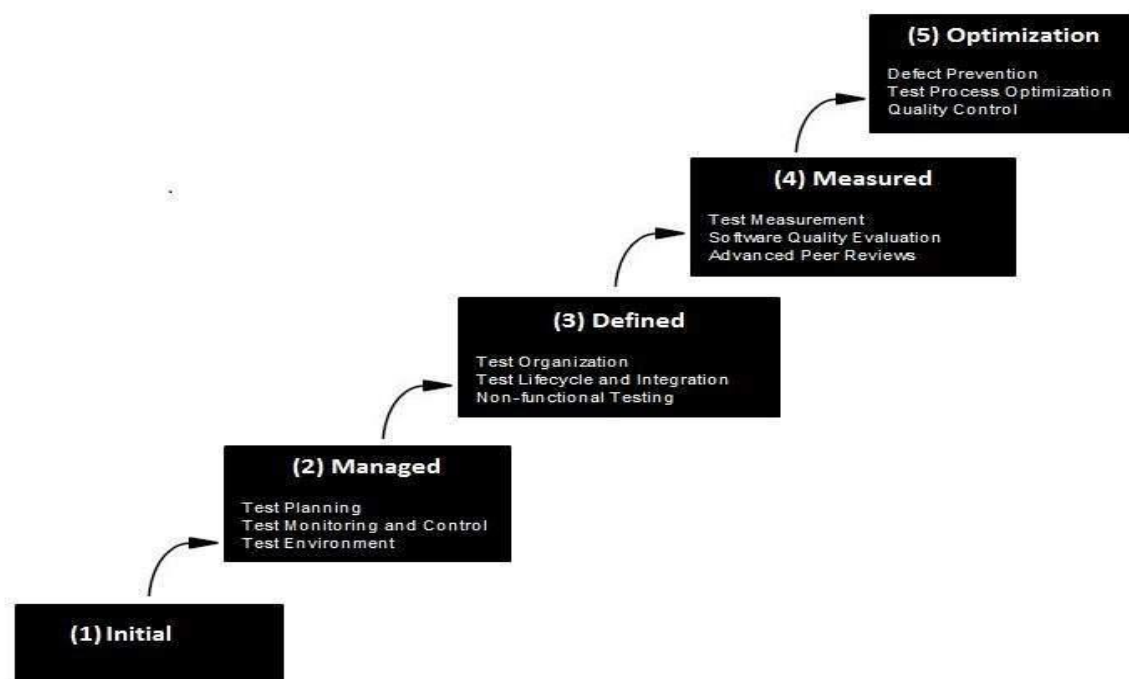
History of the Capability Maturity Model

The Capability Maturity Model came into existence during the 1980s, created to assess the ability of government-contracted software developers and implementers' processes and abilities to deliver on their contracts. The model was created and endorsed by Carnegie Mellon University's Software Engineering Institute (SEI), which was established in 1984 as an R&D center focusing primarily on the identification of software engineering and development-related issues and improvement of software engineering methods.

The SEI's core activities include the optimization and improvement of software development processes and the acquisition and maintenance of software systems for the U.S. Department of Defense (DoD), which sponsors the development center.

In 2006, the SEI developed a more improved version of the Capability Maturity Model known as the Capability Maturity Model Integration (CMMI). The CMMI framework addresses some of the limitations of the CMM, gaining immense popularity in the process.

Levels of CMM or KPA's



- **Level One : Initial** - The software process is characterized as inconsistent, and occasionally even chaotic. Defined processes and standard practices that exist are abandoned during a crisis. Success of the organization majorly depends on an individual effort, talent, and heroics. The heroes eventually move on to other organizations taking their wealth of knowledge or lessons learnt with them.
- **Level Two: Repeatable** - This level of Software Development Organization has a basic and consistent project management processes to track cost, schedule, and functionality. The process is in place to repeat the earlier successes on projects with similar applications. Program management is a key characteristic of a level two organization.

- **Level Three: Defined** - The software process for both management and engineering activities are documented, standardized, and integrated into a standard software process for the entire organization and all projects across the organization use an approved, tailored version of the organization's standard software process for developing, testing and maintaining the application.
- **Level Four: Managed** - Management can effectively control the software development effort using precise measurements. At this level, organization set a quantitative quality goal for both software process and software maintenance. At this maturity level, the performance of processes is controlled using statistical and other quantitative techniques, and is quantitatively predictable.
- **Level Five: Optimizing** - The Key characteristic of this level is focusing on continually improving process performance through both incremental and innovative technological improvements. At this level, changes to the process are to improve the process performance and at the same time maintaining statistical probability to achieve the established quantitative process-improvement objectives.

The Structure of the CMM

The CMM framework comprises five key elements:

1. **Maturity levels** (as discussed above)
2. **Primary Process areas**, which identify groups of similar activities that reach a set of important objectives when they are carried out together
3. **Goals**, which provide an overview of the key process areas
4. **Common Features**, which include procedures that execute key process areas
5. **Key Practices**, which detail the components of structure and procedures that relate to the development and establishment of the key process areas

SUPPLY CHAIN MANAGEMENT (SCM)

“Supply chain management is the management of the flow of goods and services and includes all processes that transform raw materials into final products”.

“Supply chain management is the integration of businesses from end user through original suppliers that provides products, services, and information that add value for customers.”

It involves the active streamlining of a business's supply-side activities to maximize customer value and gain a competitive advantage in the marketplace.

NATURE OF SUPPLY CHAIN MANAGEMENT

Formulate Effective Plan

Supply chain managers focus on creating effective plans and strategies for ensuring proper functioning. They develop well-defined plans by doing various analysis and forecasting that leads to deliver better results within the organization.

Acquire Raw Materials

Availability of sufficient amount of raw materials at all is must for the uninterrupted functioning of the business. Supply chain managers identify cheap sources of raw materials and ensure timely attainment of all materials in the right quantity and quality.

Supervise Production Process

Supply chain management supervises the whole production processes of business for deriving optimum results. These managers monitor all activities and ensure that each of the resources is efficiently utilized with minimum wastage.

Manages Delivery Of Goods

Supply chain management is concerned with the regulation of delivery and logistics activities of the organization. It works towards timely delivery of goods and services by bringing proper coordination in between distinct transportation channels and warehouses.

Proper Return System

It facilitates a proper return mechanism by providing an automated process on both buy and sell-side of the business. All refunds and claims of customers, distributors and suppliers are settled instantly via this process. Supply chain management ensures proper handling and inspection of defective goods.

OBJECTIVES OF SUPPLY CHAIN MANAGEMENT

Reduce Operating Expenses

Supply chain management lays emphasis on bringing down the operational expenses of a business. It reduces all types of business expenses like the cost of purchase, production and delivery by providing a proper supply chain. Holding period of both raw materials and finished goods is reduced by facilitating a smooth flow of raw materials in between the supplier and a company and movement of finished products between company and customers. This avoids losses and minimizes the overall operating cost of business.

Enhance Customer Satisfaction

Boosting the overall customer experience with business is another major role by process of supply chain management. Proper analysis of customer analysis is done before formulating any

production strategy which leads to production of right product. It keeps an eye on all activities within the organization and checks the quality of products to ensure whether they meet desired standards or not. Providing right quality of products at right price enables in providing better satisfaction to customers.

Improve Distribution Channel

Supply chain management provides an efficient supply chain to business which accelerates the whole process of distribution. Proper coordination in between various transportation channel and warehouses is achieved for facilitating faster movement of goods. This way the whole distribution system is enhanced which enables in delivering product in right time and at right location.

Strengthen Financial Position

It strengthen financial status of business by attaining better efficiency in its process. Supply chain manager prevents any shortage of materials and focuses on cutting any excessive costs. Any chance of funds blockage in inventories is avoided by facilitating a speedy movement of goods. Optimum funds are always maintained by managers within the business which leads to strengthen the financial status.

Regulate Proper Inventory

Maintainance of proper inventory is must for continued operations of business. All inventories such as raw materials, spare parts and finished product are properly recorded by managers for maintaining a right stock always. Any situations like under stocking or over stocking is avoided that leads to smooth functioning of business organization.

Promotes Better Coordination

Supply chain management aims at establishing a better coordination among all stakeholders of business. Proper channel is developed for easy communication of employees, customers and suppliers with organization. Manager can easily direct their employees and employees can also contact their supervisors via the established channel in case of any problem erupts. It promotes exchange of information among all parties and assist in bringing proper coordination within the organization.

SCOPE OF SUPPLY CHAIN MANAGEMENT

Minimises Operating Cost

Supply chain management focuses on reducing the overall operating cost of the organisation. It aims at bringing efficiency and raising the profitability of organisations. By developing a proper chain it brings down the purchasing cost, production cost and delivery cost. It enables smooth flow of raw materials from the supplier to an organisation which reduces the holding period of materials with the supplier and avoids any losses due to delay in production. Similarly,

companies are not required to hold on expensive inventories for a longer time and distribute quickly through the supply chain.

Boosts Customer Service

Supply chain management helps in providing better service to customers. All production strategies are framed in accordance with requirements of customers to manufacture right product. It properly anticipates the demands of customers before initiating the production. Supply managers monitor all operations of business and ensure that quality products are produced using best combination of resources. Right product available to right cost provide better satisfaction to customers. This will boost their confidence level in company's products.

Enhance Financial Position

Management of supply chain has an effective role on the financial position of business. It improves the efficiency of the organisation, cut down the excessive cost and avoids any shortage. Supply chain manager bring down the cost by reducing the use of fixed assets like plants, transportation vehicles, warehouses etc. Proper supply chain results in speedy flow of products which minimises the blockage of funds in inventories. It ensures that optimum funds are always available which helps in improving financial position.

Manages Distribution

Distribution of products at the right time and the right location is a complex task for every organisation. Supply chain management accelerated the overall distribution system of an organisation. It coordinates with various transportation channels and warehouses for attaining faster movement of goods. Supply chain managers ensure that all products get delivered at the right location within the time limit. By developing a proper network for movement of goods it has to ease the whole distribution system.

Bring Coordination Among Partners

Proper coordination among all partners of business increase productivity and profitability. It develops a proper channel through which employees, supplier and customers can easily interact with business. Managers can easily control the activities of their subordinates by communicating them all the required information.

Employees in case of any problem or error can contact their supervisors. Customers can also access their brands for any information through self-portals which are developed as a part of the customer support system. This enables a better exchange of information and brings coordination among partners.

Inventory Management

Maintaining an optimum inventory is a must for uninterrupted operation of every business. It keeps record of all inventories that is raw materials, spare parts and finished goods. Supply chain managers ensure that the proper amount of inventory is always maintained within the organisation. They work towards avoiding situations like understocking or overstocking. Supply chain managers frame proper strategies for procuring, producing and maintaining all inventories as per requirements.

Supplier Management

Supply chain management works on strengthening the relationships between business and suppliers. It tracks and records every interactions or transaction with the suppliers. Proper supply chain enables timely procurement of all required raw materials from suppliers. It develops a proper network through which suppliers and business can easily interact. Supply chain management solutions provide a self-service portal through which suppliers can contact the company in case of any issues or problems.

Advantages of Supply Chain Management

MARKETER'S BENEFITS;

- Improve Customer Service
- Achieve A Balance Between Costs And Service
- Reduce Costs Throughout The Supply Chain
- Improve Relationships Between Buyers And Suppliers
- Enables the streamlining of the supply chain.
- Easy analysis of demand and supply of products or services.
- Sharing of sales-related information between all the parties.
- Give competitive advantage by earning the loyalty ; of customers, shareholders and employees

CUSTOMERS'S BENEFISTS;

- Get the merchandise of their choice
- Merchandise is ready before customer entry.
- Marketers respond to their customers in real time.

Disadvantages:

- The biggest disadvantage of supply chain management is the heavy investment of time, money, resources needed to implement and overlook the supply chain.
- It is very difficult to decide, 'Outsource or Not''
- Integrating the supply chain and choosing the correct suppliers is much more difficult.

ERP**Definition**

Enterprise resource planning (ERP) is a system of integrated software applications that standardizes, streamlines and integrates business processes across finance, human resources, procurement, distribution, and other departments.

Enterprise resource planning is a modular software solution that centralizes and stores the organization's data, optimizes connectivity, and data sharing between departments of the enterprise.

Typically, ERP systems operate on an integrated software platform using common data definitions operating on a single database.

CHARACTERISTICS

ERP systems typically include the following characteristics:

- An integrated system
- Operates in (or near) real time
- A common database that supports all the applications
- A consistent look and feel across modules
- Installation of the system with elaborate application/data integration by the Information Technology (IT) department, provided the implementation is not done in small steps.
- Deployment options include: on-premises, cloud hosted, or SaaS

Types of ERP solutions

ERP systems are categorized in tiers based on the size and complexity of enterprises served. Typical tiers include:

- **Tier I ERPs** support large, global enterprises and handle all internationalization issues, including currency, language, alphabet, postal code, accounting rules, etc. For decades,

Oracle and SAP have been considered Tier I. Microsoft and Infor are more recent competitors but are frequently categorized as Tier I as well.

- **Tier I Government ERPs** support large, mostly federal, government agencies. These vendors support the nuances of government accounting, HR, and procurement. Oracle, SAP and CompuServe's PRISM are considered Tier I with Infor and CGI's Momentum close behind.
- **Tier II ERPs** support large enterprises that may operate in multiple countries but lack global reach. Tier II customers can be standalone entities or business units of large global enterprises. Most of these ERPs have some internationalization but lack Tier I breadth. Depending on how vendors are categorized there are 25 to 45 vendors in this tier.
- **Tier II Government ERPs** focus mostly on state and local governments with some federal installations. Tyler Technologies and UNIT4 fall in this category.
- **Tier III ERPs** support mid-tier enterprises. Most handle a handful of languages and currencies but only a single alphabet. Depending on how ERPs are categorized, there are 75 to 100 Tier III ERP solutions.
- **Tier IV ERPs** are designed for small enterprises and often focus on accounting.

Over the past few years, ERP vendors have created new systems designed specifically for the cloud, while longtime ERP vendors have created cloud versions of their software. Cloud ERP is becoming increasingly popular, and fall into two major types:

- **ERP as a service.** With these ERPs, all customers operate on the same code base and have no access to the source code. Users can configure but not customize the code.
- **ERP in an IaaS cloud.** Enterprises that rely on custom code in their ERP cannot use ERP as a service. If they wish to operate in the cloud, the only option is to move to an IaaS provider, which shifts their servers to a different location.

For most enterprises, ERP as a service offers three advantages: The initial cost is lower, upgrades to new releases are easier, and reluctant executives cannot pressure the organization to write custom code for their organization.

ERP implementation: The 5 major steps

Most ERP practitioners structure their ERP implementation as follows:

1. Gain approval

The executive sponsor oversees the creation of any documentation required for approval. This document, usually called a business case, typically includes the following:

- Problem definition

- Description of the program's objectives and scope
- Assumptions
- Implementation costs
- Implementation schedule
- Development and operational risks
- Projected benefits

Once the business case is complete, the executive sponsor presents the business case to the appropriate group of senior executives for formal approval to spend money and direct staff to implement the ERP.

2. Plan the program

The high-level timeline created for the business case is then refined into a work plan, which should include the following steps:

- **Finalize team members.** Key internal individuals should be identified by name. Other required staff should be identified by role. External partners need to be selected. Typical partners include: ERP implementation specialists, organizational change management specialists and technical specialists.
- **Complete contracts.** Contracts for new software, technology, and services should be finalized.
- **Plan infrastructure upgrades.** On-premises ERP systems frequently require faster processors, additional storage, and improved communications. Some organizations can minimize infrastructure upgrades by using a cloud ERP. But even cloud ERPs can require infrastructure upgrades.
- **Create a work plan and timeline.** Tasks, dependencies, resources, and timing need to be made as specific as possible.

3. Configure software

This is the largest, most difficult phase. Major steps include:

- **Analyze gaps.** Understanding the gaps in current business processes and supporting applications helps the project team determine how to change business processes to conform to the software.
- **Configure parameters.** Parameters in the ERP software are set to reflect the new business processes.

- **Complete required programming.** Ideally, no changes are needed for the ERP software. However, some programming may be required for interfaces to other systems or for data migration.
- **Migrate data.** The team standardizes data definitions and examines existing files for data completeness, quality, and redundancy. Finally, existing data is cleansed and migrated to the new ERP.
- **Test system.** The system is tested to ensure it delivers the needed functionality and required responsiveness.
- **Document system.** Required functional and technical documentation is created. Typically, the vendor has documentation that can be tailored to enterprise standards.
- **Upgrade infrastructure.** Complete any required upgrades.

4. Deploy the system

Prior to the final cutover when the new system is in production, multiple activities have to be completed. These include:

- **Train staff.** All staff need to be trained to operate the system and be given access rights.
- **Plan support.** A support team will be needed to answer questions and resolve problems after the ERP is operational.
- **Test the system.** The new system must be thoroughly tested to ensure it is secure, responsive, and delivers the functionality described in the business case.
- **Make the “Go live” decision.** Once the executive sponsor is confident the new ERP is ready, the enterprise needs to switch from the old system to the new system.

5. Stabilize the system

Following ERP deployment, most organizations experience a dip in business performance as staff learn new roles, tools, business processes, and metrics. In addition, poorly cleansed data and infrastructure bottlenecks will cause disruption. All impose a workload bubble on the ERP deployment and support team.

Features of ERP:

Integration of different functions of business

A business, irrespective of size, must deal with various functions like purchase management, sales management, material management, finance, accounting payroll etc. Having a separate system to manage each of these functions is a nightmare. Here is why the integrated solution of ERP software will be helpful.

Accounting and finance management

Accounting and managing finance is one of the important features of ERP software. With inbuilt automation and powerful capabilities, ERP software helps you to manage accounting and finance needs efficiently.

Purchase and sales processing

ERP software helps in increasing the process efficiency by streamlining the order management system right from order-to-Invoice-to-cash. Also, with complete tracking ability, ERP software simplifies the procurements process of stock and other supplies.

Inventory management Efficient inventory management is key for success of any business. The better you manage it, the better the business results. Just like Cash flow, it can make or break your business. Here is why ERP software plays a key role. By automating the inventory management process, ERP software helps you manage optimum inventory levels.

Cost control and analysis

Just like a small leak will sink a great a ship, even the little expenses when you put together has the potential to impact your business in several ways. ERP software helps you track the cost incurred at different projects, units, departments etc. and allows you to set the budget using which one can track variances.

Insightful reports

By integrating all functions and process in a single system, ERP software provides complete information that helps you and your team to take the right decision. By automating the report generation, ERP software eliminates the time needed to manually collect data, reconcile it and then putting into a readable format.

Access to data on the go

With the evolution of technology, the need to access the business data outside the business premises has accelerated. Keeping the growing needs, most of the ERP software's are designed with powerful net capabilities that allow you to access the data remotely including the crucial information on mobile.

Main Advantages and Disadvantages of ERP System

The advantages presented by the ERP are:

1. Optimization of business processes.
2. Accurate and timely access to reliable information.
3. The ability to share information between all components of the organization.

4. Elimination of unnecessary operations and data.
5. Reduction of time and costs of litigation
6. Then, as each module of the ERP system enters the same real-time database, another advantage is that no duplicate records or playback operations, ie, redundancy is avoided.
7. The performance of all work units that make up their business because better use time is increased. If you previously had to make reports and take them from one place to another, now the time is spent on other activities.
8. To improve performance and save time, optimize the control and analysis of management decisions there in the long term, reduced costs for the company.
9. Another obvious advantage is in terms of customer service, because the response time is reduced attention to them.
10. When a company has an ERP system is more competitive in the environment in which it operates.

Disadvantages of ERP are:

1. The installation of the ERP system is costly. ERP consultants are very expensive take approximately 60% of the budget.
2. The success depends on the skills and experience of the workforce, including education and how to make the system work properly.
3. Resistance in sharing internal information between departments can reduce the efficiency of the software.
4. The systems can be difficult to use.
5. Change of staff, companies can employ administrators who are not trained to manage the ERP system of the employing company, proposing changes in business practices that are not synchronized with the system.
6. Having an ERP system has many advantages, but does not guarantee the total success of the company. Organizational culture, know how to involve staff and anticipate changes that will suffer the organization using this system of administration, are important elements for the completion of the implementation.
7. The effectiveness of the ERP system may decrease if there is resistance to share information between business units or departments. Due to strong changes that implementation of the ERP system brings in the culture of work, there may be poorly trained or disinterested in making use of the same staff...

8. The benefits of having an ERP system are not presented immediately with the implementation of the software, they will be evident long after the system is running.
9. The culmination of the implementation depends on the ability and skill of the workforce, also involves education and training, to make the system is correctly applied.

PERFORMANCE MANAGEMENT

Definition: Performance Management can be defined as a process which continuously **identifies, measures and develops** the performance of the workforce in the organization. And to do so, each individual's performance and objectives are connected with the overall mission and goals of the enterprise. Hence, the two key elements of performance management are:

- Continuous process
- Link to mission and goals

In performance management, the managers try to figure out, the existing performance level of the employees and works on improving that level. It is **a systematic assessment of the performance of an employee** and using the assessment to better the performance over time.



Performance Management Cycle

Performance management is commonly misconstrued with **performance appraisal**, which is a process of evaluating the performance of the workforce and includes **feedback/review** on an employee job performance. However, getting feedback is not sufficient. There are three more steps involved in the accomplishment of the process, which are:

1. Performance Interview
2. Archiving Performance Data

3. Use of Appraisal Data

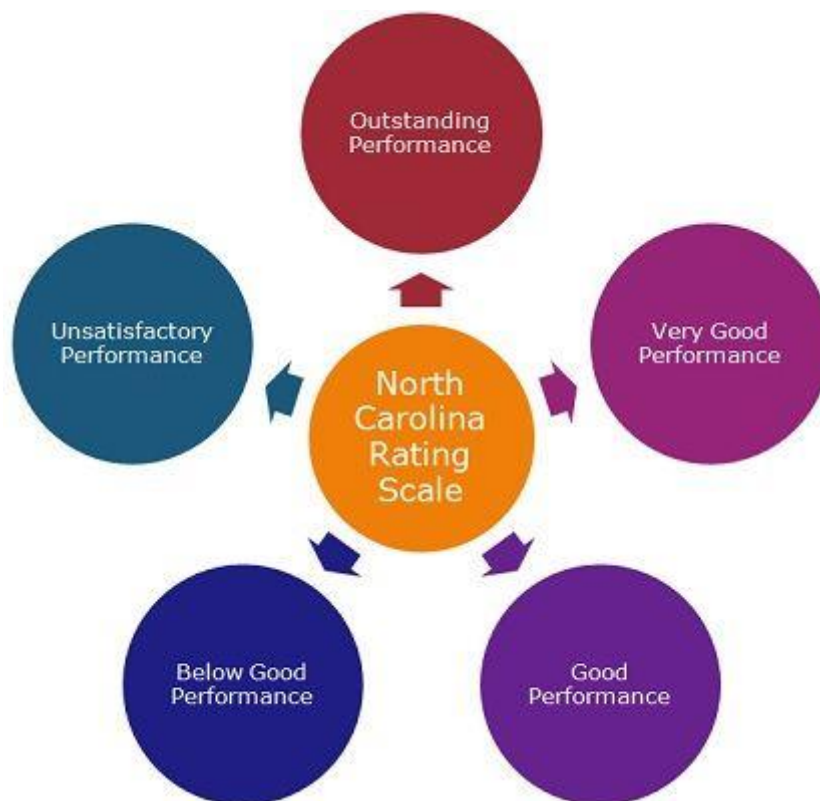
These three activities along with the feedback are the essence of the process, which when undertaken in an organized way, the process turns out as **Performance Management System (PMS)**.

Concept of Performance Management

As a communication system, it is developed to assist employees in succeeding. It not just requires direction, from the end of managers and supervisors, but also needs active participation on the part of employees.

It makes sure that employees are known about their key job functions, aware of the performance level expected, get constant feedback on their performance, have opportunities for training and development, get ratings on their performance and reward thereon, in just and fair manner.

North Carolina Rating Scale



North Carolina Rating Scale

North Carolina Rating Scale is a five point performance management rating scale, which guides the performance functions, expectations, and appraisals.

1. **Outstanding Performance (O):** Performance level supersedes the expected level. The employee is doing an outstanding job and the performance goes beyond expected level, because of the employee's efforts and skills.

2. **Very Good Performance (VG):** Performance level fulfills established job expectations and in many ways exceeds it.
3. **Good Performance (G):** Performance level of the employee, is in alignment with the job expectations set by the organization, the employee is said to be doing a good job.
4. **Below Good Performance (BG):** The level of performance of employee meets a few job expectation, but not all the parameters are met. It is said that employee's performance is at a minimal level and improvements are required.
5. **Unsatisfactory Performance (U):** The level of performance is below a certain level, i.e. he/she fails to fulfill the established standards. It is said that the employee is not performing well at the level expected and requires high supervision and direction.

The primary focus of the performance management is towards the achievement of results. It highlights the difference between **being engaged and producing results**, meaning that, when one is busy, it does not mean that outcome is generated. It is often seen that employees seem to be very busy, but when it comes to performance, the results are against what is expected. So, performance management, is a must for every organization.

Performance Management Objectives

The major objectives of performance management system are enlisted below:

1. To emphasise on career planning and future growth opportunities for employees;
2. It is ensure to raise the efficiency and productivity of employees;
3. To encourage belongingness, team spirit and devotions among employees with the job;
4. To provide feedback about HR planning and potentialities to implement the planning facts;
5. To identify systematically the need and requirements of some learning and training aspects;
6. To promote better and high work culture in the organisation;
7. To focus more on systems approach to perform appraisals rather than to make any formalities;
8. To foster a positive relationship between managers and employees through a two way communication process;
9. To appreciate, recognise and to give reward and compensate employees for achievement of performance objectives successfully in a more objective, transparent and justified way.

Performance Management – Utility/Uses

- (1) It translates corporate goals into individual, team, departmental, and divisional goals,
- (2) It helps to clarify corporate goals,

- (3) It is a continuous and evolutionary process in which performance improves over time,
- (4) It is realized on consensus and cooperation, rather than control or coercion. It encourages the self-management of individual performance. It requires a management style that is open and encourages two-way communication between supervisors and subordinates,
- (5) It requires continuous feedback, and
- (6) It measures and assesses all performance against jointly agreed goals. It applies to all staff, and it is not primarily concerned with linking performance to financial rewards.

Performance Management – 7 Important Dimensions: Result and Output, Input, Time, Focus, Quality, Cost and Output

It involves thinking through various facts of performance, identifying critical dimensions of performance, planning, reviewing, developing, enhancing performance and related competencies.

Performance has many dimension:

Dimension # 1. Result and Output:

The most acceptable and measurable dimension of performance is result and output. It describes the conditions of inputs which included raw material, working conditions, process capabilities and talent of employees in the final form of product or service. It is necessary to plan all the performance activities in a scientific and systematic manner so that the desired result or output may be obtained.

Dimension # 2. Input:

This dimension deals with the activities to be accomplished by the employee. Performance can be achieved if the nature of inputs can be managed without mistake, because performance is a function of three sets of factors – ability, motivation and organizational support. If anyone among these three factor is less the performance is to be poor.

Employee Performance = Employee Competence + Employee Motivation + Organisational Support

Dimension # 3. Time:

Time is precious and very important dimension of performance. In the current scenario of world, the performance management is time bound otherwise the survival of organisation is not possible in the future. Performance of an employee in relation to a given role during particular period of time under the set of circumstances operating at that point of time. Therefore, time may become the target.

Dimension # 4. Focus:

Performance also has a focus dimension. For example in case of sales, profits and new areas. Focus means attention, not only on own activities but should also keep close watch on related activities.

Dimension # 5. Quality:

‘Quality is not destination but a journey’. Quality refers to doing the things right from the first time rather than making and correcting mistakes in order to achieve total customer satisfaction. It means quality is conformance to customer requirements, not goodness. Higher is the quality greater is the satisfaction of customers. It is the responsibility of each and every employee as well as management to build a quality standard which provides reasonable customer satisfaction at economical cost. Quality is the core dimension of performance management.

Dimension # 6. Cost:

The ultimate principle of purchasing is the low cost with best quality. Therefore cost effectiveness is another dimension of performance management. It implies the capacity of a business unit to produce a given commodity at a lower cost through more effective utilization of existing resources. It is the process of cost reduction by improving efficiency of operations.

Dimension # 7. Output:

Output relationships and analysis- It is relevant and essential to understand the input-output relationships and analysis. The purpose of input-output analysis is to find out the interdependencies and complexities of the economy in order to determine the conditions for maintaining balance between demand and supply. It describes the inputs required to produce the outputs of different sectors of the economy. It also involves the study of the exchange of goods and services among industries.

The performance appraisal is the part of performance management because the moment assessing process is started there is an appraisal taking place. Therefore, performance appraisal is static in nature while performance management is a dynamic process. Performance management with performance appraisal should lead to increased performance. The effort should be to make the performance management process more effective and productive using appraisal system.

Performance Management – Significance

The significance or importance of performance management in any organisation are given here:

1. Defining Objectives and Targets – It helps to define the objective, aims, targets, missions, vision, strategy and values of the organisation in order to enable them to achieve in proper way.

2. Opportunities for Learning – Performance management system provide some worthwhile opportunities to create and develop learning aspects and employment advancements. It may be identify the competencies as required for high performance.
3. Fulfil the Managerial Commitment – It is helpful to fulfil the required commitment and assurances as given by the management. These are in the form of work facility, job security, rewards and compensation promotion and career orientation programmes.
4. Fair and Justified Treatment with Workers – Within the purview of performance management, the employees can get fair, justified and optimum behaviour from the management. There is a pre-required condition is that there is an utmost need to make fair and justified treatment with employees.
5. Human Resource Planning – Performance appraisal information provides a valuable input for skills inventories and human resource planning (HRP). By providing information about the human resource strengths and weaknesses of the organisation, the performance appraisal system helps determine the promotability and potential of all employees. It, therefore, constitutes an important information base for developing succession plans, HR programmes and creating new positions in the organisation.
6. Recruitment and Selection – It is used to validate or evaluate the approaches and decisions relating to employees g recruitment and selection. It aims to determine the effectiveness of them on a particular job as well as on organisation as a whole.
7. Training and Development – With a comprehensive role of performance management, it can be assess the need and requirement of training and development. It can also identify performance deficiencies at the individual as well as organisation level.
8. Motivation – A good plan of performance management aims to motivate work appearances with better work culture. It motivate the employees to better work culture, as well as to develop their efficiency for more and better work performance.
9. Increase in Efficiency and Productivity – An optimum performance plan ensure to raise the efficiency and productivity of employees. A fair, optimistic and rational performance evaluation system motivate the employees to raise their better work performance.
10. Decrease in Turnover of Employees – The performance plan also aims to stabilise the existence of employees as well as to reduce the employees' turnover. It has the signifying role to reduce a wide range of turnover in the enterprise.
11. Career Planning and Development – The performance management helps in identifying employee potential and in planning future growth opportunities for the employee. Information about the strengths, weaknesses and potential of employees can be used to assist them in developing and implementing realistic career plans.

12. Compensation and Reward – A fair and objective performance appraisal system helps in making differential reward decisions, such that the most productive workers or teams are rewarded accordingly. When rewards and compensation are linked to performance, it reinforces the belief that pay raises should be linked to.

13. Better Employee Relations – Performance management aims to maintain better organisational cultural. Any dissatisfaction or grievances can be overcome or managed by using performance data. As such, by using optimum system of appraisal the better and most amicable relations with employees can be developed.

Performance Management – Implementation (With Steps)

In any organisation, the performance management has a wider scope and it includes the merit rating, assessment system, performance appraisal and online evaluation etc. Within HR scenario it serves both evaluative and the development purposes.

For specified design and implementation of a wide and broader form of performance management, certain steps to be followed in the development of its various systems are as follows:

1. To determine the objectives, plans, programmes and procedural to manage and organise the performance evaluation systems;
2. To determine the type or nature of jobs, skills, human resources and their performance that the enterprise will require to achieve its objectives;
3. To determine and develop the performance standards for each performance dimensions. These can specify the expected levels of performance. The standards are determined on the basis of quality and quantity as perceived by operating targets;
4. To collect and using performance management informations to make managerial and operating decisions, identify and overcome the problems and develop action plans for performance improvement;
5. Select and choose appropriate methods of performance appraisal which are based on employees' skill, performance status, job required, nature and types of job etc.;
6. Developing a plan to monitor and control over the employee performance. It is useful in tracking corrective actions at the right time; and
7. Determining who will conduct performance evaluation like manager, supervisor or any superior among the employees.

BUSINESS PROCESS OUTSOURCING (BPO)

Business process outsourcing (BPO) or Outsourcing is a method of subcontracting various business-related operations to third-party vendors.

Outsourcing, the name itself suggest its meaning, i.e. source from outside. It refers to a practice of contracting out the non-core and sometimes core activities of the organisation to a captive unit or a third party.

It is a process, in which business-houses handover their regular or peripheral work to the external organisation, who has got experience and expertise in the respective field and can perform it in a better way.

Outsourcing aims at freeing cash, time, manpower and other resources, for the activities, in which the organisation has got a competitive advantage. It covers both domestic as well as foreign contracting.

Need for Outsourcing

The key factors that cause the firms to rethink at business processes are the **best quality products at the minimum price, ever-demanding customers** and **emerging technologies**. These determinants are the reason for the advancement of outsourcing as an emerging business mode. And now, the company resort to outsourcing, not because of compulsion, but because of choice.

The main reasons for outsourcing are provided as under:



1. **Concentrating on primary activities:** Outsourcing of lower-level work or non-core activities can help the company to focus on the activities in which they have an edge over others.
2. **Quest for excellence:** With the help of outsourcing, the firm can gain excellence in two ways; first the firm itself perform those activities in which they are specialised, due to

limited focus and next they excel by outsourcing the rest of the activities to the external service provider who excels in performing them.

3. **Cost savings:** A savings in cost can be marked, by transferring work to the third party. This is due to economies of scale, as the third-party service provider performs the same work for many organisations. It may also be due to low labour cost, i.e. the labour is available at cheap rates in developing countries.
4. **Fewer regulations:** Companies generally outsource their businesses to developing countries, where the regulatory restrictions are low. It results in the low cost of operations and an increase in productivity.
5. **Flexibility:** As the non-core activities of the company are outsourced, the firm can stay stress-free and easily adapt to change.
6. **Helpful in economic development:** It encourages entrepreneurship, employment and export in the countries in which outsourcing is performed, i.e. host country, which results in the overall economic growth.

Outsourcing of activities like human resources, marketing, data processing, manufacturing, customer support, technical support, accounting, legal and the like, are in vogue so that the company can focus on the activities, at which they excel.

In this way, the company can reduce the average cost per unit and improve efficiency, by assigning their work to the outside organisation instead of performing it internally.

Top 3 Features

(i) Outsourcing Involves Contracting Out:

It involves contracting out an activity to an outside specialised agency which undertakes complete responsibility to handle it using its own manpower.

For example, many companies have started outsourcing sanitation and housekeeping functions, which were earlier performed by their in-house staff.

(ii) Generally Non-Core Business Activities are Outsourced:

Outsourcing is done of non-core activities, such as housekeeping or security in case of school.

For example, a school may tie-up with some computer training institute to impart computer education to its students.

(iii) Processes may be Outsourced to a Captive Unit or a Third Party:

A number of processes such as recruitment, selection, training etc., may also be outsourced to third party service providers.

In the outsourcing terminology, there are three types of third party service providers:

- (a) Captive BPO Unit – It provides services of a given kind to only one firm, e.g., General Electric (GE) is the largest captive BPO unit in India for providing certain kinds of services to the parent company in the US.
- (b) Horizontals – Persons/firms which provide their services to a wide base of clients, cutting across industries.
- (c) Verticals – They may specialise in one or two industries and scale up to doing a number of processes from non-core to core.

Business Process Outsourcing (BPO) – Types of Outsourcing

Type # 1. Transactional:

High value in focusing internal information system group on value added applications development and integration.

Hire more information system people or contractors.

- (i) Outsourcing all or portions of the new applications.
- (ii) Outsource legacy applications maintenance.

Type # 2. Best of breed:

- (i) In the past, there have been many “Mega-deal” or “Soup-to-nuts” sourcing contracts.
- (ii) Best-of-breed outsourcing contracts with the consortium of providers.

Type # 3. Offshore:

A firm based in the USA can choose to outsource its non-core activities to a group based in America or it can give this work to another group in an offshore location, be it India or China. Offshore outsourcing leads to huge cost reductions, with most firms claiming a savings of between forty to fifty percent. But cost is not the only reason.

Type # 4. Shred Services:

This involves creating an “in sourcing” unit that:

- (i) Operates as a separate business unit.
- (ii) Charges for services at full cost.
- (iii) Competes and works with other ESPs.

Type # 5. Business Process:

- (i) Outsourcing an entire business process that is typically non-core.
- (ii) This sounds new but in actual fact has been common for years e.g. payroll processing.
- (iii) Have been relabeled and also moved more towards core process e.g. Billing, Procurement etc.

Type # 6. E-Business:

- (i) This is basically the same as business outsourcing, but is carried out electronically.
- (ii) Outsourcing internet/interact/extranet i.e. web related matter as opposed to other forms of outsourcing.

The major differences between e-business outsourcing and traditional IT outsourcing is that there is no need to buy machines from the client, no personnel has to be moved, and no software licenses have to be transferred.

Advantages:

Typically company considers outsourcing to provide a number of benefits.

Some of the important benefits are given below:**i. Cost Saving:**

Cost is the most important factor in the outsourcing decision. The outsourcer should compare the cost of outsourcing a function vis-a-vis performing it by itself. The cost as a result of outsourcing should be reasonable one. Cost saving helps business enterprises to gain competitiveness.

ii. Concentration on Core Competency:

There is a change in the business strategy of performing all business activities in house. The present day competitive and high technology environment requires the customer orientation. This requires consistent efforts to improve the productivity of assets of the organization. Outsourcing helps the management to concentrate on core issues and leave non-core activities to outsiders.

iii. Speed to Market:

Outsourcing alone generates savings by increasing cash-flow, accelerating growth in market share and concentration of resources of the company towards product/service design and customer service.

iv. Cost Deferral or Other Financial Engineering:

Outsourcing can include some financial engineering, which shifts costs from one accounting period to another. Many outsourcing deals are driven by a desire to avoid having to purchase an upgrade in computers, networks or other capital equipment at the expiration of either useful life or the contractual right to use old infrastructures.

v. Diversion of Problem Area:

The organization may go for outsourcing simply for the reason that it is prepared to pay a bit more to divert itself from problem area and hand over the problem to somebody else.

vi. Human Resource Management:

Traditional activities relating to human resource management such as recruitment, training, supervision, development and annual appraisal etc. can be minimized through outsourcing. Outsourcing of desired personnel for desired activities for the required duration will improve the corporate profitability and efficiency.

vii. Technological Change:

Outsourcing helps improving the existing technology to achieve the objectives for better quality products at minimum cost. Access to latest technological or process sophistication becomes easy and assured in outsourcing. Outsourcing provides a firm to avail the opportunity of using world class external sources.

viii. Flexibility:

Outsourcing will bring in flexibility in scope of services, volumes of outputs pricing, geographical extension etc. and the company make least capital investment.

ix. Higher Stock Market Price:

Since stock market analysts measure the revenue per employee, the reduced head count due to outsourcing result in increased per employee revenue and it will lead to increase in stock price.

x. Shared Risk:

With the changing environment the risk in investment is high. Outsourcing is more dynamic to accept change to produce maximum returns on investment. In other words, there are no compulsion in investment in the own back dated production to receive under rated return for the investors.

Drawbacks:

The outsourcing concept is criticized for the following drawbacks:

- (a) Reduction in Control – The organization may lose some control and become dependent on the suppliers.
- (b) Staff Issues – Outsourcing may lead to low employee morale and reduce performance of other staff.
- (c) Exploitation – All the labour is cheap in the developing countries. Many organizations shift their manufacturing units to such countries from the advanced countries.
- (d) Information Security – Outsourcing may lose the information security which may lead the competitors to a front-foot.

BUSINESS PROCESS REENGINEERING

Definition:

The **Business Process Reengineering** or **BPR** is the analysis and redesign of core business processes to achieve the substantial improvements in its performance, productivity, and quality. The business process refers to the set of interlinked tasks or activities performed to achieve a specified outcome.

The **Business Process Reengineering** is the examination and redesign of business process to achieve the dramatic improvement in its performance, productivity, and quality.

Simply, the business process reengineering means to change the way an individual performs the work such that better results are accomplished. The purpose of business process reengineering is to redesign the workflows in order to dramatically improve the customer service, achieve higher levels of efficiency, cut operational costs and become a world-class competitor.

The business process is required to be reengineered because of the following reasons:

- The processes the company is using might have become outdated or hold no relevance in the current market scenario.
- Often, the sub-divisions in the organization aim at improving their respective division performance and overlook the resultant effects on the other departments. This might lead to the underperformance of the firm as a whole.
- Due to the departmentalization, each employee focuses on the performance of his respective department and may overlook the critical issues emerging in other areas of the firm, and therefore, the need for re-engineering arises so that the role of the employees could be broadened and shall be made more responsible towards the firm.

- The existing business process could be lengthy, time-consuming, costly, obsolete, therefore, is required to be redesigned to match it with the current business requirements.
- The technology keeps on updating and in order to catch up with it, reengineering is a must.

Thus, the business process reengineering focuses on obtaining the quantum gains in terms of cost, time, output, quality and responsiveness towards customers. Also, it emphasizes on simplifying and streamlining the business process by eliminating the unnecessary or time-consuming business activities and speeding up the workflow by making the use of high-tech systems.

The Business process reengineering comprises of following steps:



1. **Define Objectives and Framework:** First of all, the objective of re-engineering must be defined in the quantitative and qualitative terms. The objectives are the end results that the management desires after the reengineering. Once the objectives are defined, the need for change should be well communicated to the employees because, the success of BPR depends on the readiness of the employees to accept the change.
2. **Identify Customer Needs:** While, redesigning the business process the needs of the customers must be taken into prior consideration. The process shall be redesigned in such a way that it clearly provides the added value to the customer. One must take the following parameters into the consideration:
 - Type of Customer and customer groups.
 - Customer's expected utilities in product and services
 - Customer requirements, buying habits and consuming tendencies.

- Customer problems and expectations about the product or service.
3. **Study the Existing Process:** Before deciding on the changes to be made in the existing business process, one must analyze it carefully. The existing process provides a base for the new process and hence “what” and “why” of the new process can be well designed by studying the right and wrongs of the existing business plan.
 4. **Formulate a Redesign Business Plan:** Once the existing business process is studied thoroughly, the required changes are written down on a piece of paper and is converted into an ideal re-design process. Here, all the changes are chalked down, and the best among all the alternatives is selected.
 5. **Implement the Redesign:** Finally, the changes are implemented into the redesign plan to achieve the dramatic improvements. It is the responsibility of both the management and the designer to operationalise the new process and gain the support of all.

Thus, the business process reengineering is collection of interrelated tasks or activities designed to accomplish the specified outcome.

Characteristics of BPR

Re-engineering varies from one business to another. However, Michael Hammer and James Champy observed some common characters in the re-engineering process of their client firms.

1. **Several jobs are combined into one:** The feature of re-engineered process is absence of an assembly line. The formerly distinct tasks/ jobs are combined and compressed into one. The jobs are combined mostly based on the needs and preference of the customer.
2. When **jobs are integrated**, the chances of errors are reduced, eliminates misunderstandings, delays and reworking are minimized.
3. **Workers make decisions** as they are required to do so. Jobs are combined, both horizontally but also vertically. In other words jobs are compressed based on job enlargement and job enrichment. Vertical integration incorporates the tasks of decision making in the top ladders of the hierarchy. In addition, the workers in the re-engineering are empowered. The empowered workers are motivated and self determined to make decision. Decision-making is part of the work of the workers.
4. **Compressing the work both horizontally and vertically** reduces delays, overhead costs and betterment of response and satisfaction of customers.
5. The **steps in the process are performed in a natural order** straight-line sequence is avoided in the re-engineering. Activities are performed not in artificial order but in natural order. This process is termed as ‘De-linearising’ which allows performing of many jobs simultaneously. This process reduces the process time and thereby delay.

Advantages of BPR

- It gives an appropriate focus to business as it revolves around customer needs.
- BPR helps in building a strategic view of operational procedures by making radical inquiries about how processes are improved and how things could be done.
- It eliminates unnecessary activities and thereby helps in reducing organizational complexity.
- It coordinates and integrates several functions immediately.
- Provides improved viability and adequacy to an organization by eliminating the delay and unessential phases of operations and management.
- Reduced the number of checks/controls and reconciliation processes.
- It helps overcome short-sighted approaches that usually emerge from excessive concentration on functional boundaries.

Dis-Advantages of BPR

- It doesn't suit every business need as it depends on factors like size and availability of resources. It usually benefits large organizations.
- In some cases, the efficiency of one department was improved at the expense of the overall process.
- This BPR approach does not provide an immediate resolution. It concentrates significantly upon long haul income collaborations of a business which not only takes some effort to take shape but are hard to gauge as well
- It might require a substantial investment in IT along with proper planning, fantastic teamwork, and exceptional implementation.
- It can replace humans when it comes to getting the job done error-free hence posing as a real threat to jobs.

BENCHMARKING

Definition: Benchmarking, is a tool of strategic management that allows the organization to set goals and measure productivity, on the basis of the best industry practices. It is a practice in which quality level is used as a point of reference to evaluate things by making a comparison.

The process helps in comparing and gauging the processes, programs, strategies and performance metrics with the standard measurements or to other similar companies. It is concerned with the analysis of three major dimensions:

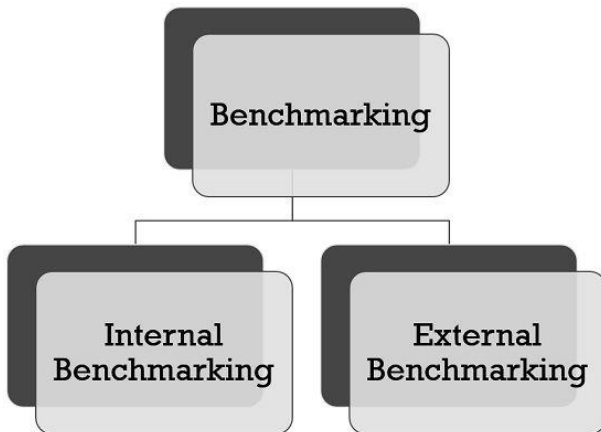
- Quality
- Time
- Cost

It is a useful technique for enhancing the organisation's performance by identifying and implementing the finest process and practices, for achieving them.

The process involves repeatedly evaluating the aspects of performance with the similar measurements of its peers, identifying the gaps, discovering new methods for filling gaps and also for excelling the condition, so that the gaps might prove positive for the organisation.

Types of Benchmarking

There are two types of Benchmarking, discussed as under:



- **Internal Benchmarking:** When measurement and comparison of key operations between teams, groups and individuals are made within the organization, the benchmarking is said to be internal.
- **External Benchmarking:** When measurement and comparison of key operations are made with the competitors, then, it is called as external benchmarking.

Further, the process is sub-classified as:

1. Process Benchmarking
2. Performance Benchmarking
3. Strategic Benchmarking

Application of Benchmarking

The process entails looking outside the organization to study what others do to achieve their performance level and also the processes they use. So, the approach helps in determining the processes behind the exceptional performance. It can be applied in the following areas:

- Human Resource Management
- Product Development
- Product Distribution
- Maintenance Operations
- Plant utilisation levels
- Customer Services

Process of Benchmarking

- Identifying the need for Benchmarking
- Understanding the existing process and practices
- Obtaining support and approval from the top management.
- Identifying best practices.
- Making a comparison between the firm's processes and performance with those of rivals.
- Preparation of report, regarding the differences in standard and actual results.
- Implementing steps necessary for filling gaps in performance.
- Evaluation and review

Benchmarking does not provide a solution to all the problems rather it analyses the situations and processes and helps in improving the performance.

It is a continuous improvement process. Hence, the benchmarking exercises are applied appropriately and performed regularly, so as to gain competitive advantage and also refining performance in the major areas of business.

In this process, a firm's major operations are measured and compared with the rivals and acknowledged leaders of the industry.

Advantages of Benchmarking

- The main advantage is that it brings the focus to the areas which require special attention. This is how it sets the base for improvement. It forces the organization to adopt change, the other name of which is progress.
- While conducting this exercise, a company discovers various new ideas and way of working. It gets a feel of the strategy adopted by its competitors.

- Improvement based on the past performance of the company is not the right measure. Comparison with competitors is required because survival has to be fought with these rivals. This is what exactly benchmarking provides.

Limitations of Benchmarking

- Benchmarking simply helps you to spot areas which need improvement. It does not contribute to solving the issues at hand. Benchmarking can just be the first of many steps to improve a company's performance.
- It simply compares the numbers. It does not take into account the micro and macro factors that led to your competitor or industry leader to succeed or fail.

BALANCED SCORECARD (BSC)

The term balanced scorecard (BSC) refers to a strategic management performance metric used to identify and improve various internal business functions and their resulting external outcomes.

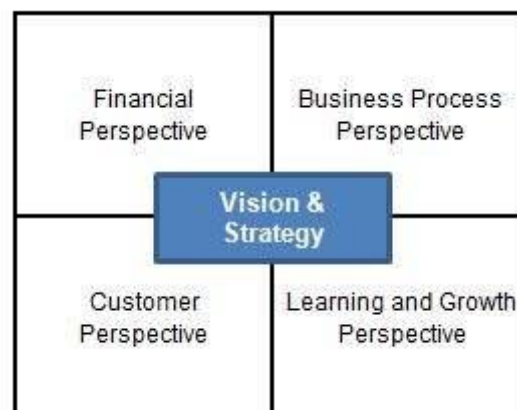
Used to measure and provide feedback to organizations, balanced scorecards are common among companies in the United States, the United Kingdom, Japan, and Europe.

Data collection is crucial to providing quantitative results as managers and executives gather and interpret the information.

Company personnel can use this information to make better decisions for the future of their organizations.

The Basics of Balanced Scorecard

Following is the simplest illustration of the concept of balanced scorecard. The four boxes represent the main areas of consideration under balanced scorecard. All four main areas of consideration are bound by the business organization's vision and strategy.



The balanced scorecard is divided into four main areas and a successful organization is one that finds the right balance between these areas.

Each area (perspective) represents a different aspect of the business organization in order to operate at optimal capacity.

- **Financial Perspective** - This consists of costs or measurement involved, in terms of rate of return on capital (ROI) employed and operating income of the organization.
- **Customer Perspective** - Measures the level of customer satisfaction, customer retention and market share held by the organization.
- **Business Process Perspective** - This consists of measures such as cost and quality related to the business processes.
- **Learning and Growth Perspective** - Consists of measures such as employee satisfaction, employee retention and knowledge management.

The four perspectives are interrelated. Therefore, they do not function independently. In real-world situations, organizations need one or more perspectives combined together to achieve its business objectives.

For example, Customer Perspective is needed to determine the Financial Perspective, which in turn can be used to improve the Learning and Growth Perspective.

FEATURES OF BALANCED SCORECARD

From the above diagram, you will see that there are four perspectives on a balanced scorecard. Each of these four perspectives should be considered with respect to the following factors.

When it comes to defining and assessing the four perspectives, following factors are used:

- **Objectives** - This reflects the organization's objectives such as profitability or market share.
- **Measures** - Based on the objectives, measures will be put in place to gauge the progress of achieving objectives.
- **Targets** - This could be department based or overall as a company. There will be specific targets that have been set to achieve the measures.
- **Initiatives** - These could be classified as actions that are taken to meet the objectives.

Characteristics of the Balanced Scorecard Model (BSC)

Information is collected and analyzed from four aspects of a business:

1. **Learning and growth** are analyzed through the investigation of training and knowledge resources. This first leg handles how well information is captured and how effectively employees use that information to convert it to a competitive advantage within the industry.

2. **Business processes** are evaluated by investigating how well products are manufactured. Operational management is analyzed to track any gaps, delays, bottlenecks, shortages, or waste.
3. **Customer perspectives** are collected to gauge customer satisfaction with the quality, price, and availability of products or services. Customers provide feedback about their satisfaction with current products.
4. **Financial data**, such as sales, expenditures, and income are used to understand financial performance. These financial metrics may include dollar amounts, financial ratios, budget variances, or income targets.¹

These four legs encompass the vision and strategy of an organization and require active management to analyze the data collected.

The Need for a Balanced Scorecard

Following are some of the points that describe the need for implementing a balanced scorecard:

- Increases the focus on the business strategy and its outcomes.
- Leads to improvised organizational performance through measurements.
- Align the workforce to meet the organization's strategy on a day-to-day basis.
- Targeting the key determinants or drivers of future performance.
- Improves the level of communication in relation to the organization's strategy and vision.
- Helps to prioritize projects according to the timeframe and other priority factors.

Advantages of BSC:

Following are some advantages of BSC:

- (i) BSC adopts a balanced and comprehensive approach for judging and controlling an organization's performance; by setting objectives and performance measures in four key perspective viz. financial, customer, business and internal processes and learning and growth.
- (ii) BSC facilitates communication and understanding of business goals and strategies, at all levels of an organization.
- (iii) BSC brings organization's strategy and vision, to the centre of management focus; so that management may never deviate from these.
- (iv) BSC integrates financial and non-financial goals and performance measures into a single system – a thing which traditional controlling techniques never consider.

Limitations of BSC:

BSC approach to controlling suffers from serious limitations, some of which are as follows:

- (i) BSC bases its approach to analysis around four perspectives (viz. financial, customer, business and production processes and learning and growth) only. In fact, there may be many more perspectives more important than these e.g. managerial development perspectives, social responsibility perspective and so on. As such, the so called Balanced Score Card really turns into an imbalanced and imperfect score card.
- (ii) BSC is a vague concept and approach, to controlling an organization's success; as there are neither any set of standard goals nor any set of standard performance measures, for each of the four perspectives, which form the core of BSC.
- (iii) BSC just considers organizational performance from four perspectives. It suggests nothing about what should be done to better performance in each of these perspectives. Its job, it seems, is just counting casualties, after the battle is over.